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HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

FACULTY OF COMPUTER SCIENCE AND ENGINEERING



PROJECT REPORT

Software Engineering

SUBMISSION #2

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1 UI Design

1.1 Authentication and Hub & Identity

1.1.1 UC-A01 — Sign In and Authenticate

Students and Operators enter their account credentials to access the portal. The interface presents the sign-in options and directs the user into the relevant portal after authentication.

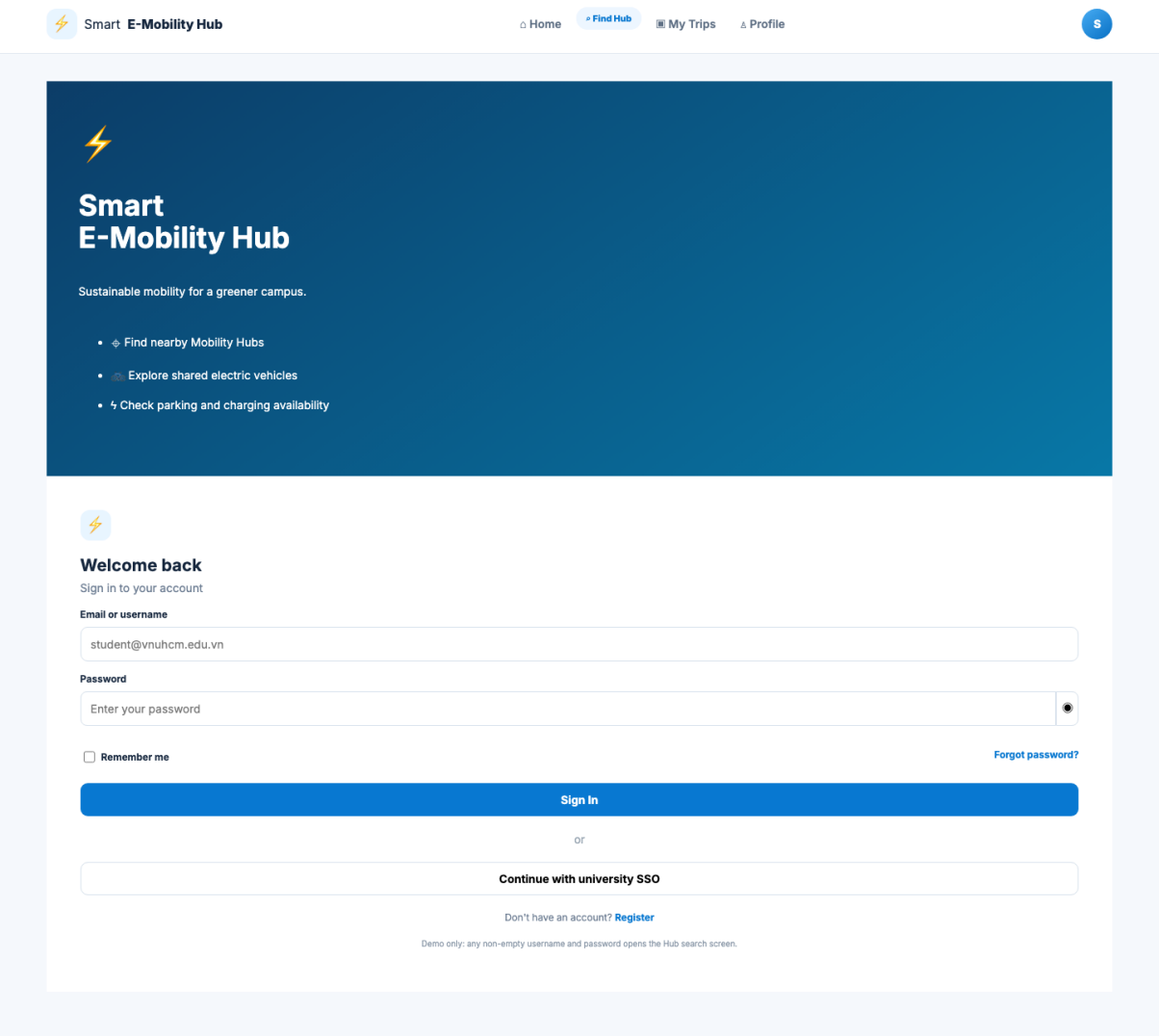


Figure 1: Student Portal sign-in screen (UC-A01).

1.1.2 UC-H01 — Manage Personal Profile

A signed-in student reviews identity information and edits permitted contact fields. The profile screen keeps account identity and role read-only while providing save and cancel actions.

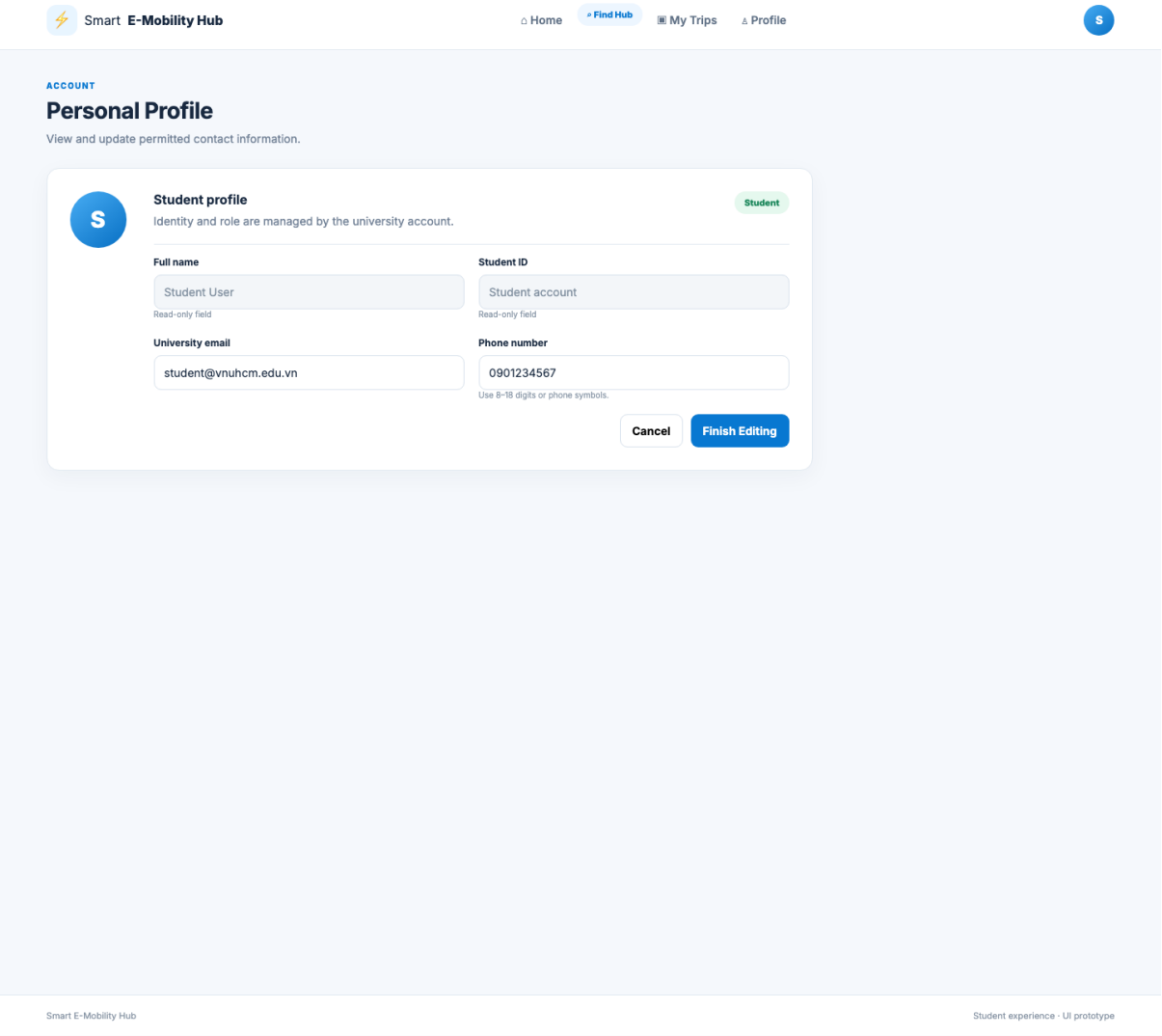


Figure 2: Personal profile with permitted contact fields (UC-H01).

1.1.3 UC-H02 — Find a Mobility Hub

Students search near a location and refine results by services and distance. The screen combines search controls, nearby Hub results, and a map view for selecting a Hub.

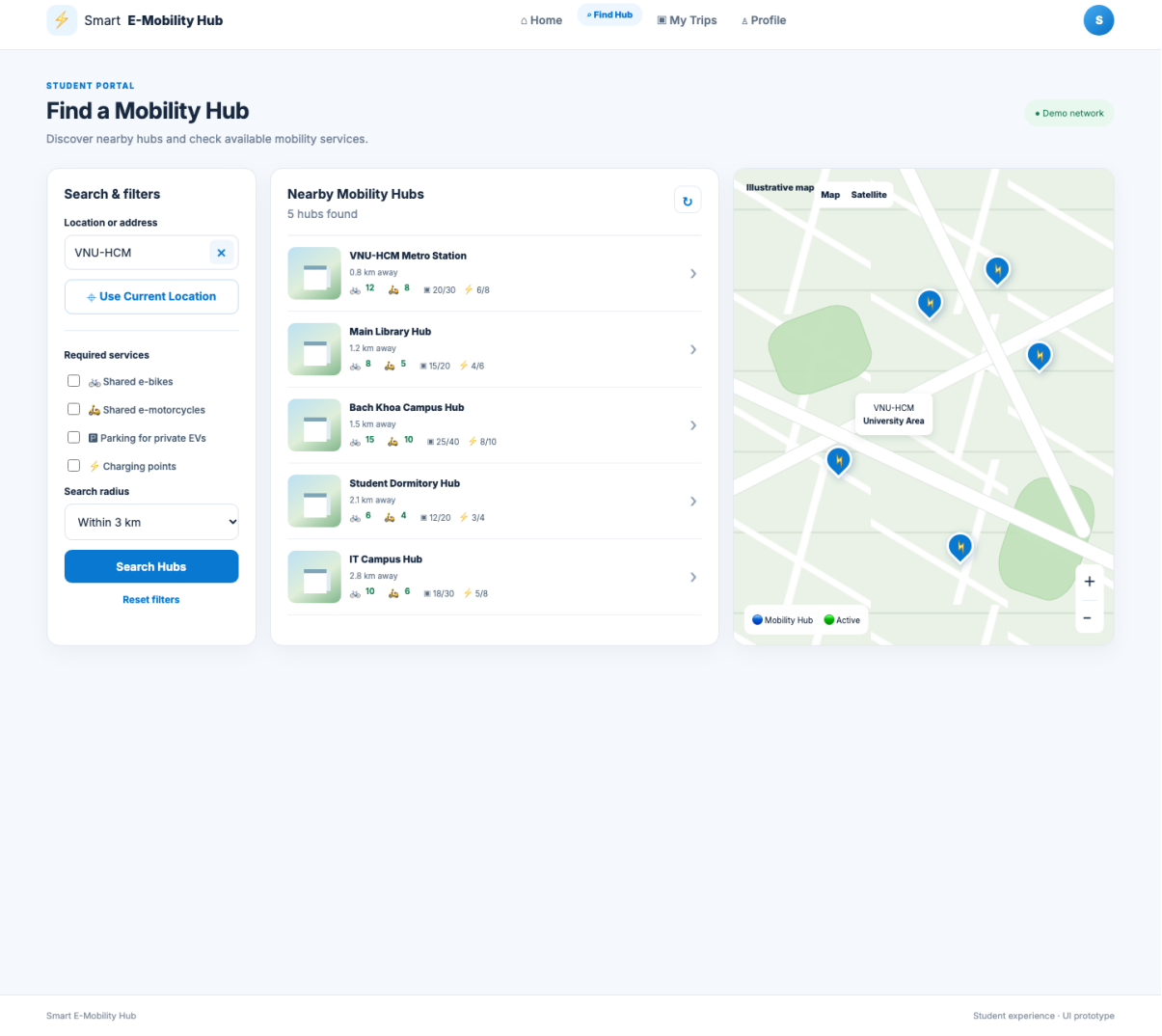


Figure 3: Mobility Hub search, service filters, results, and illustrative map (UC-H02).

1.1.4 UC-H03 — View Hub Status

Students inspect a selected Hub's current resource availability and operating status. The screen summarizes shared vehicles, parking, charging points, battery levels, and data freshness.

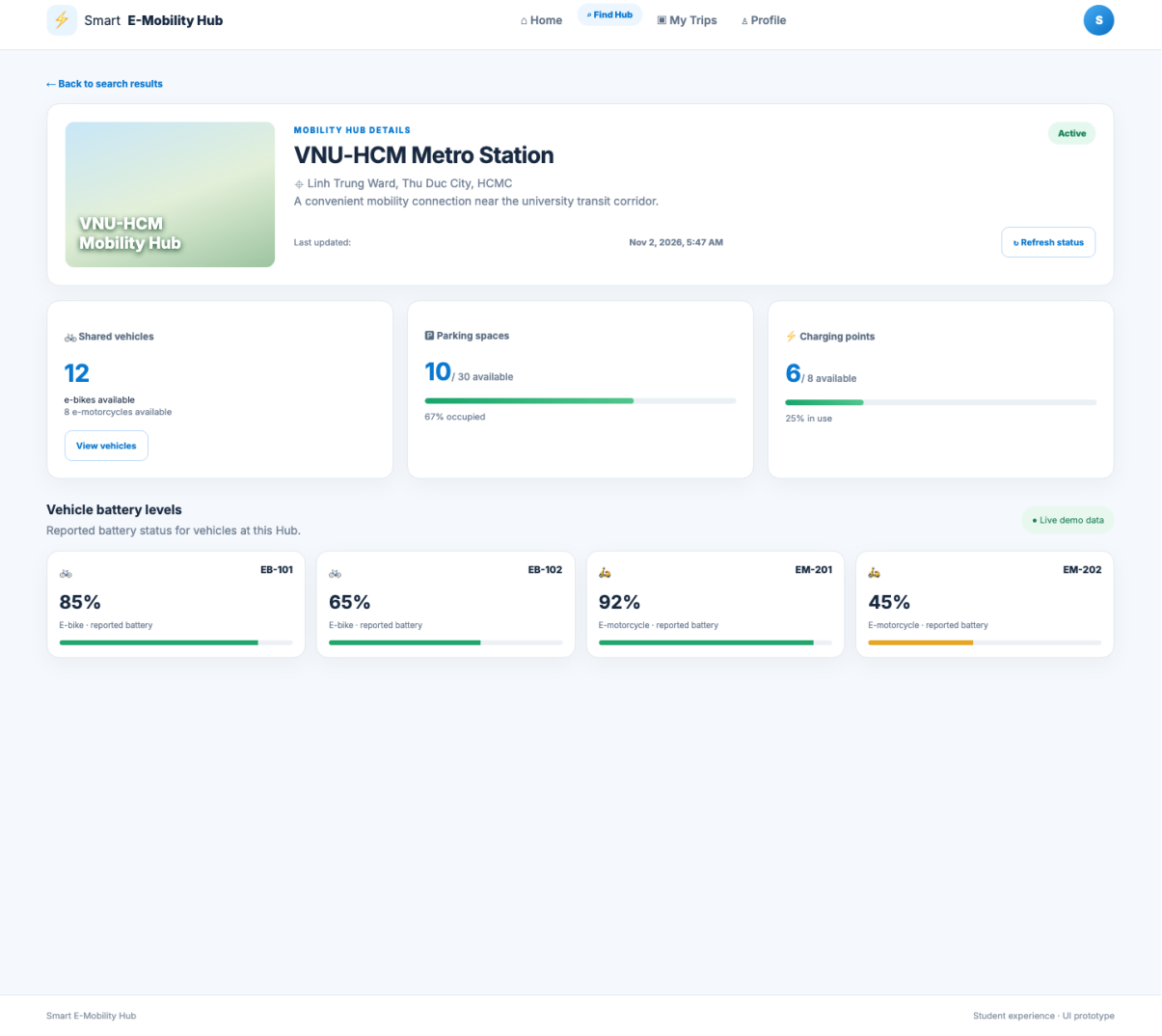


Figure 4: Mobility Hub status with vehicle, parking, charging, and battery information (UC-H03).

1.2 Vehicle Rental

1.2.1 UC-R01 — Browse Shared Vehicles

Students browse shared vehicles and narrow the list by Hub, vehicle type, and battery level. The screen helps them choose a vehicle that fits the intended trip.

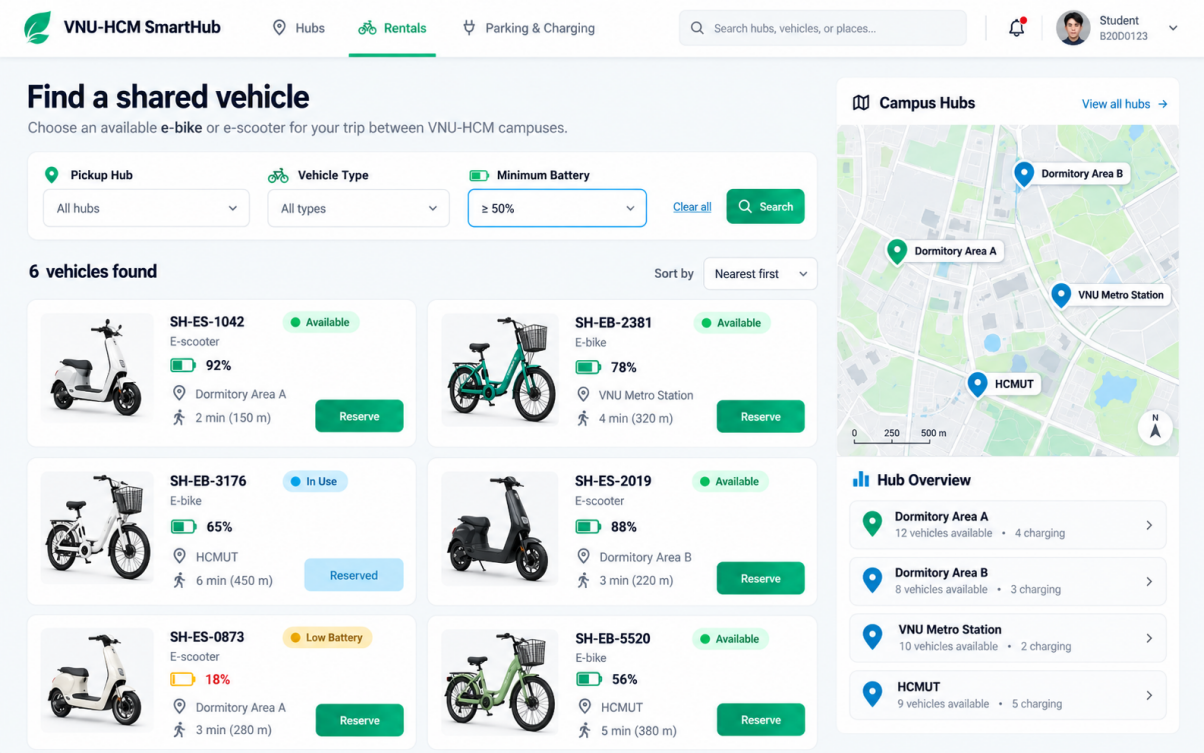


Figure 5: Shared vehicle list with Hub, type, and battery filters (UC-R01).

1.2.2 UC-R03 and UC-R04 — Reserve and Pick Up a Shared Vehicle

The student reserves an available vehicle and receives a time-limited pickup window. The reservation screen keeps the countdown and QR or Bluetooth unlock instructions visible before pickup.

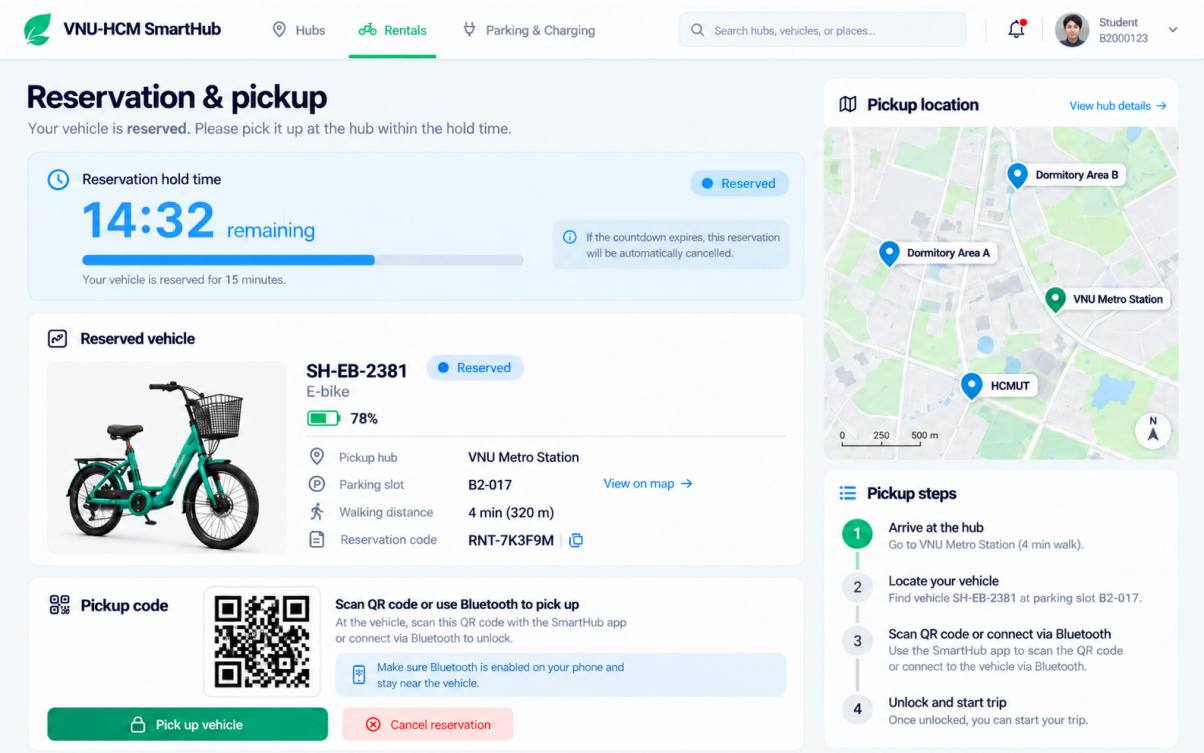


Figure 6: Vehicle reservation and pickup instructions with hold countdown (UC-R03, UC-R04).

1.2.3 UC-R04 and UC-R05 — Use and Return a Shared Vehicle

The student reviews an active trip and completes the return at an eligible Hub. The trip screen presents journey details and return actions needed to close the rental.

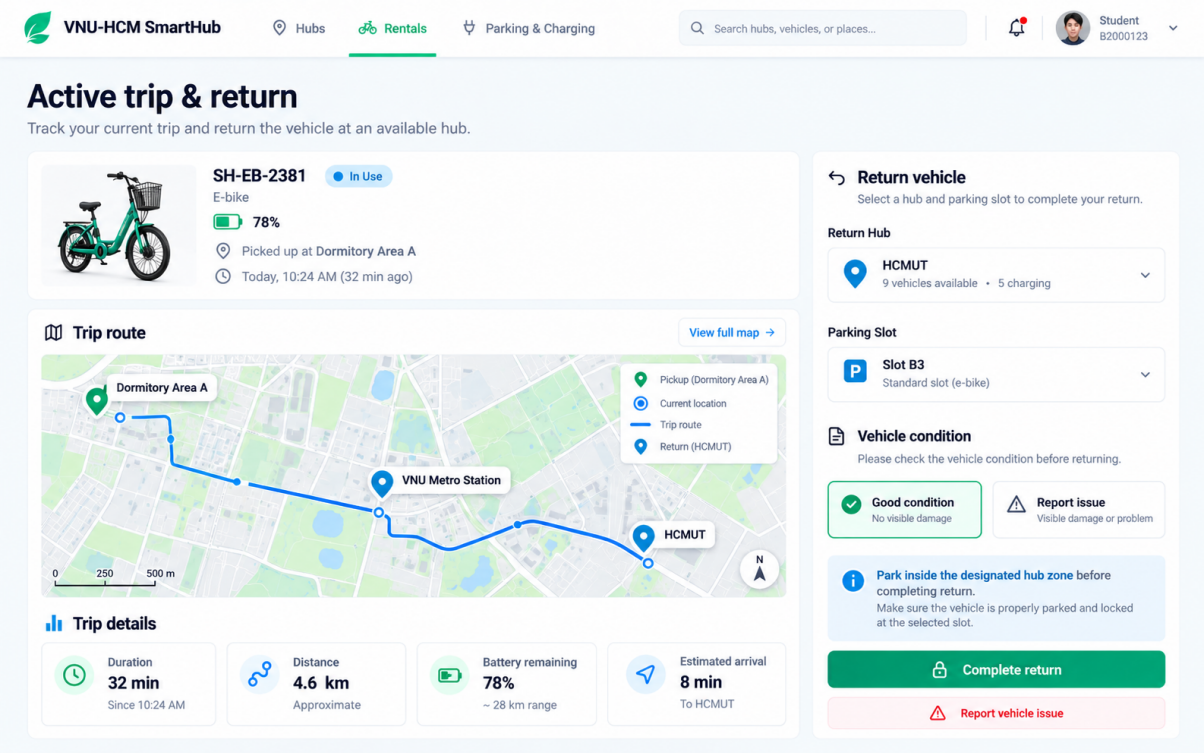


Figure 7: Active rental trip and vehicle return screen (UC-R04, UC-R05).

1.3 Parking and Private EV

1.3.1 UC-P01 — Reserve a Parking Space

A student selects a Hub and a requested parking interval. The interface presents availability across the selected time blocks before the reservation is submitted.

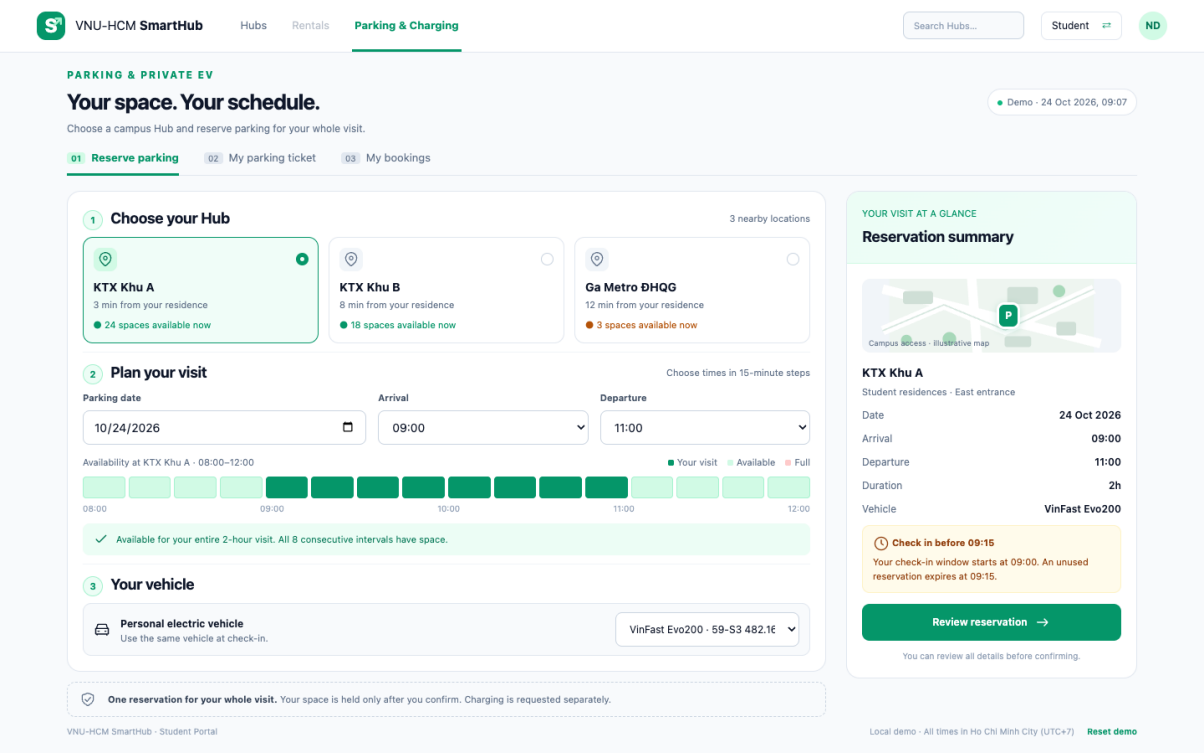


Figure 8: Parking reservation by date and continuous time blocks (UC-P01).

1.3.2 UC-P02 and UC-P06 — Submit Charging Request and Check In

A private EV owner submits a charging request and uses the electronic parking ticket at arrival. The ticket screen provides the QR code and booking information needed for Hub check-in.

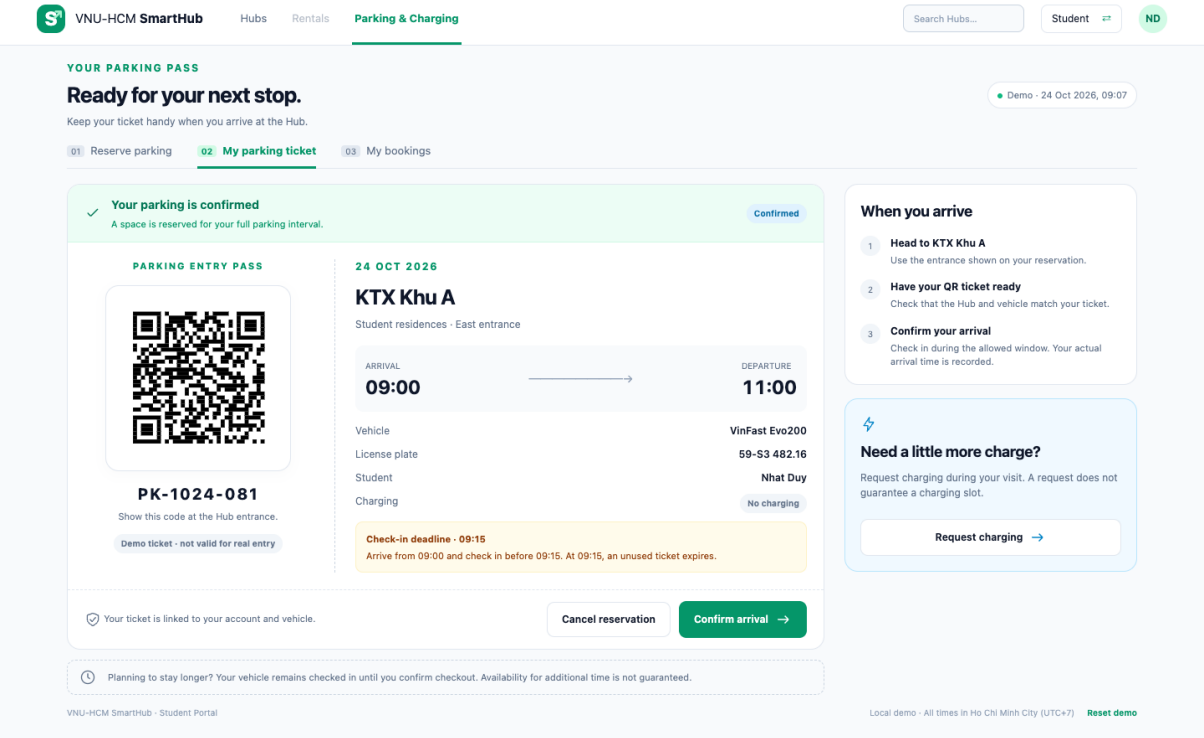


Figure 9: Electronic parking ticket and QR check-in code (UC-P02, UC-P06).

1.3.3 UC-P03, UC-P04, and UC-P07 — View and Manage Parking Bookings

Students review current and upcoming parking or charging bookings. The screen exposes the relevant cancellation, check-in, and checkout actions for each booking state.

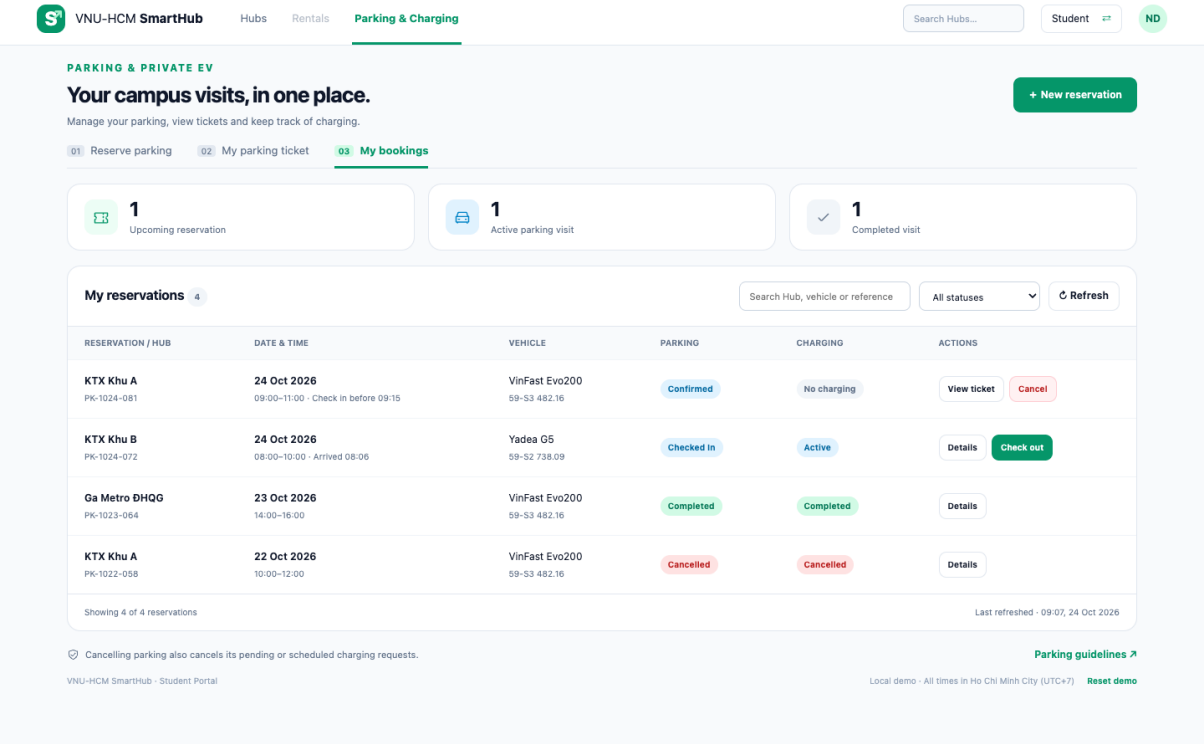


Figure 10: Parking and charging bookings with booking-management actions (UC-P03, UC-P04, UC-P07).

1.4 Fleet Monitoring and Incident Handling

1.4.1 UC-F01 and UC-F02 — Monitor the Hub Network and Inspect Vehicles

Operators review network-wide Hub capacity, resource availability, vehicle distribution, and operational alerts. The overview supports identifying a Hub or vehicle that needs closer inspection.

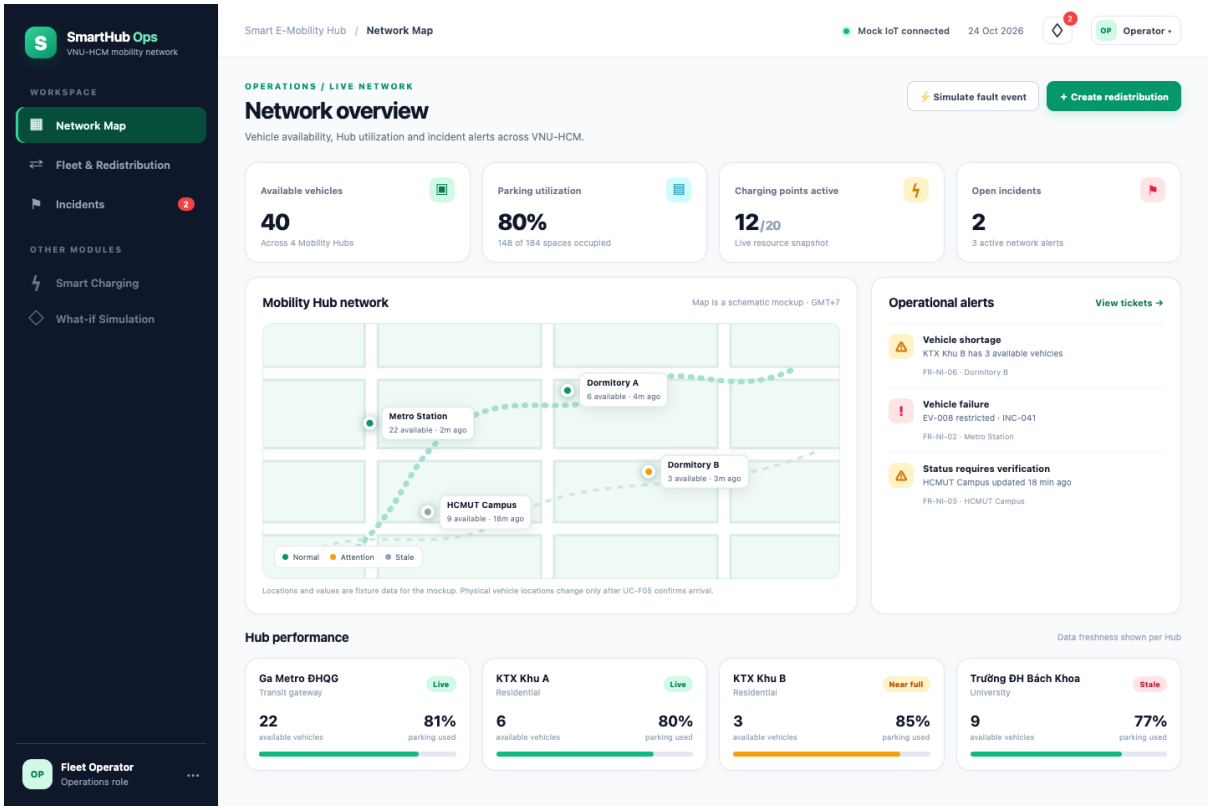


Figure 11: Operator network overview with capacity, availability, and freshness indicators (UC-F01, UC-F02).

1.4.2 UC-F04 and UC-F05 — Coordinate and Complete Vehicle Redistribution

Operators select a feasible vehicle transfer and record confirmed arrivals at the destination Hub. The screen distinguishes assignment planning from physical completion of the transfer.

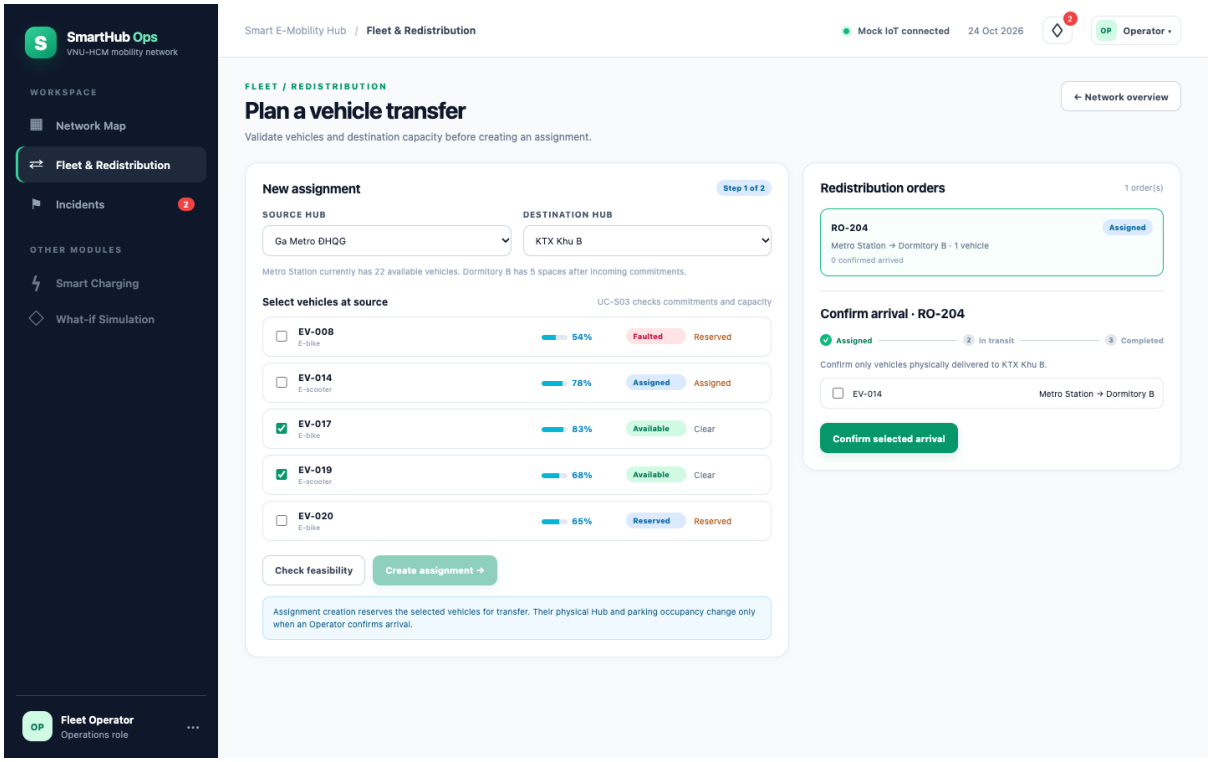


Figure 12: Fleet redistribution planning and arrival confirmation (UC-F04, UC-F05).

1.4.3 UC-I02 — Handle a Vehicle Failure

Operators investigate a reported vehicle failure, review affected commitments, and record corrective actions. Vehicle restoration is available after the incident and repair information has been reviewed.

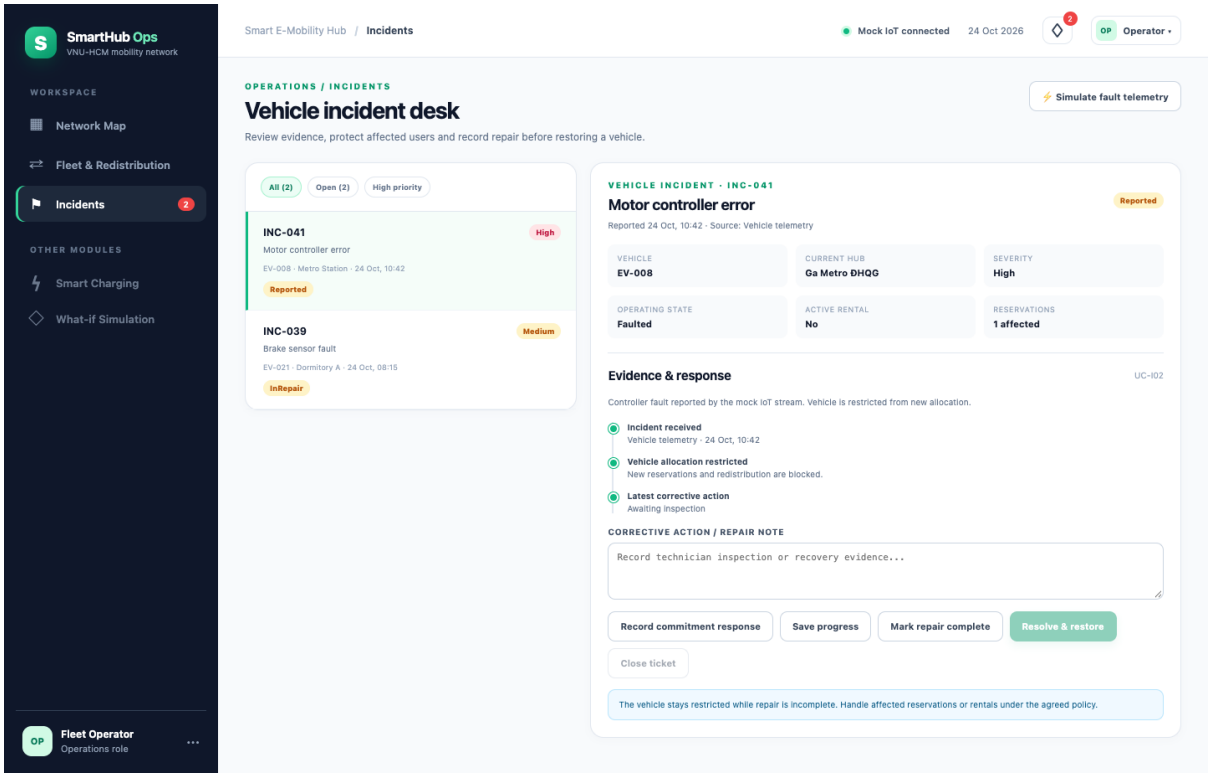


Figure 13: Incident desk with evidence, affected commitments, and repair progress (UC-I02).

1.5 Smart Charging Scheduling

1.5.1 UC-C01 — View the Charging Queue and Schedule

Operators review queued requests alongside charging-point occupancy and scheduled intervals. The queue presents requests in FCFS order so the next charging work is easy to identify.

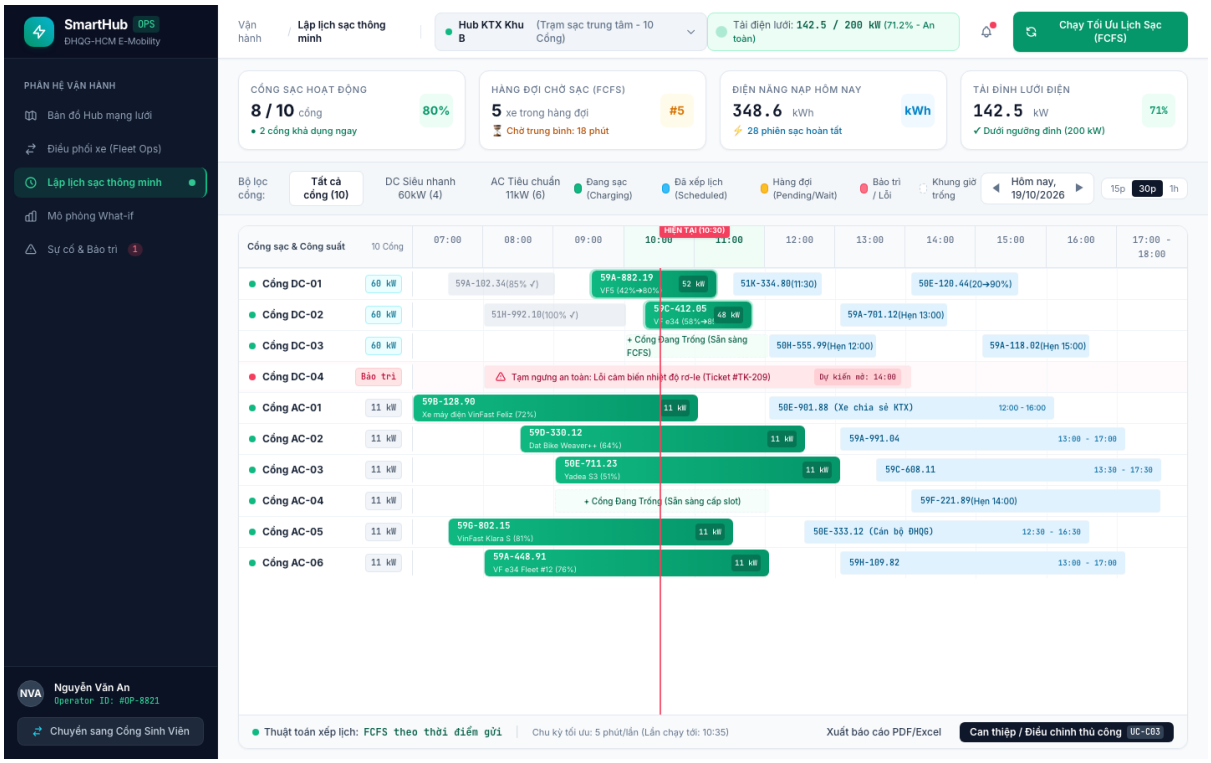


Figure 14: Charging-point schedule and Gantt view (UC-C01).

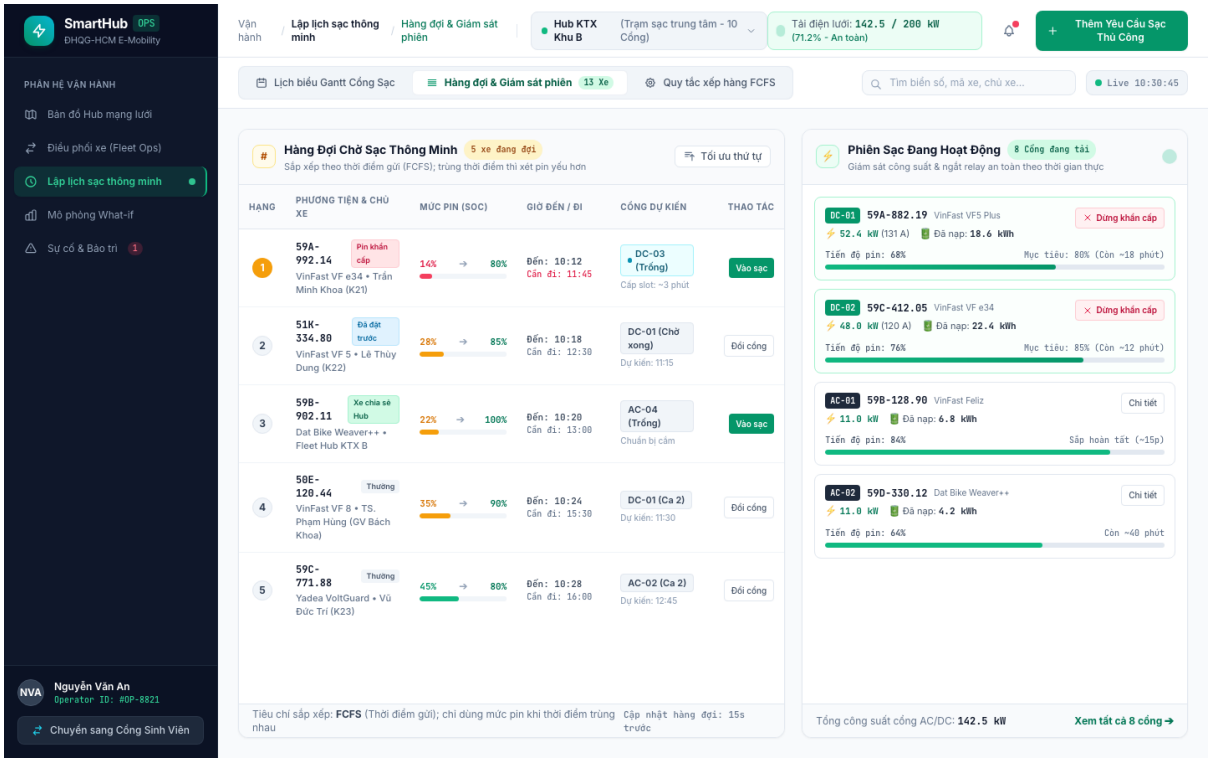


Figure 15: FCFS charging queue and active charging sessions (UC-C01).

1.5.2 UC-C03 — Adjust a Charging Schedule

Operators change a request's charging point or time interval through a manual adjustment form. The interface shows the selected request and supports checking the proposed change before saving.

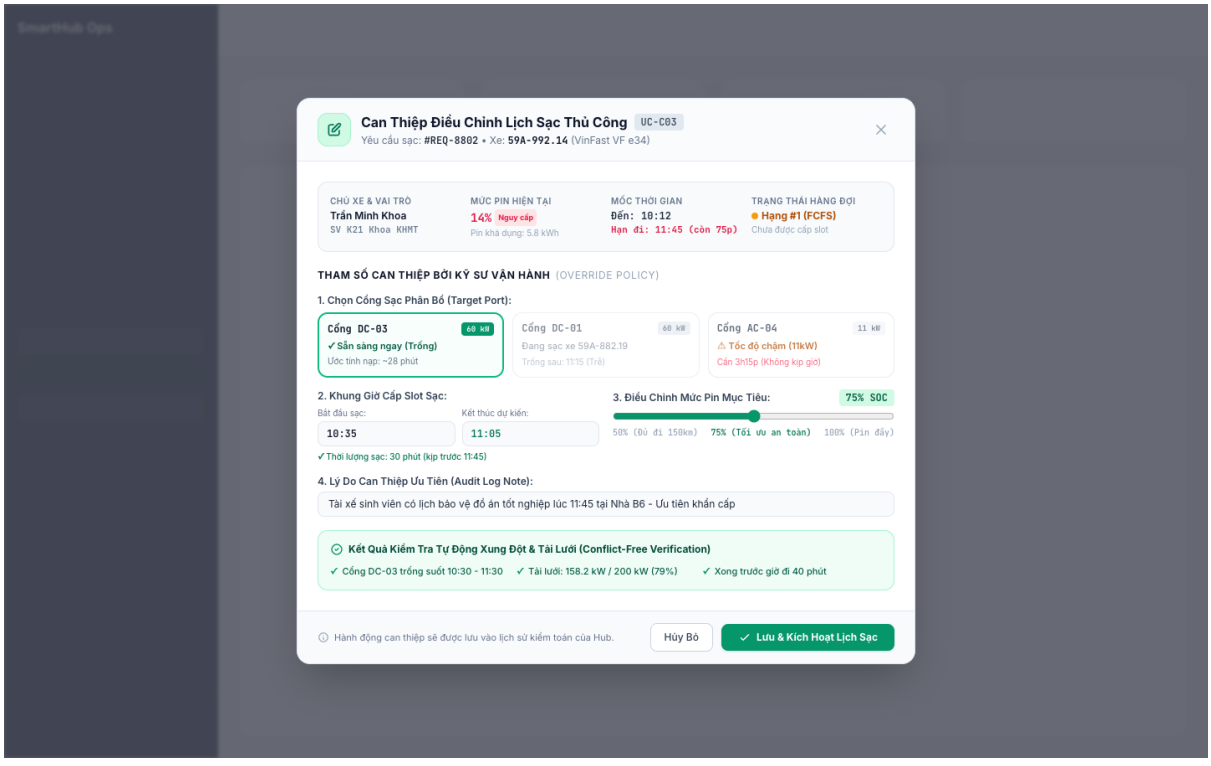


Figure 16: Manual charging schedule adjustment form (UC-C03).

1.6 What-if Simulation

1.6.1 UC-W01 — Configure a What-if Scenario

Operators define hypothetical conditions, affected resources, and a simulation period. The configuration screen organizes scenario parameters and presents them for validation before execution.

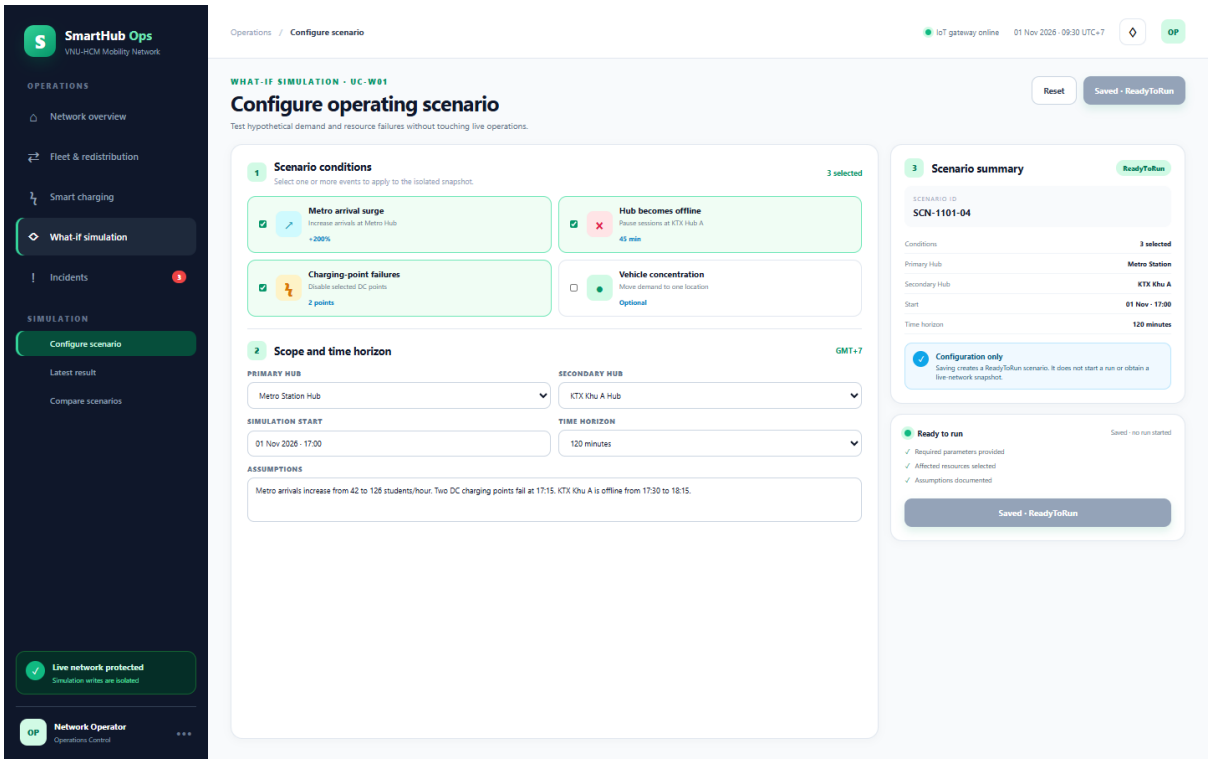


Figure 17: What-if scenario configuration form (UC-W01).

1.6.2 UC-W02, UC-W03, and UC-W04 — Run a Simulation and Review Results

The operator runs a configured scenario against a network snapshot and reviews predicted effects. The results view presents impact measures and coordination recommendations for review.

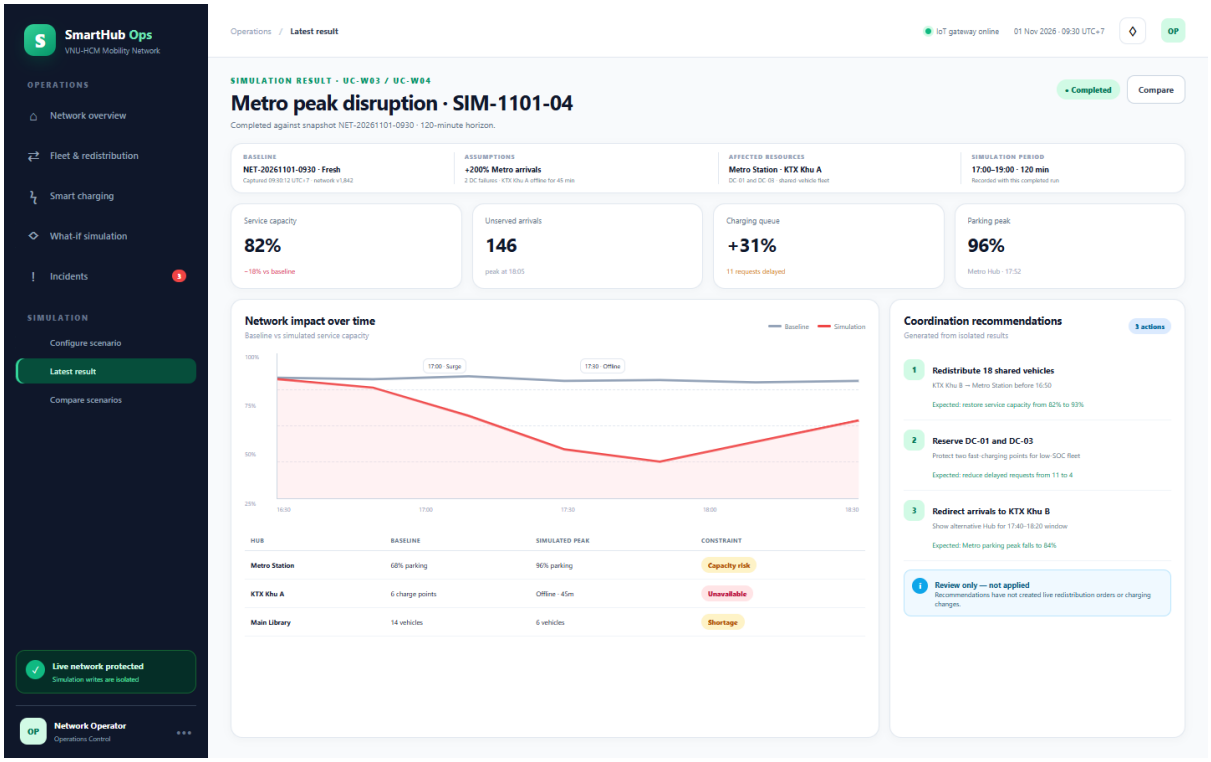


Figure 18: Simulation results and coordination recommendations (UC-W02, UC-W03, UC-W04).

1.6.3 UC-W05 — Compare What-if Scenarios

Operators compare completed simulation runs or a run against its baseline. The comparison screen presents common measures with the context needed to interpret differences.

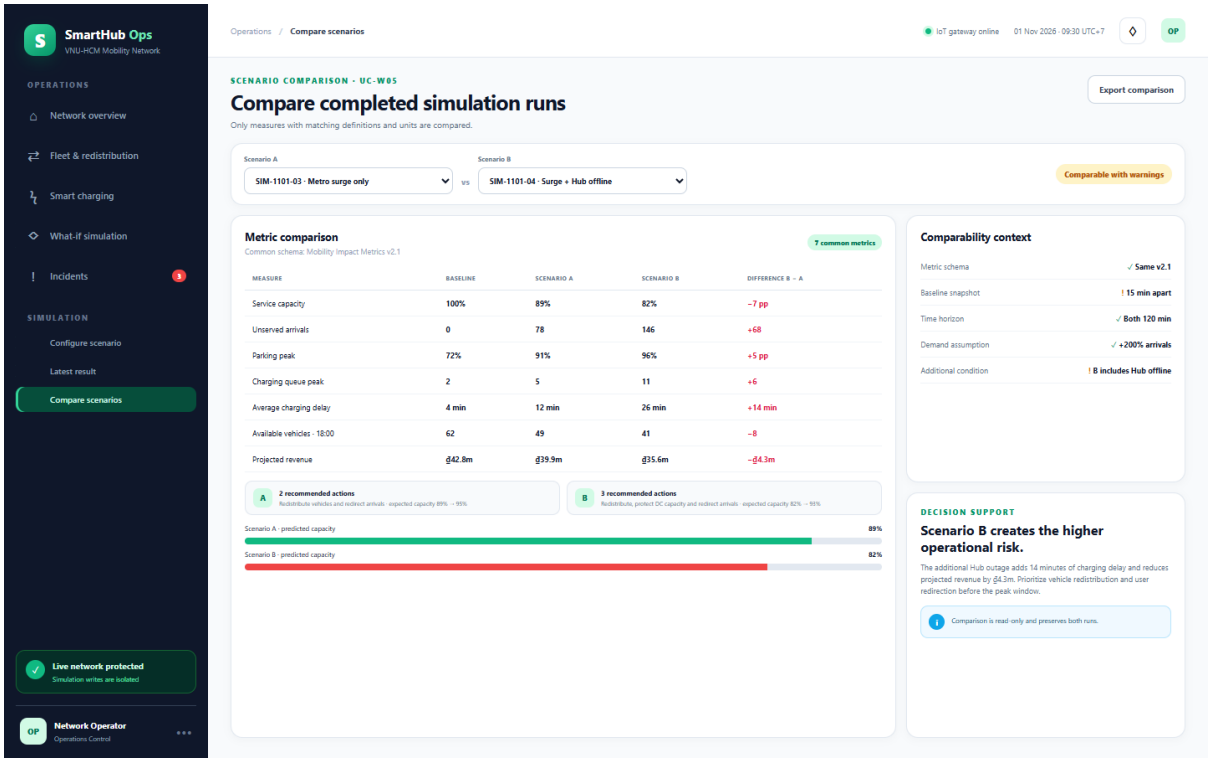


Figure 19: Side-by-side what-if scenario comparison (UC-W05).

2 Sequence Diagrams

2.1 Authentication and Hub & Identity

2.1.1 UC-H01 — Manage Personal Profile

The student opens a profile and views permitted personal information. When details are edited, the controller validates the changes and saves only allowed fields.

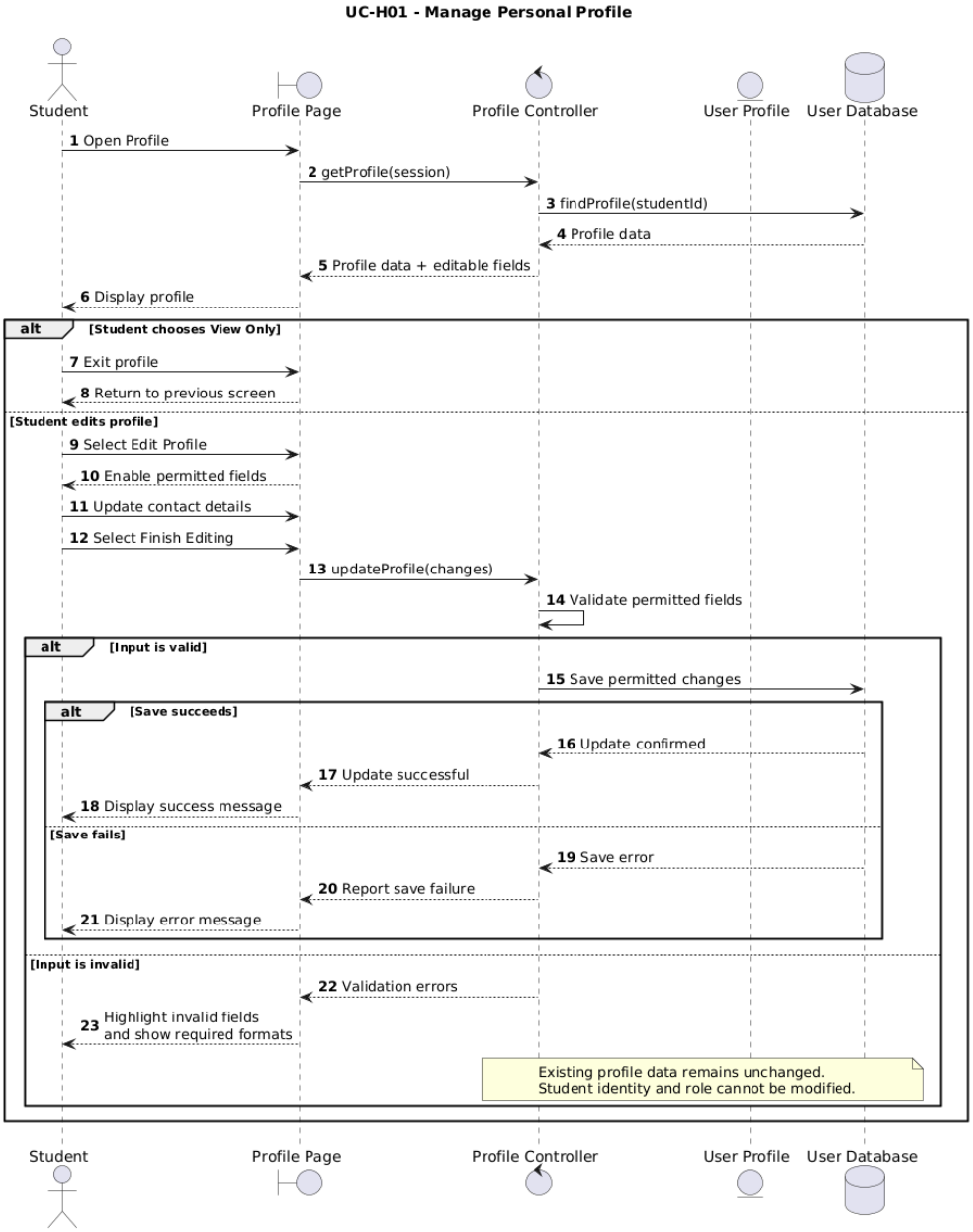


Figure 20: Sequence diagram for managing a personal profile (UC-H01).

2.1.2 UC-H02 — Find a Mobility Hub

The student searches by current or manually entered location and selected services. The system returns matching Hubs, handles unavailable location data, and supports refining an empty result.

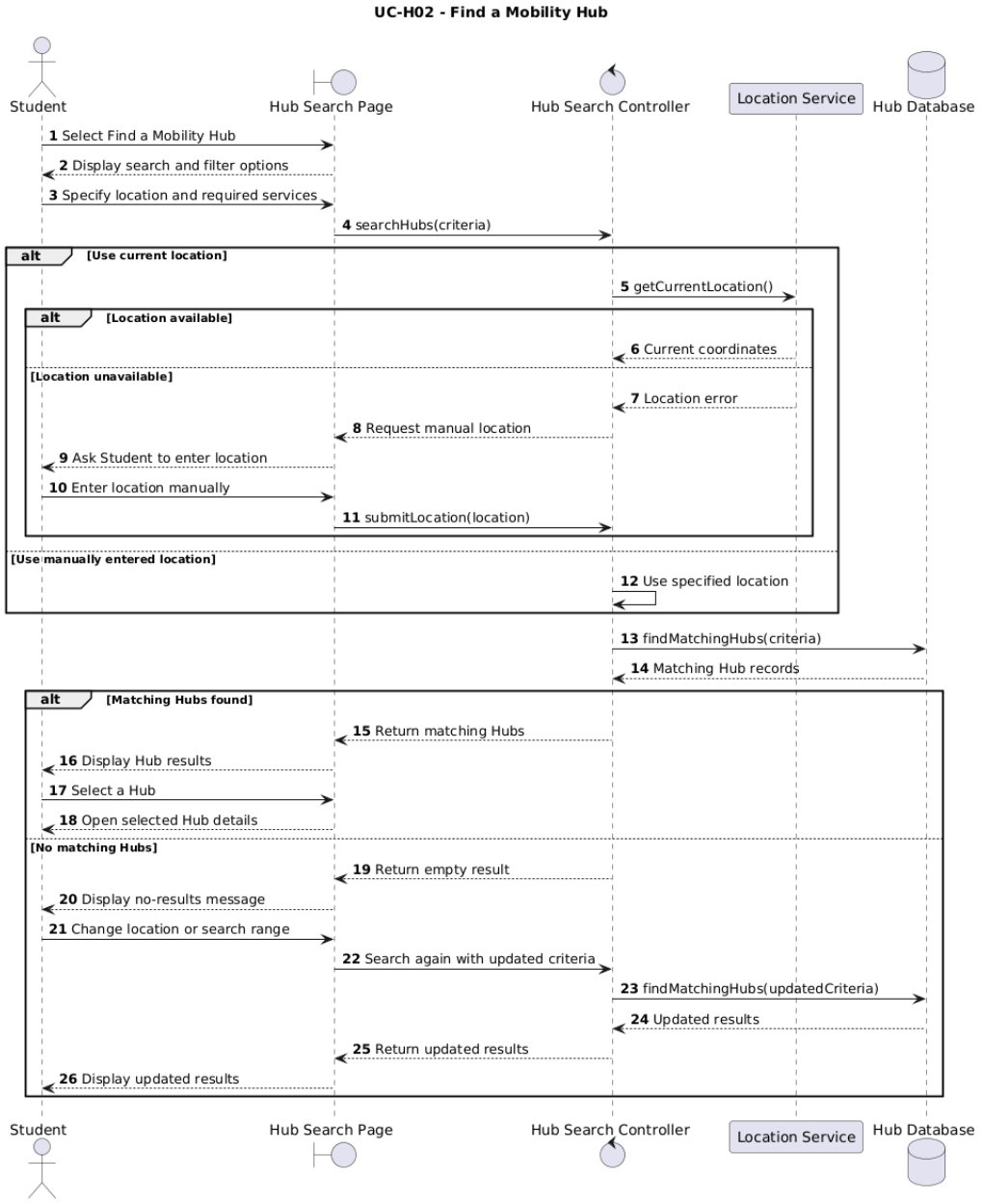


Figure 21: Sequence diagram for finding and filtering Mobility Hubs (UC-H02).

2.1.3 UC-H03 — View Hub Status

The student selects a Hub and requests its latest operational information. The controller gathers vehicle, parking, and charging status, adds freshness information, and returns available data or a retrieval message.

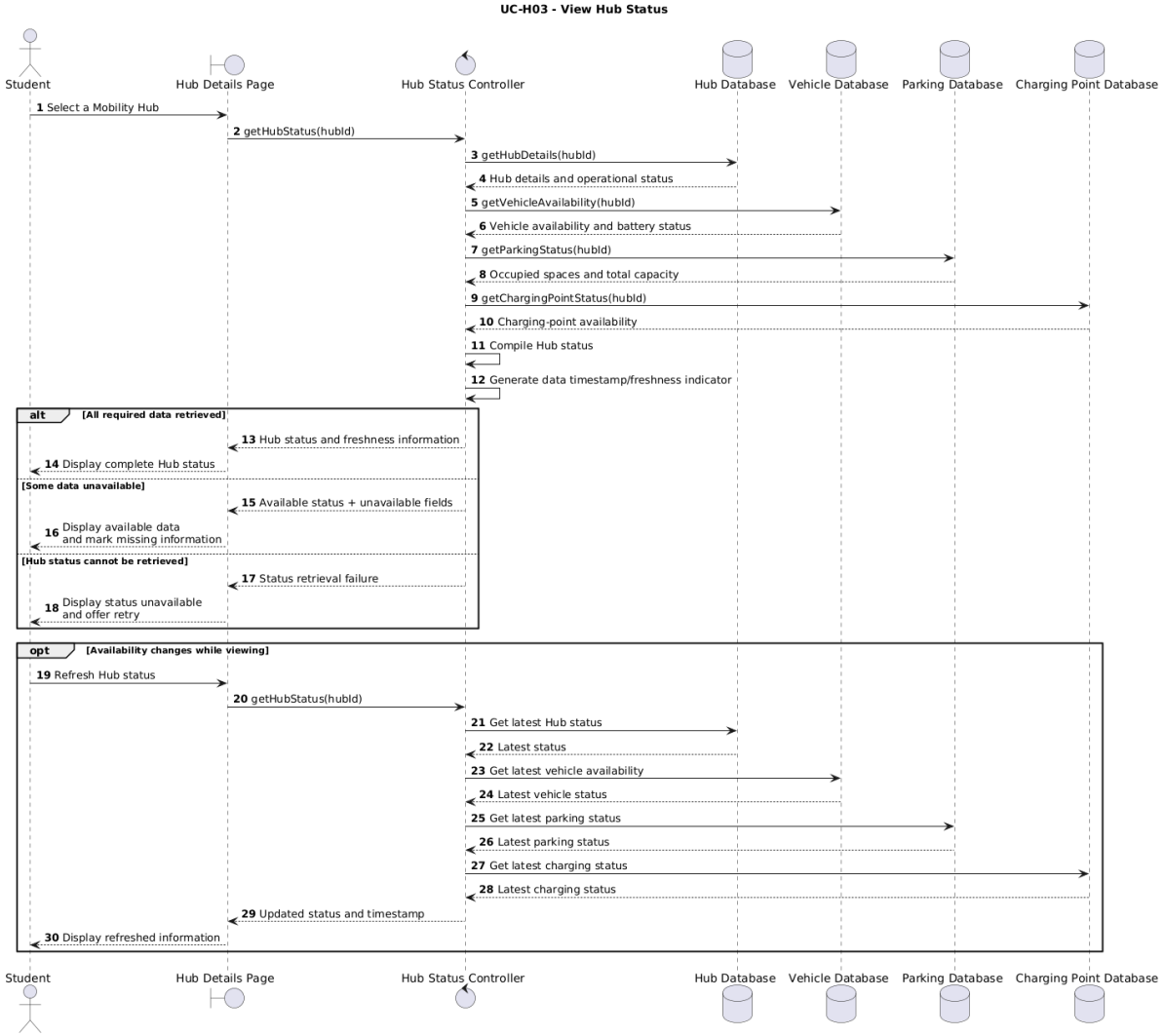


Figure 22: Sequence diagram for viewing aggregated Hub status (UC-H03).

2.2 Vehicle Rental

2.2.1 UC-R03 — Reserve a Shared Vehicle

The student selects an available shared vehicle and requests a reservation. The system validates availability, creates a time-limited hold, and returns the reservation details or an alternative when the request cannot be fulfilled.

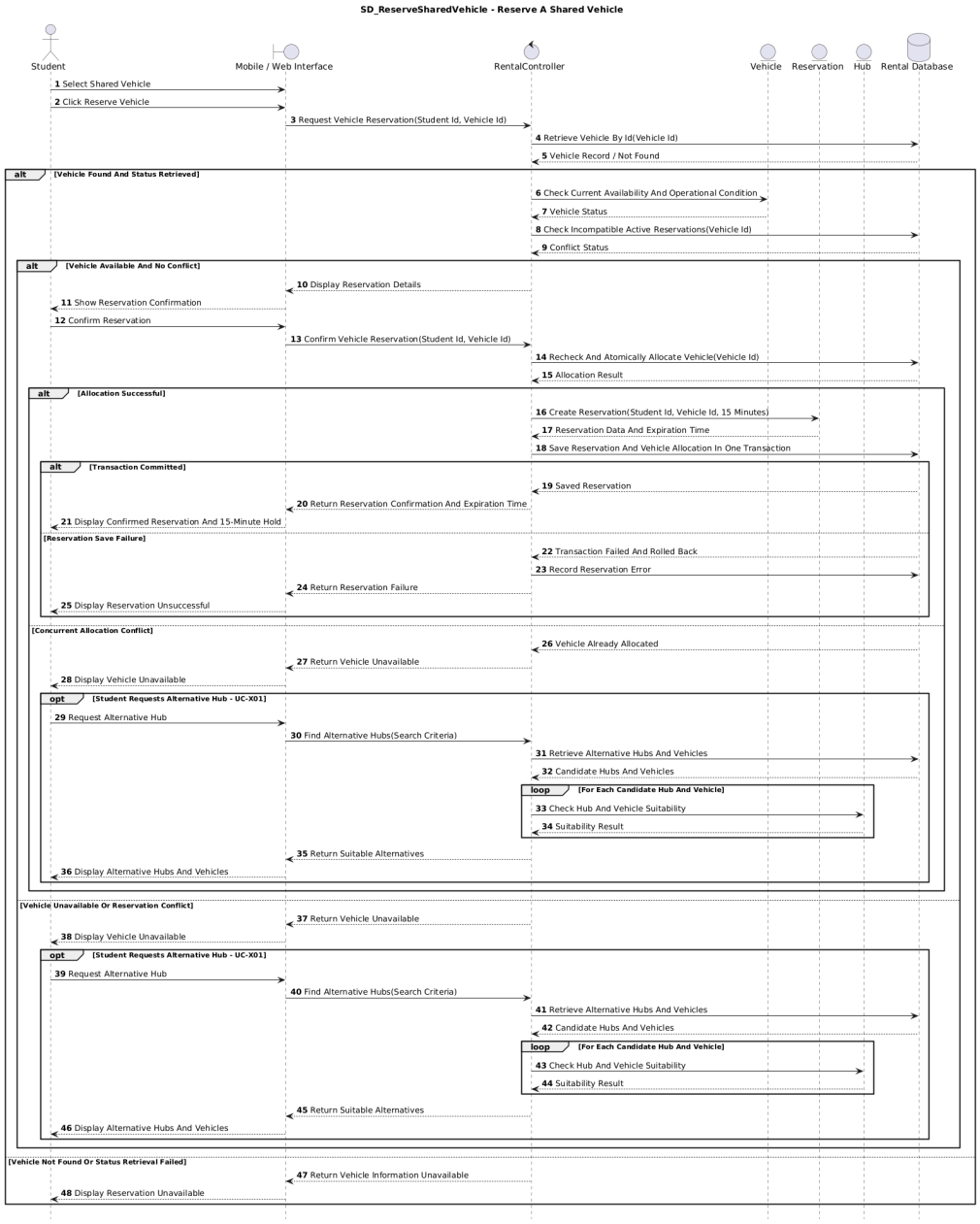


Figure 23: Sequence diagram for reserving a shared vehicle (UC-R03).

2.2.2 UC-R04 — Pick Up a Shared Vehicle

At the Hub, the student presents the reservation through QR or Bluetooth. The system verifies the reservation, commands the vehicle unlock, and records the pickup state.

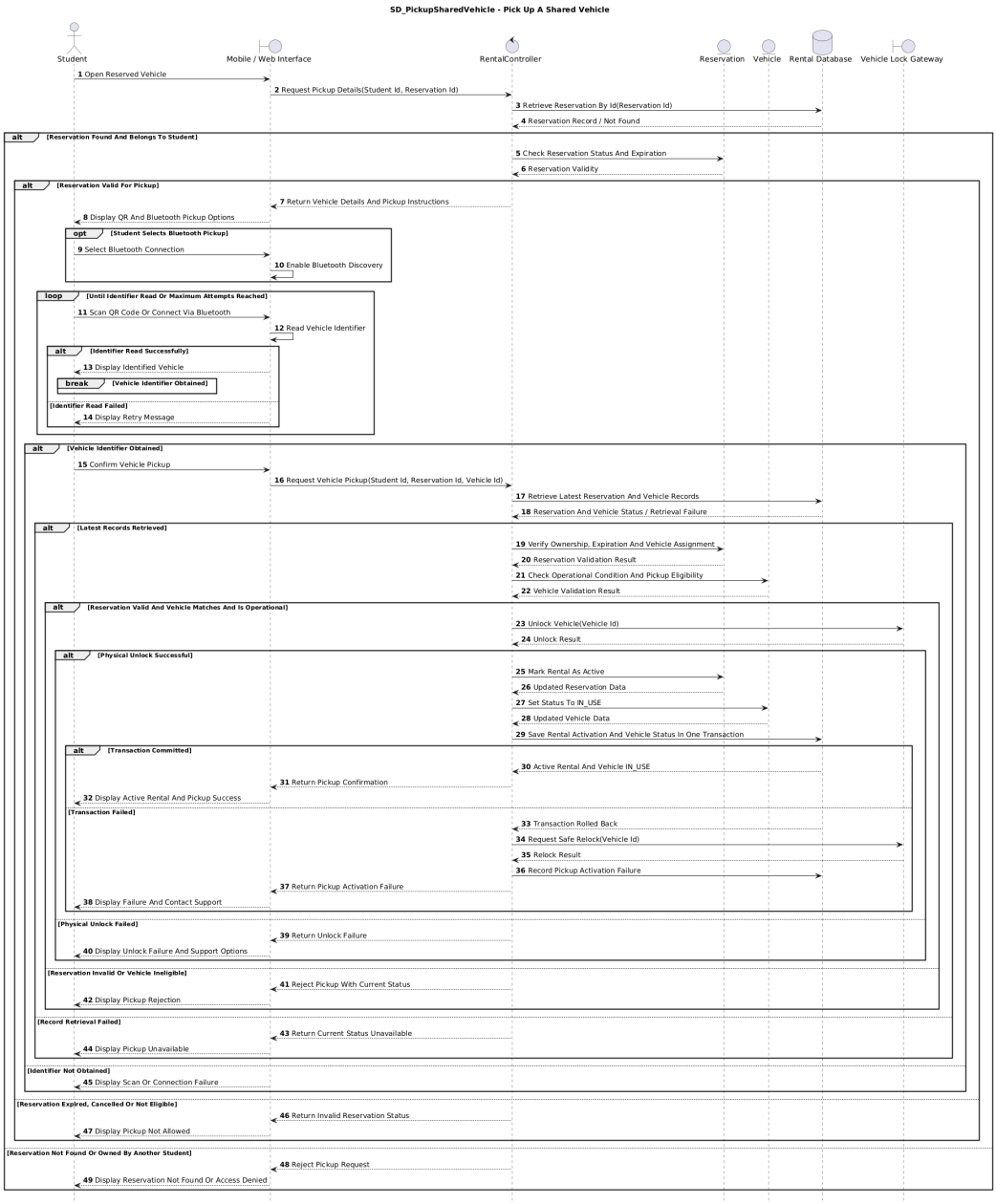


Figure 24: Sequence diagram for picking up and unlocking a shared vehicle (UC-R04).

2.2.3 UC-R05 — Return a Shared Vehicle

The student returns the vehicle at an eligible Hub and ends the trip. The system locks the vehicle, records trip details, and updates vehicle and Hub availability.

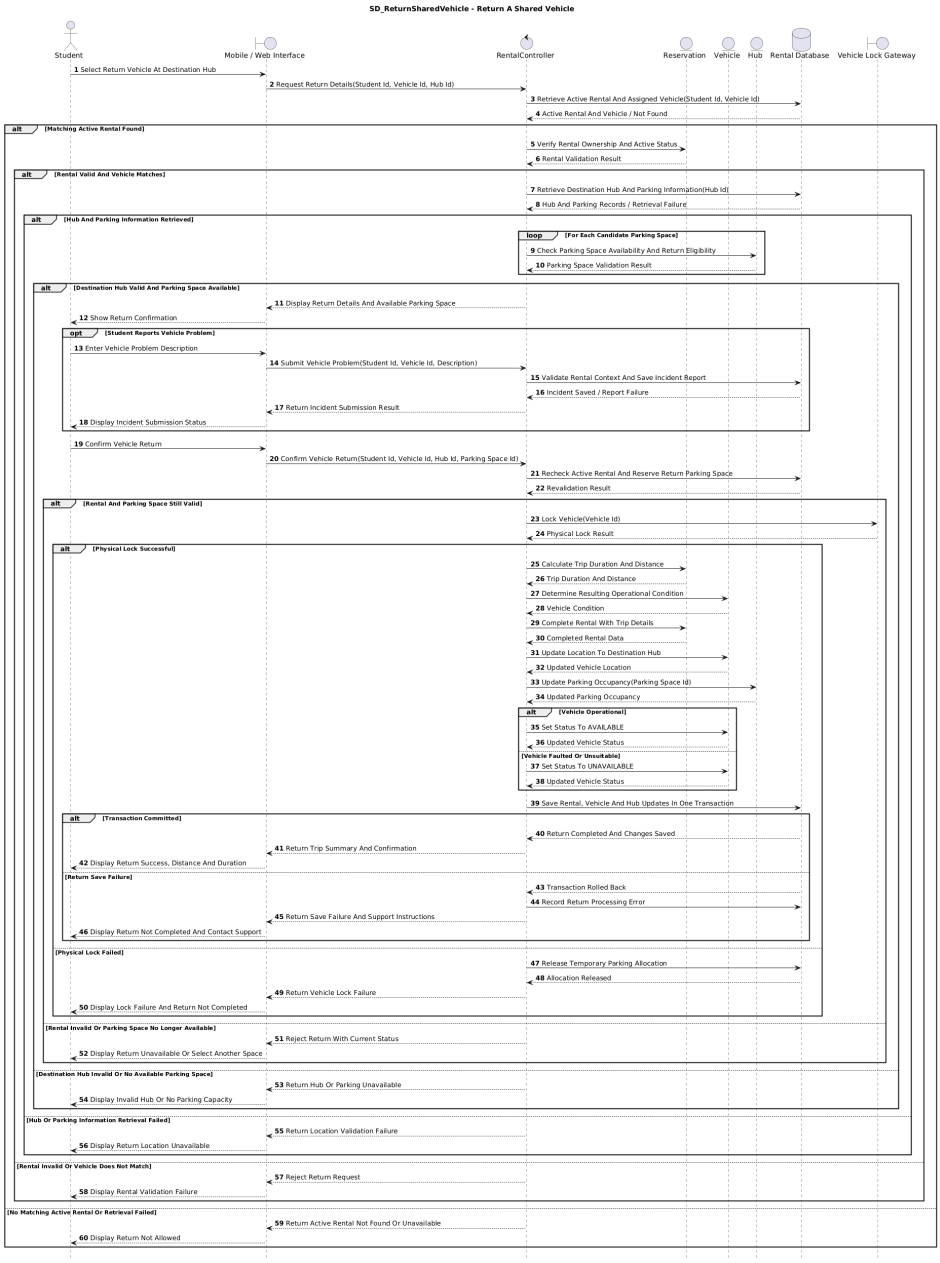


Figure 25: Sequence diagram for returning a shared vehicle (UC-R05).

2.3 Parking and Private EV

2.3.1 UC-P01 — Reserve a Parking Space

The student requests parking for a selected Hub and time interval. The system checks capacity across every interval block and records the reservation only when the full request is feasible.

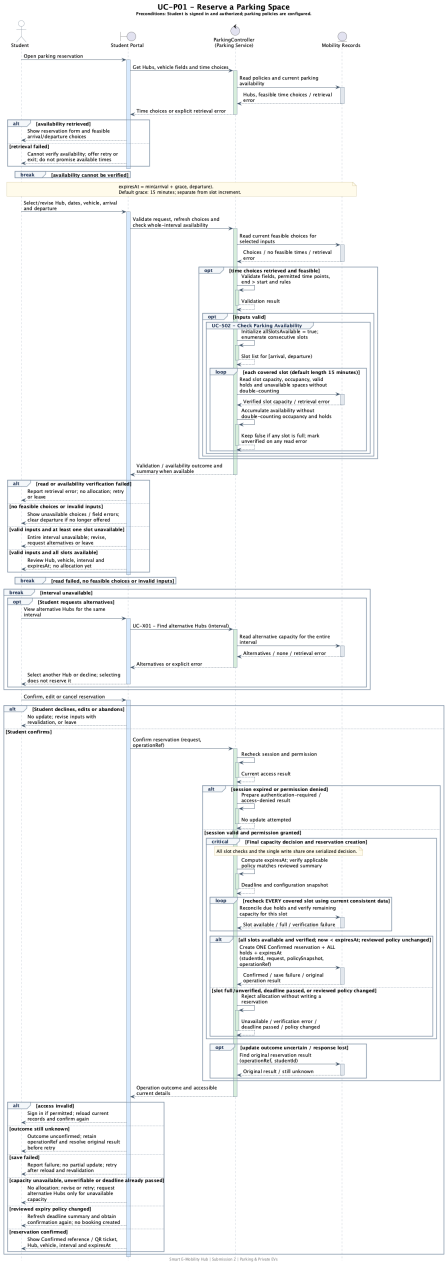


Figure 26: Sequence diagram for reserving a parking space (UC-P01).

2.3.2 UC-P02 — Submit a Private-EV Charging Request

The private EV owner selects a parking booking and submits charging requirements. The system validates the request against the booking and charging resources before saving it.

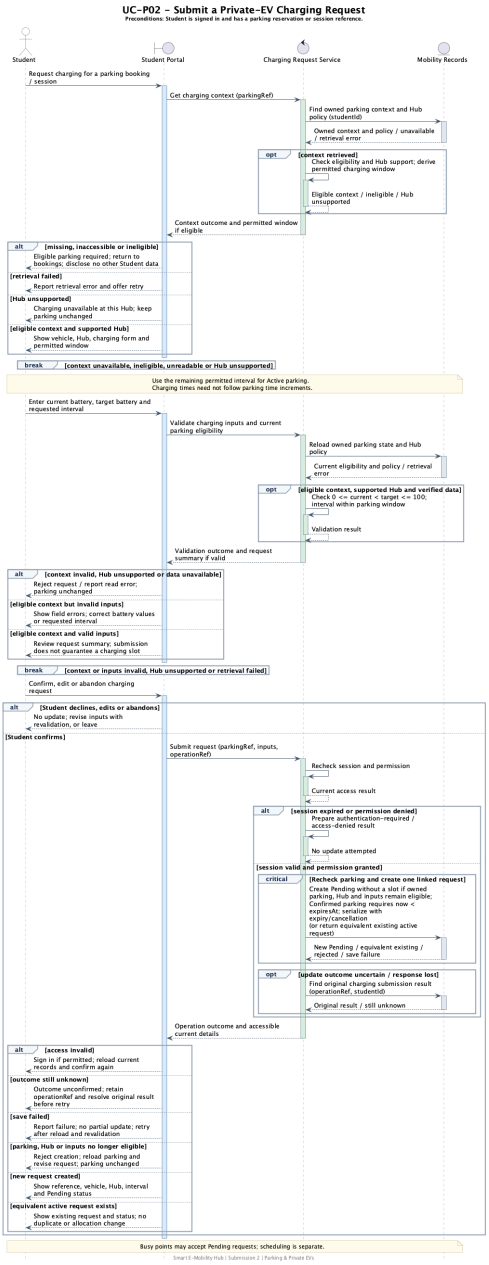


Figure 27: Sequence diagram for submitting a private-EV charging request (UC-P02).

2.3.3 UC-P03 — View Parking and Charging Bookings

The student requests a list of personal parking and charging bookings. The system retrieves the related records and returns their status and available actions.

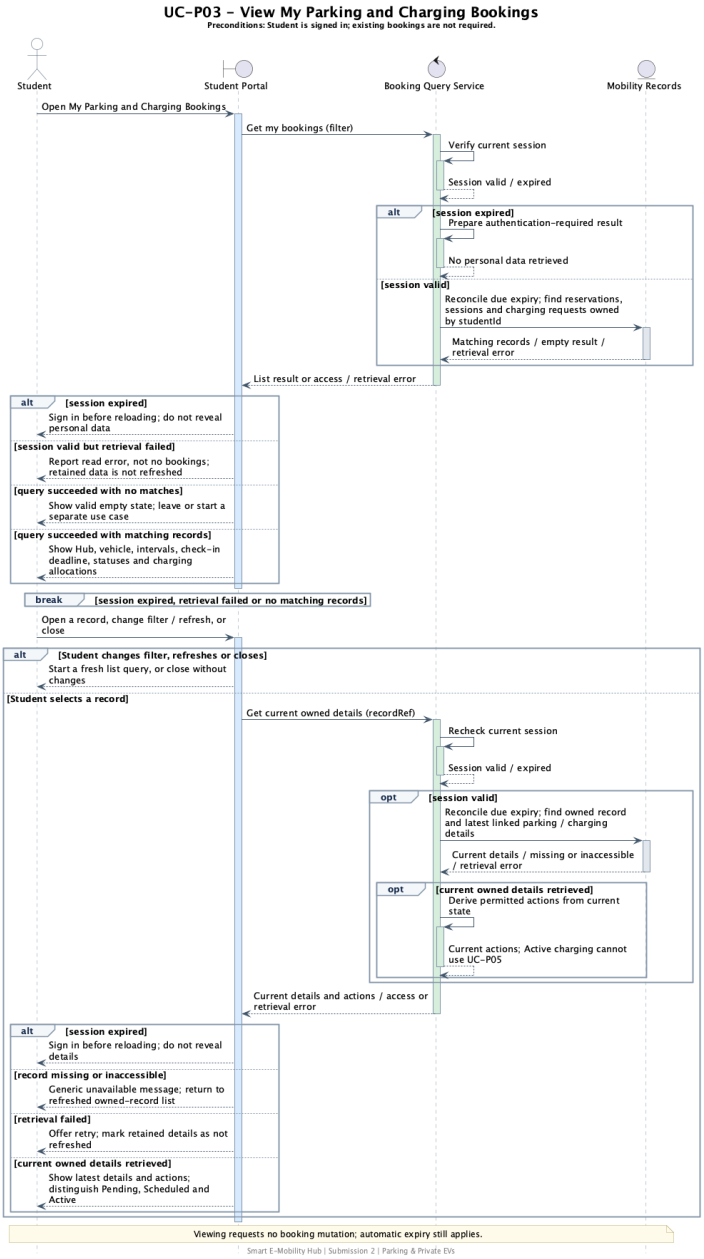


Figure 28: Sequence diagram for viewing parking and charging bookings (UC-P03).

2.3.4 UC-P04 — Cancel a Parking Reservation

The student requests cancellation of a parking reservation. The system verifies that cancellation is permitted, updates the reservation, and releases the associated capacity.

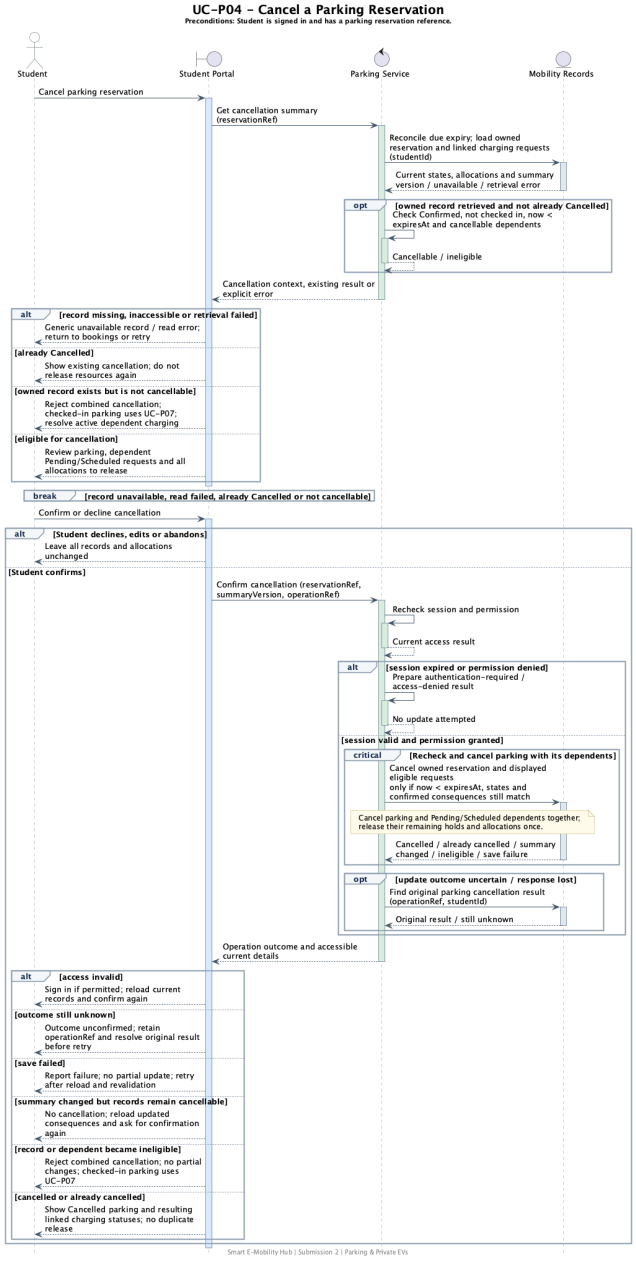


Figure 29: Sequence diagram for cancelling a parking reservation (UC-P04).

2.3.5 UC-P05 — Cancel a Charging Request

The student requests cancellation of a charging request. The system checks the request's current state and updates the request and charging schedule when cancellation is allowed.

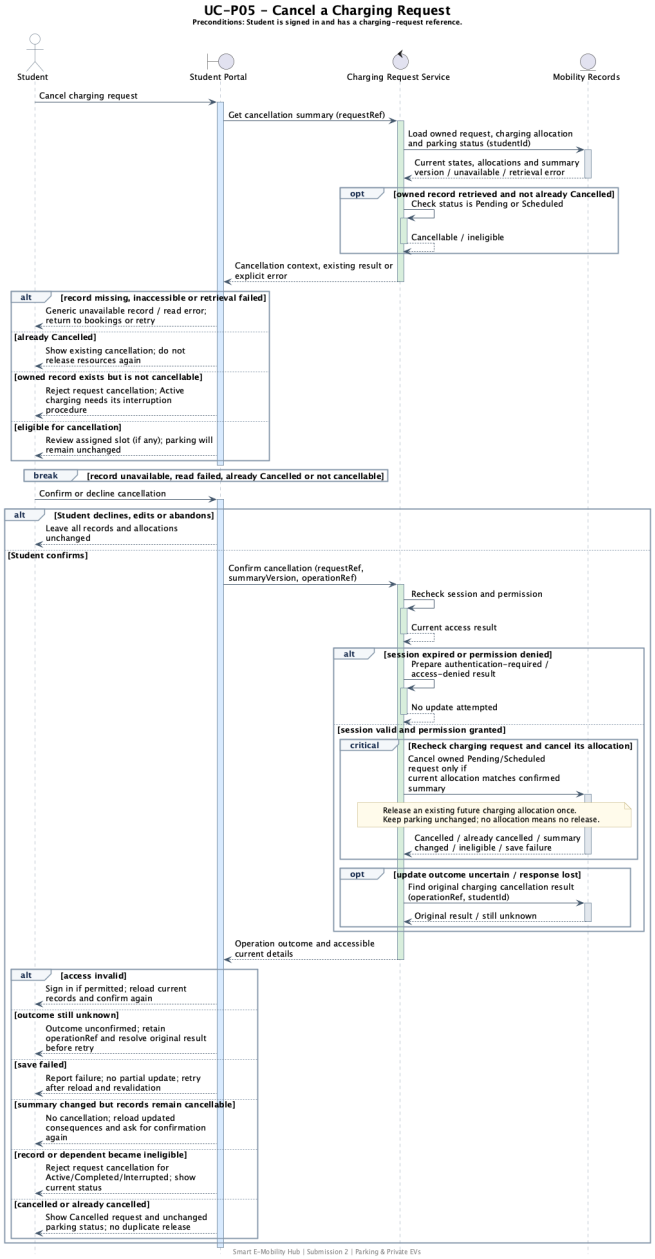


Figure 30: Sequence diagram for cancelling a charging request (UC-P05).

2.3.6 UC-P06 — Check In a Private EV

At the Hub, the student presents the parking ticket for check-in. The system validates the booking, records arrival, and updates actual occupancy without counting the reservation twice.

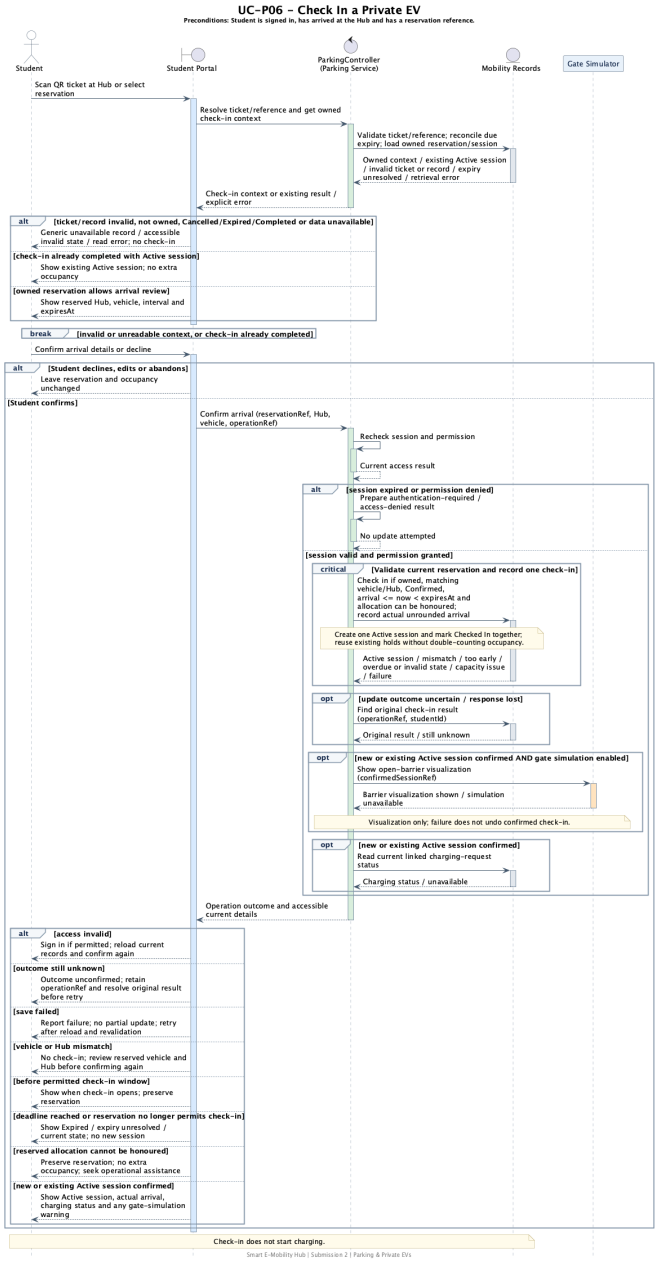


Figure 31: Sequence diagram for checking in a private EV (UC-P06).

2.3.7 UC-P07 — Check Out a Private EV

The student checks out when leaving the Hub. The system records departure, finalizes any applicable charging or overstay information, and releases occupied capacity.

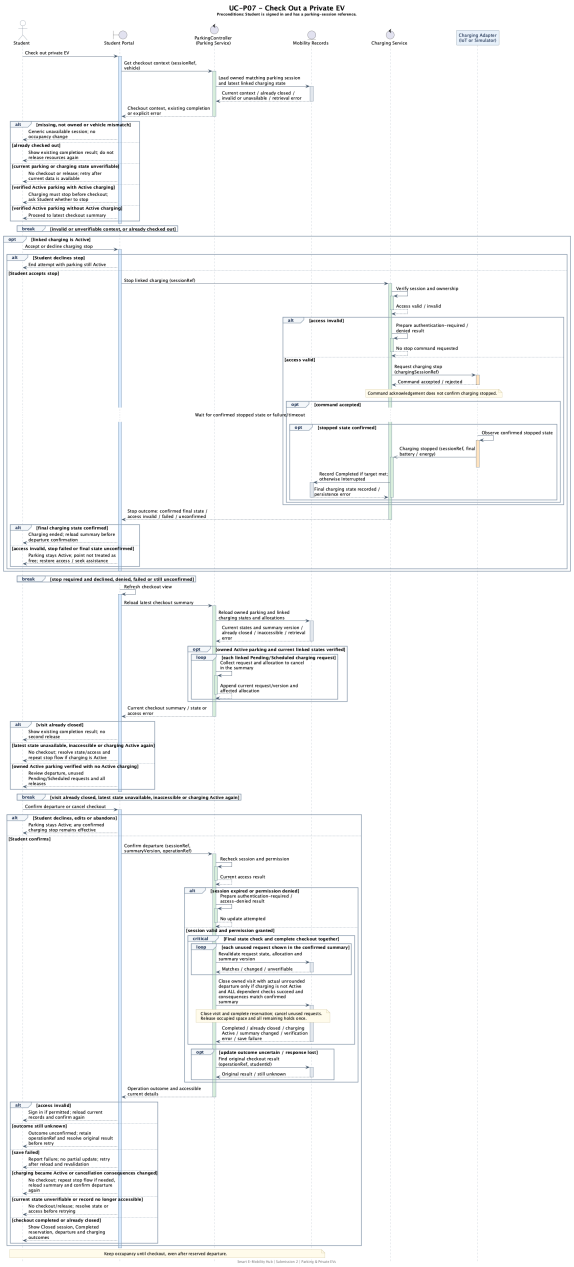


Figure 32: Sequence diagram for checking out a private EV (UC-P07).

2.4 Fleet Monitoring and Incident Handling

2.4.1 UC-F01 — Monitor the Mobility Hub Network

The operator requests a network overview and the system checks authorization before gathering Hub, vehicle, incident, and charging data. It calculates operational indicators and freshness, then returns the overview or marks unavailable data.

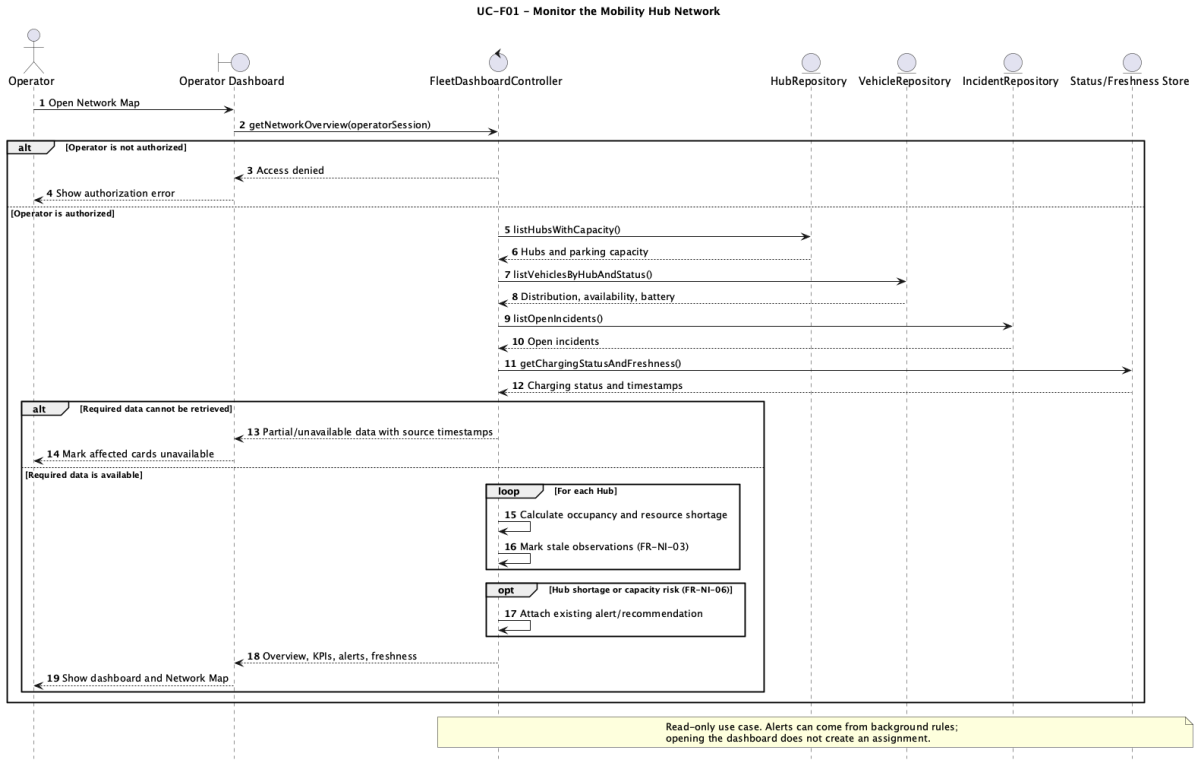


Figure 33: Sequence diagram for monitoring the Hub network (UC-F01).

2.4.2 UC-F04 — Coordinate Vehicle Redistribution

The operator selects vehicles and source and destination Hubs for a transfer. The system checks vehicle commitments and Hub capacity, revalidates the proposal, and saves a feasible assignment.

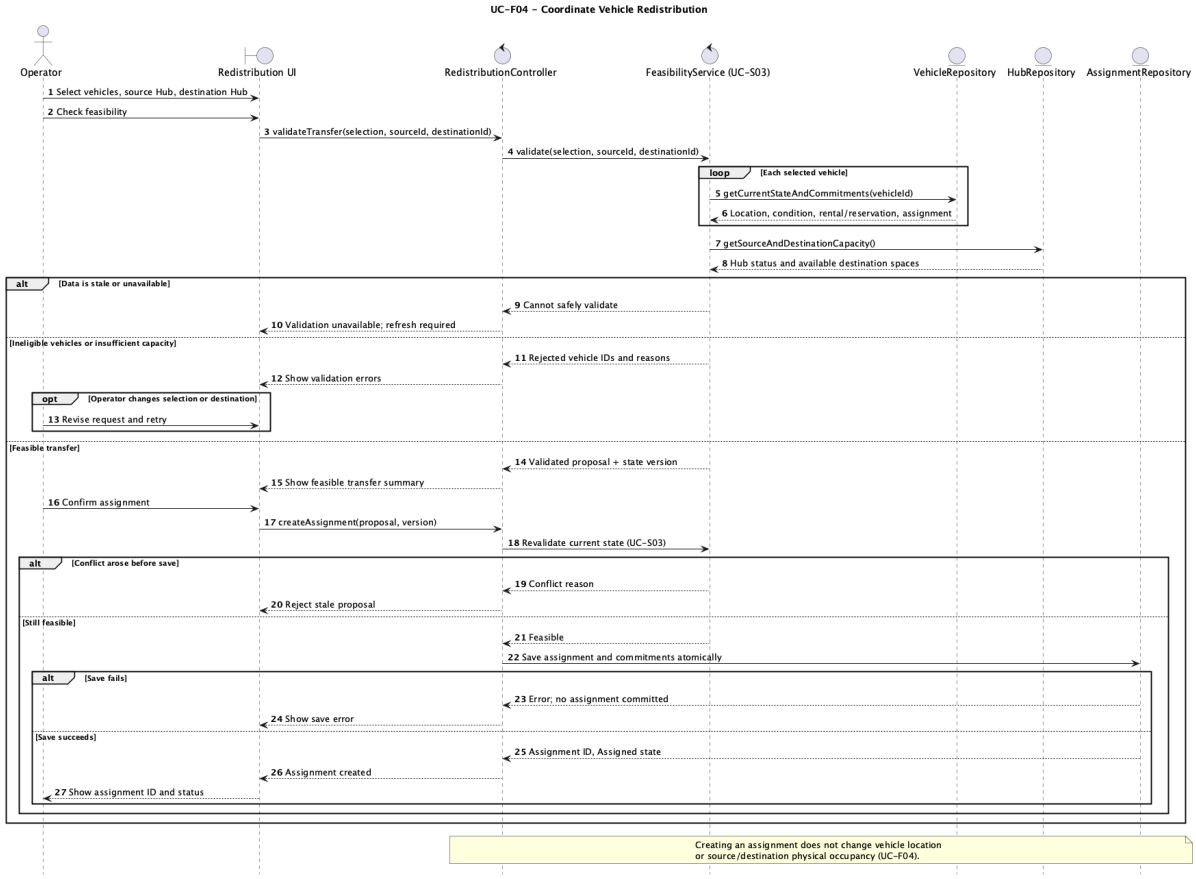


Figure 34: Sequence diagram for coordinating vehicle redistribution (UC-F04).

2.4.3 UC-F05 — Record Redistribution Completion

The operator confirms which assigned vehicles have arrived. The system verifies each vehicle and records only confirmed arrivals while keeping assignment progress and Hub occupancy consistent.

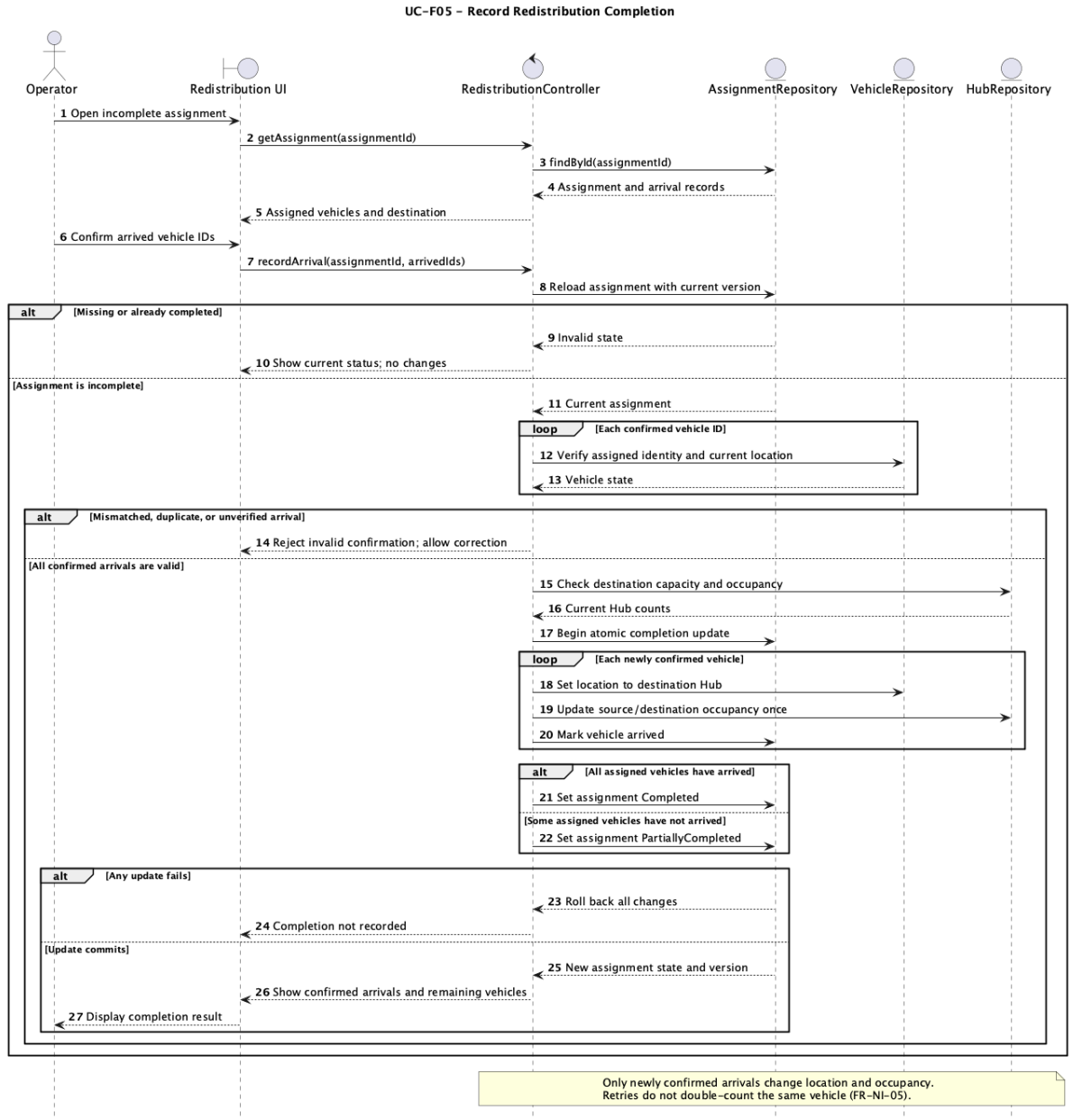


Figure 35: Sequence diagram for recording redistribution arrivals (UC-F05).

2.4.4 UC-I02 — Handle a Vehicle Failure

The operator investigates a vehicle incident, checks affected reservations or rentals, and records the response and repair action. The system keeps the vehicle restricted until restoration checks succeed.

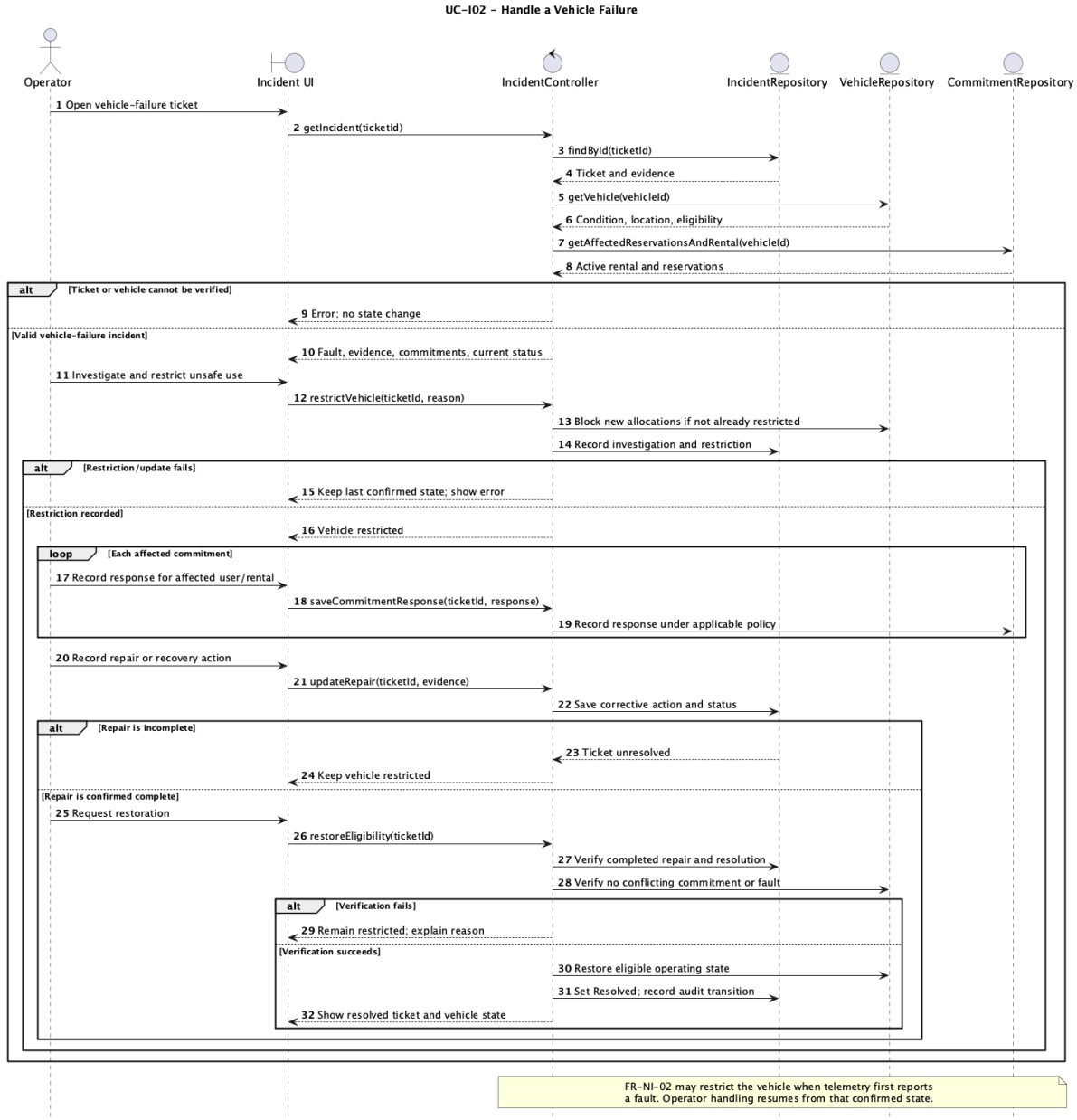


Figure 36: Sequence diagram for handling a vehicle failure (UC-I02).

2.4.5 Supporting Flow — Process Automatic Vehicle-Fault Telemetry

A telemetry provider sends vehicle condition data without an operator action. The system validates event freshness, updates vehicle eligibility, links fault evidence to an incident, and alerts affected assignment workflows.

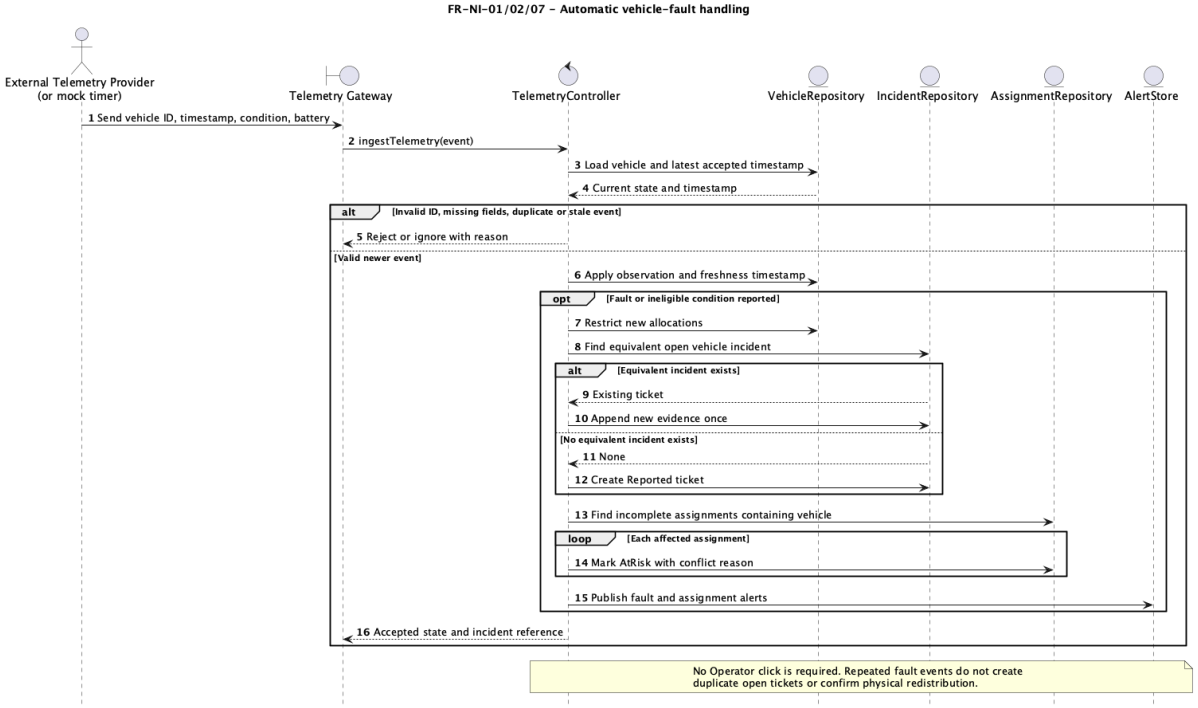


Figure 37: Sequence diagram for automatic vehicle-fault handling (FR-NI-01, FR-NI-02, FR-NI-07).

2.5 Smart Charging Scheduling

2.5.1 UC-C01 — View the Charging Queue and Schedule

The operator requests the current charging queue and schedule. The system retrieves requests, charging-point allocations, and active sessions, then displays queued and scheduled work in FCFS order.

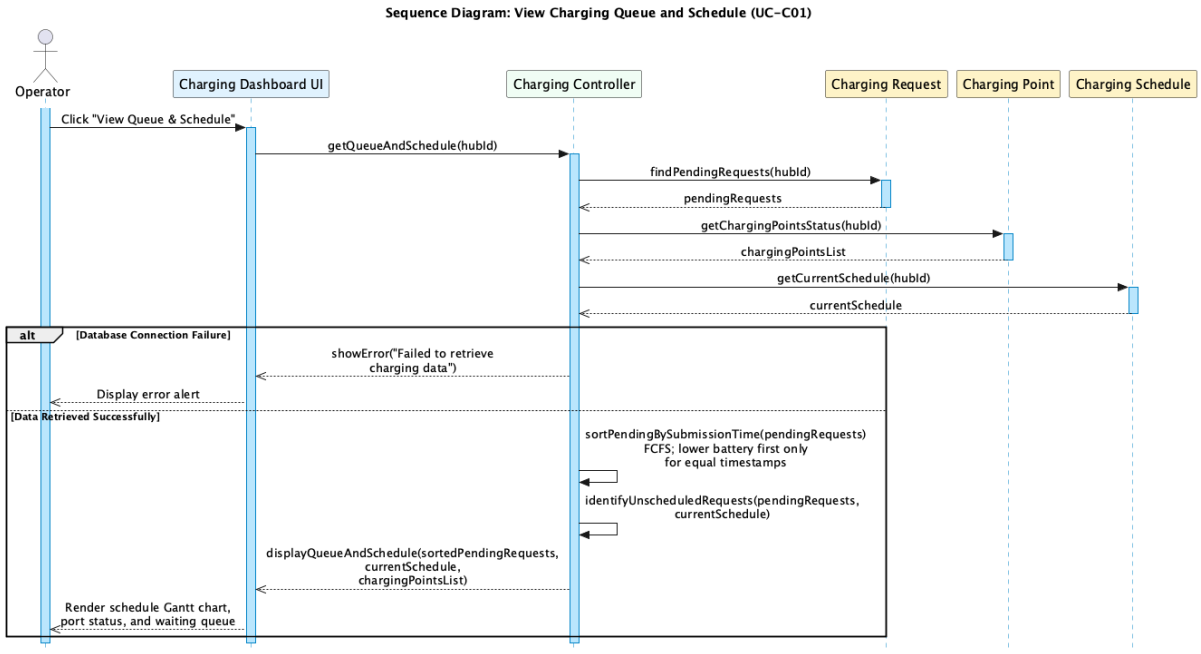


Figure 38: Sequence diagram for viewing the charging queue and schedule (UC-C01).

2.5.2 UC-C02 — Generate a Charging Schedule

The operator starts schedule generation and the system retrieves valid requests and charging-point availability. It processes requests in FCFS order, applies the lower-battery tie-break only for identical arrival timestamps, assigns feasible slots, and reports unscheduled requests.

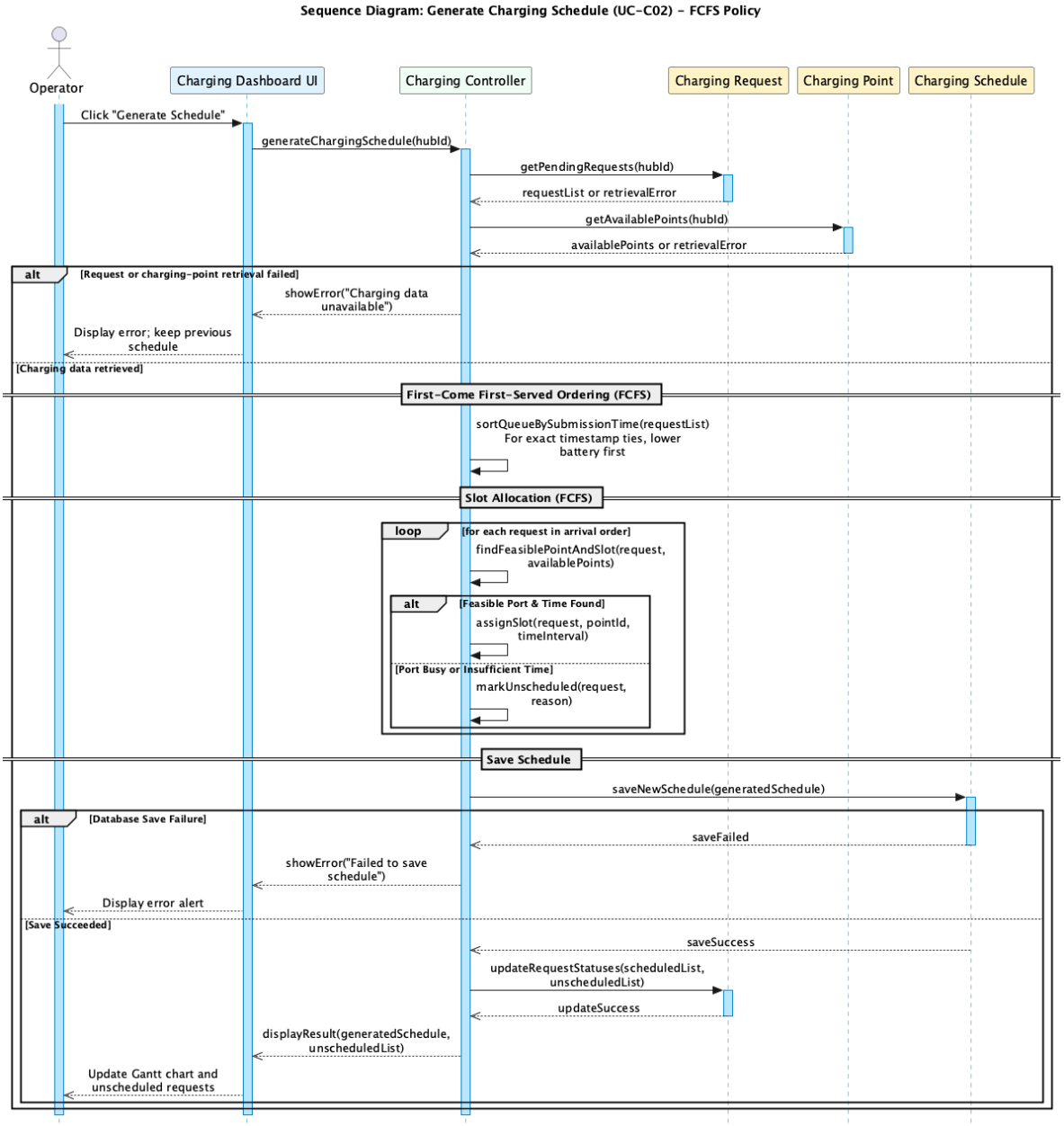


Figure 39: Sequence diagram for generating an FCFS charging schedule (UC-C02).

2.5.3 UC-C03 — Adjust a Charging Schedule

The operator changes a request's assigned port or interval. The system checks the proposed change for availability and conflicts, then saves the adjustment or returns a reason it cannot be applied.

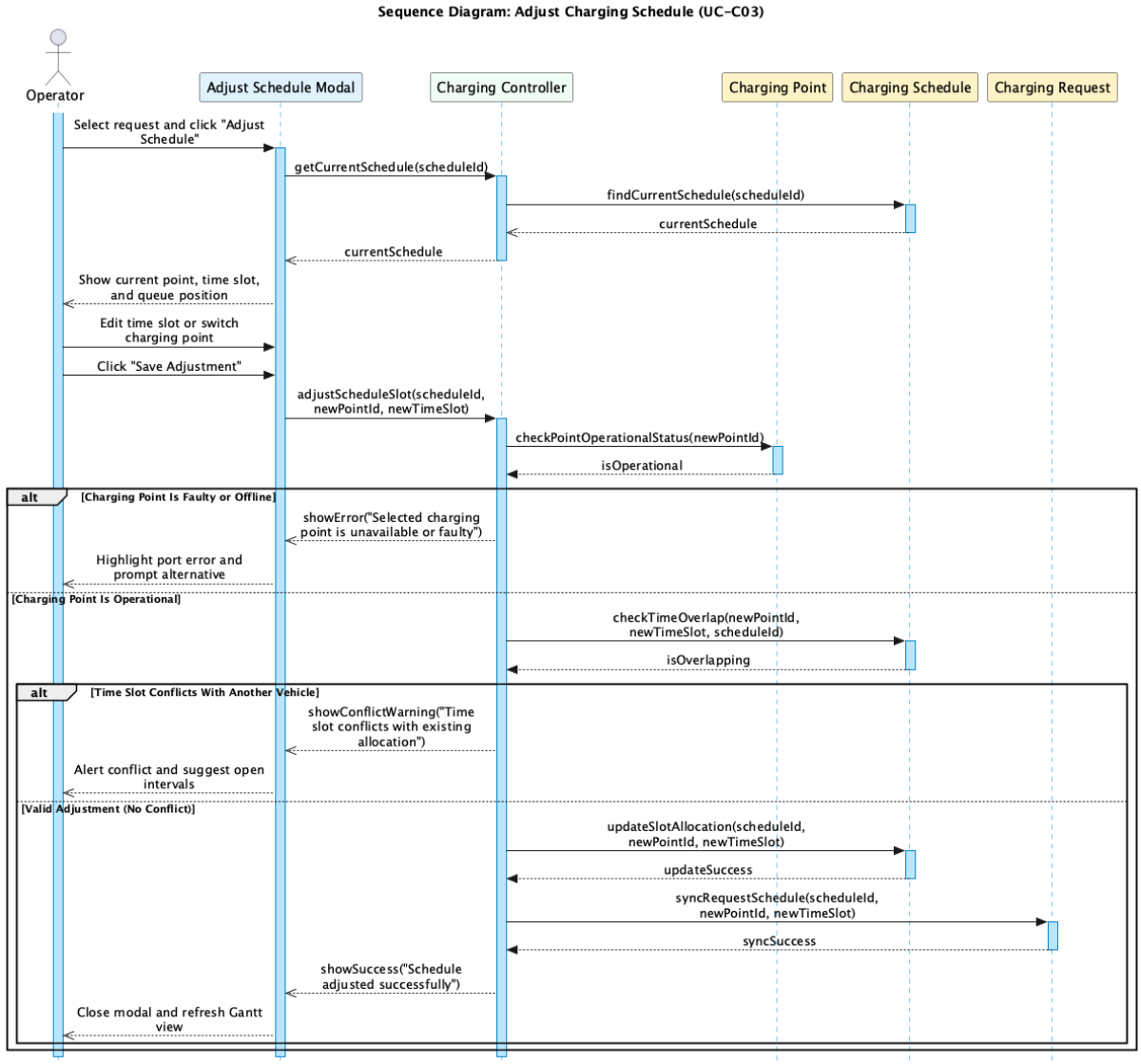


Figure 40: Sequence diagram for adjusting a charging schedule (UC-C03).

2.6 What-if Simulation

2.6.1 UC-W01 — Configure a What-if Scenario

The operator defines hypothetical conditions, affected resources, assumptions, and a simulation period. The system validates the configuration and resource versions before saving a scenario ready to run.

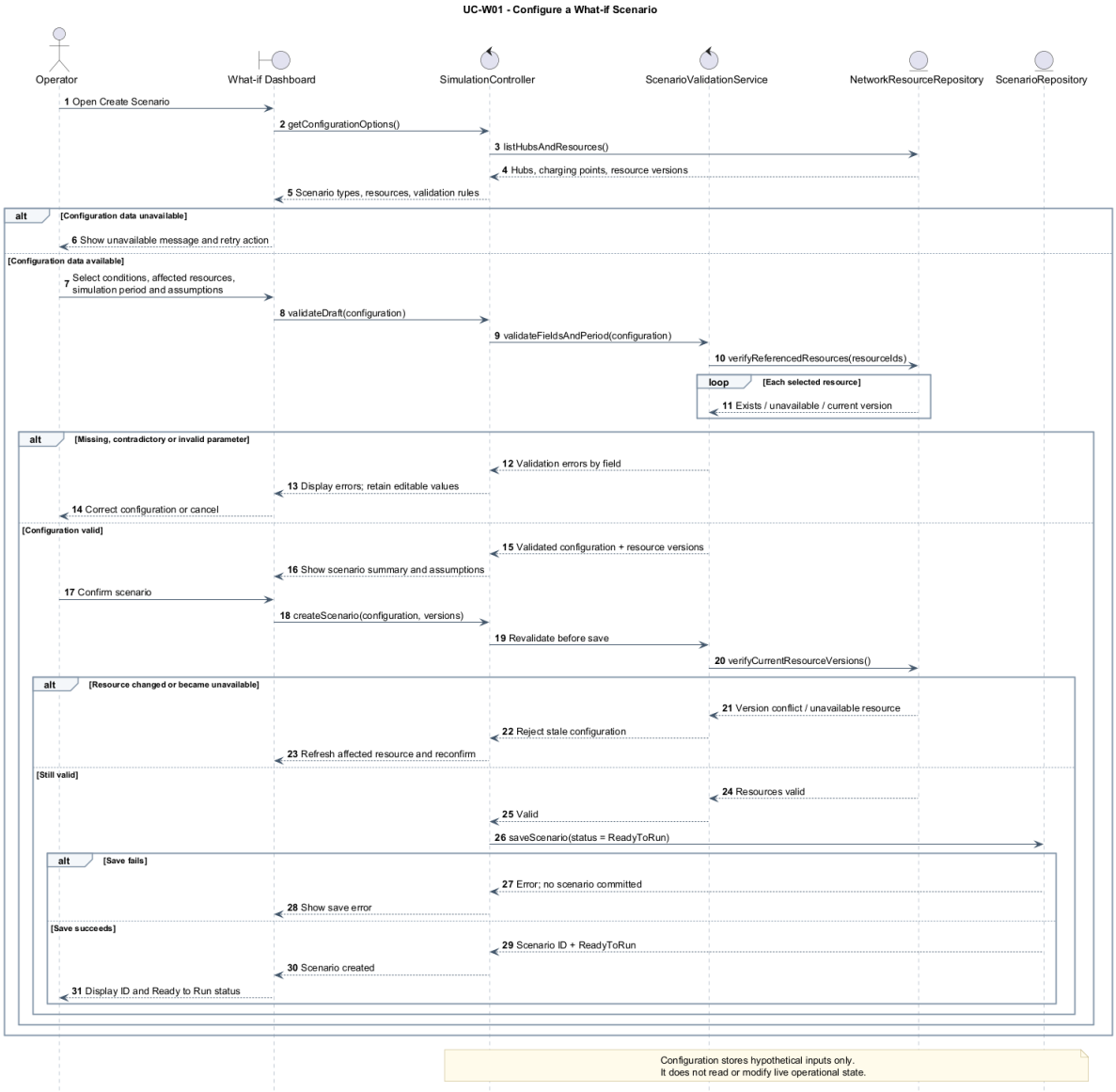


Figure 41: Sequence diagram for configuring a what-if scenario (UC-W01).

2.6.2 UC-W02 — Run a What-if Simulation

The operator requests a run of a validated scenario. The system obtains a consistent network snapshot, executes against an isolated state, evaluates impact, generates recommendations, and stores the completed run without changing live operations.

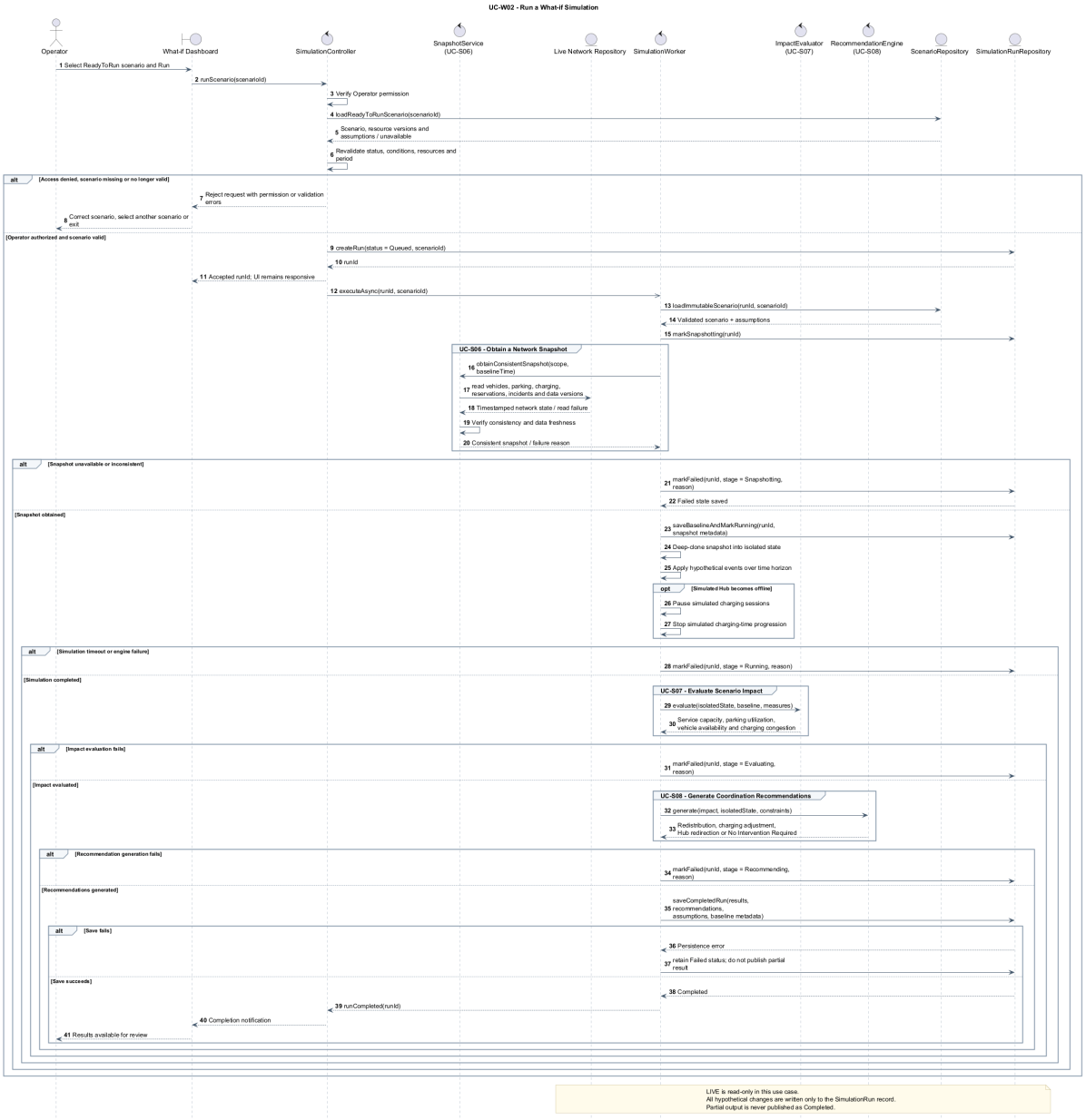


Figure 42: Sequence diagram for running an isolated what-if simulation (UC-W02).

2.6.3 UC-W05 — Compare What-if Scenarios

The operator selects completed runs or a run and its baseline for comparison. The system checks metric compatibility and context, calculates comparable differences, and presents warnings where snapshots or assumptions differ.

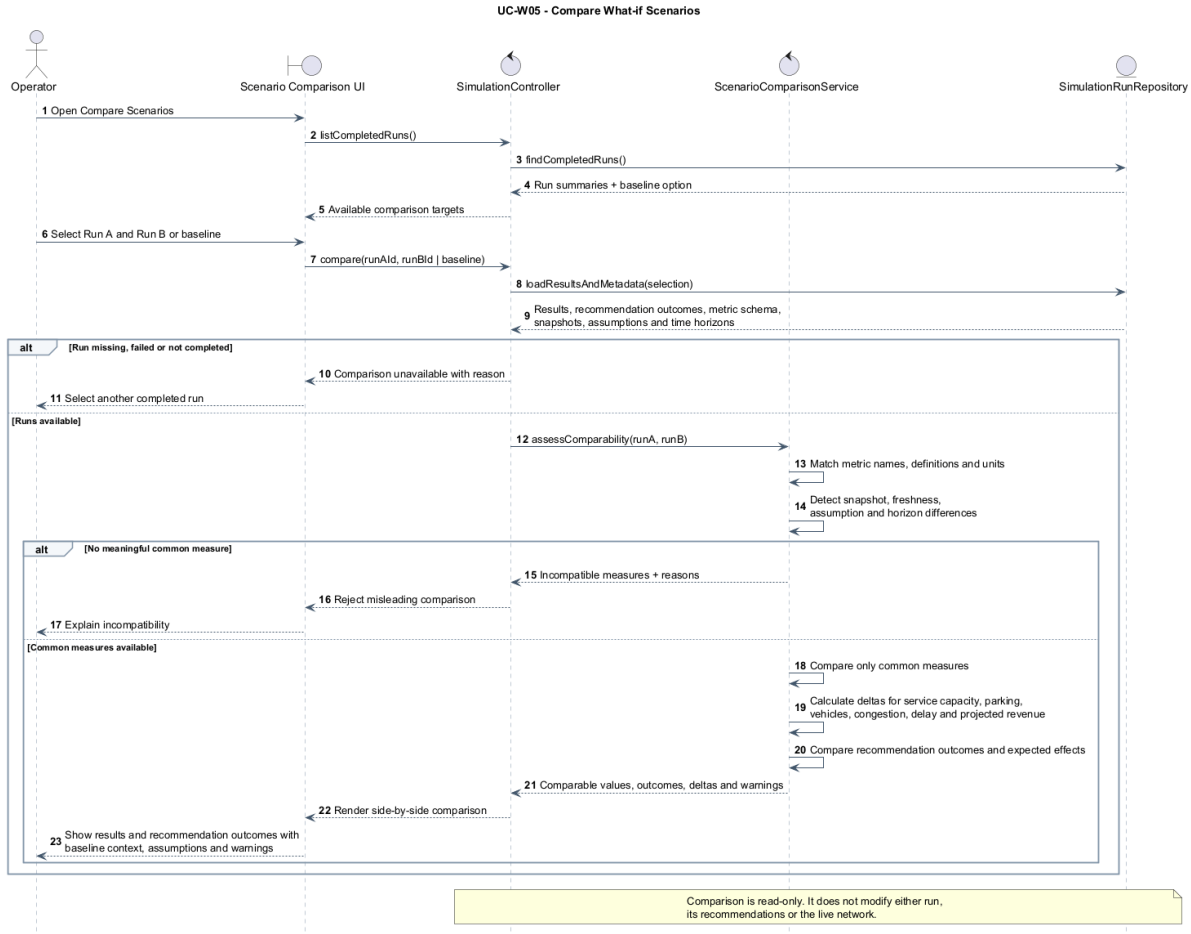


Figure 43: Sequence diagram for comparing what-if scenarios (UC-W05).

2.7 External Telemetry

2.7.1 UC-T01 — Update Vehicle Telemetry

An external telemetry source sends a vehicle observation to the system. The system validates and records the observation while preventing stale or duplicate events from overwriting newer state.

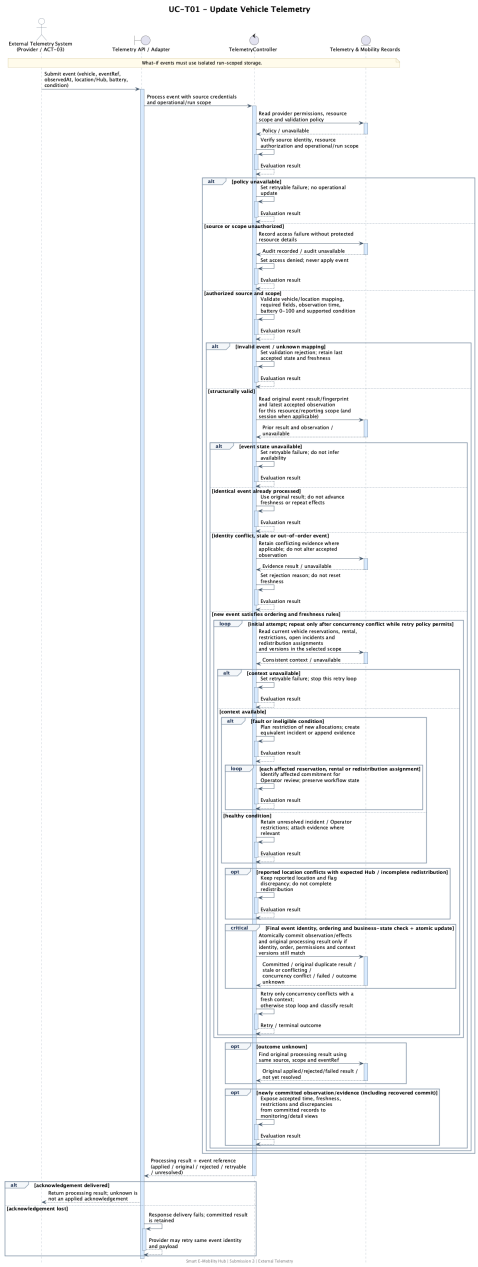


Figure 44: Sequence diagram for updating vehicle telemetry (UC-T01).

2.7.2 UC-T02 — Update Parking Occupancy

A parking sensor reports an occupancy change for a Hub. The system validates the event, reconciles the reported count with current records, and makes the refreshed status available to Hub and parking workflows.

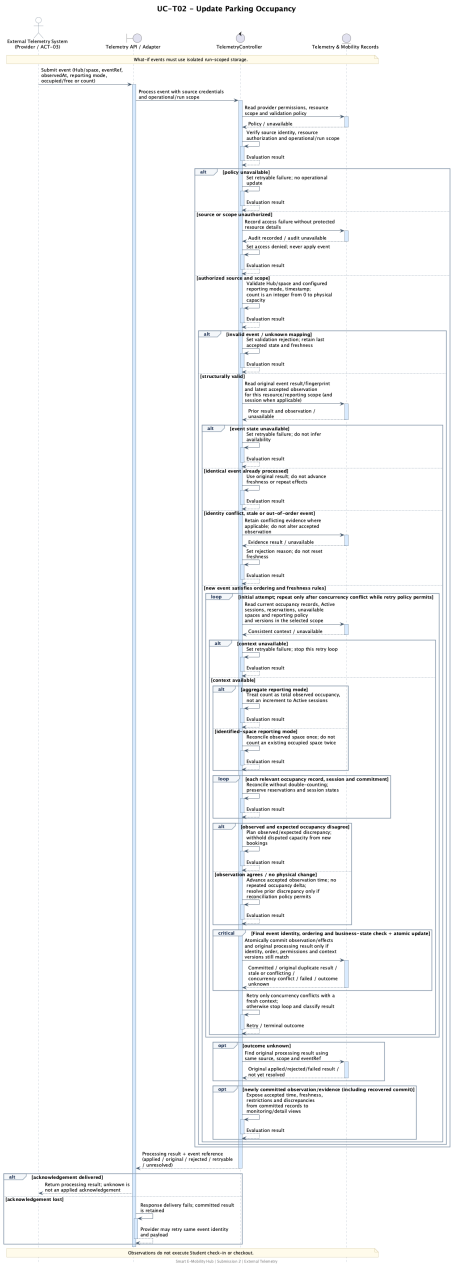


Figure 45: Sequence diagram for updating parking occupancy (UC-T02).

2.7.3 UC-T03 — Update Charging-Point Status

A charging-point telemetry source reports availability or session state. The system validates the observation and updates operational status used by charging queue and schedule workflows.

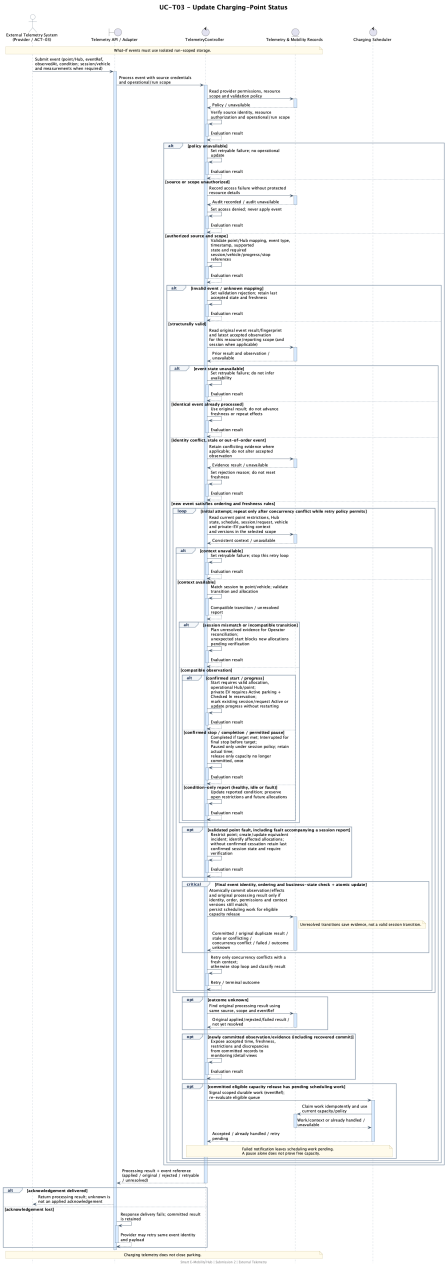


Figure 46: Sequence diagram for updating charging-point status (UC-T03).

3 Activity Diagrams

3.1 Authentication and Hub & Identity

3.1.1 UC-A01 — Authenticate and Authorize a User

The user enters credentials and the system validates the account and its status. A valid Student or Operator receives the matching access, while invalid credentials, inactive accounts, or unknown roles end with an access message.

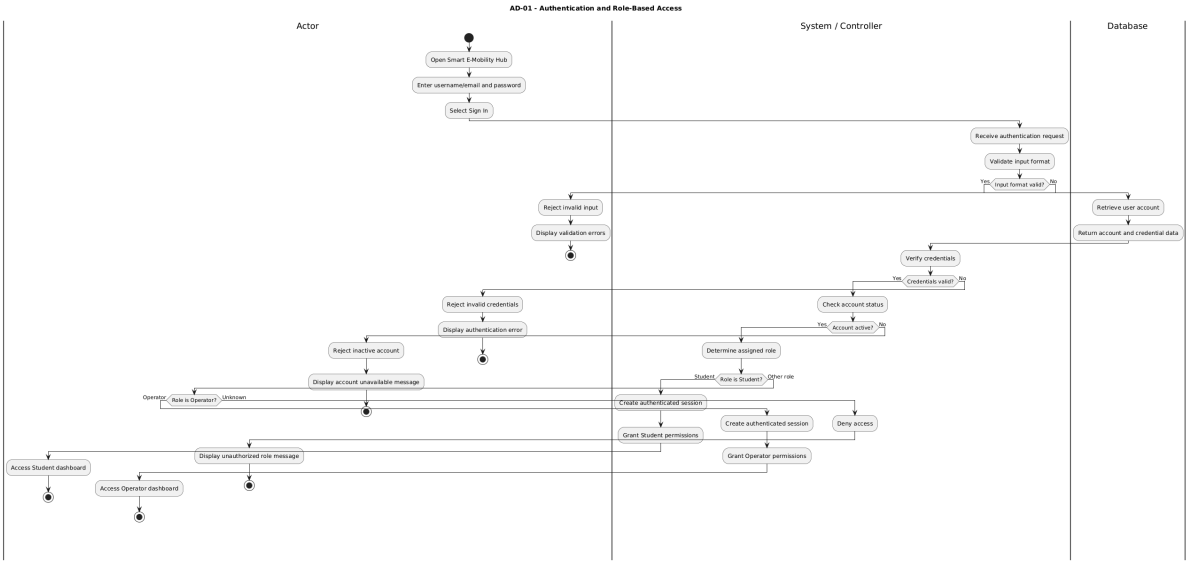


Figure 47: Activity diagram for authentication and role-based access (UC-A01).

3.1.2 UC-H02 — Find and Filter a Mobility Hub

The student chooses GPS or a manually entered location and selects service filters. The system searches the Hub records, displays matches, or allows the criteria to be changed when no result is found.

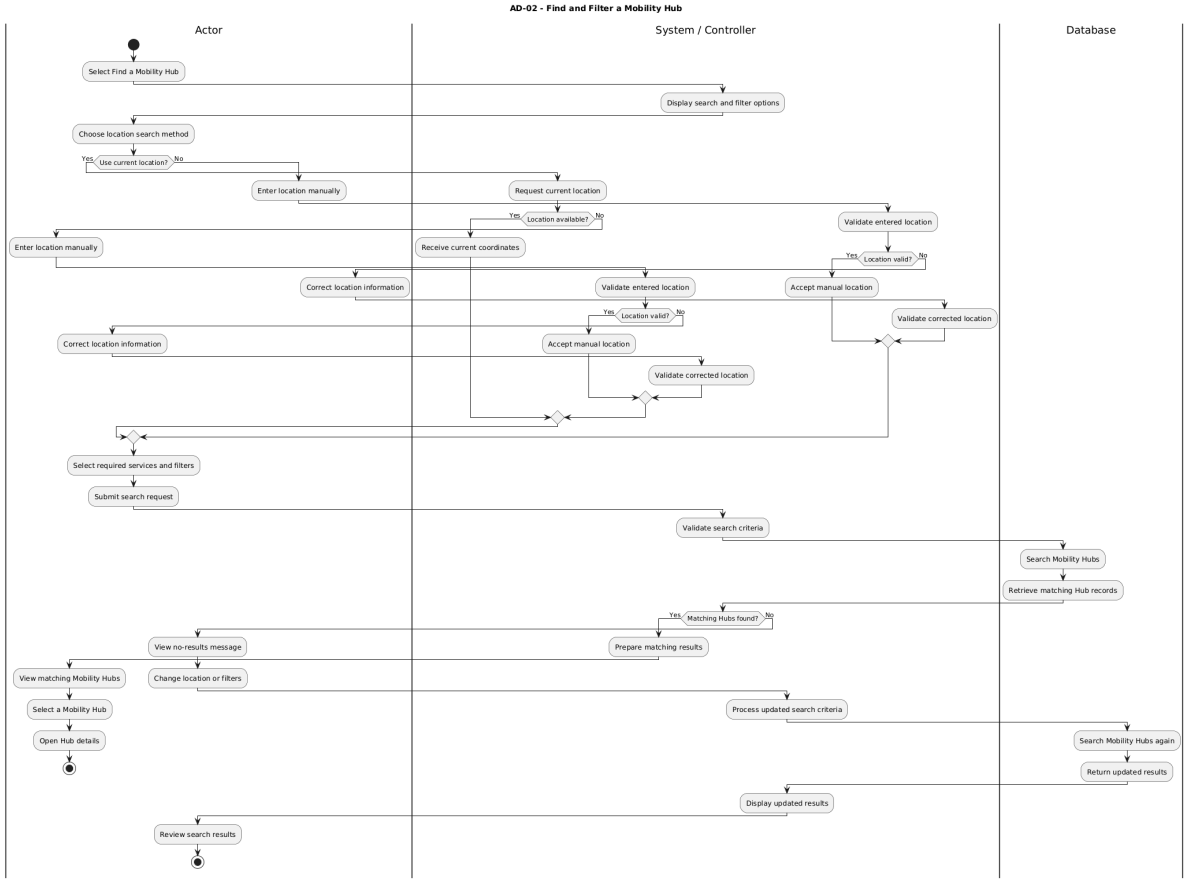


Figure 48: Activity diagram for finding and filtering Mobility Hubs (UC-H02).

3.2 Vehicle Rental

3.2.1 UC-R03 — Reserve a Shared Vehicle with Timeout

The student requests an available shared vehicle and the system creates a temporary reservation after validation. A countdown governs pickup; if the hold expires, the reservation is released and the vehicle becomes available again.

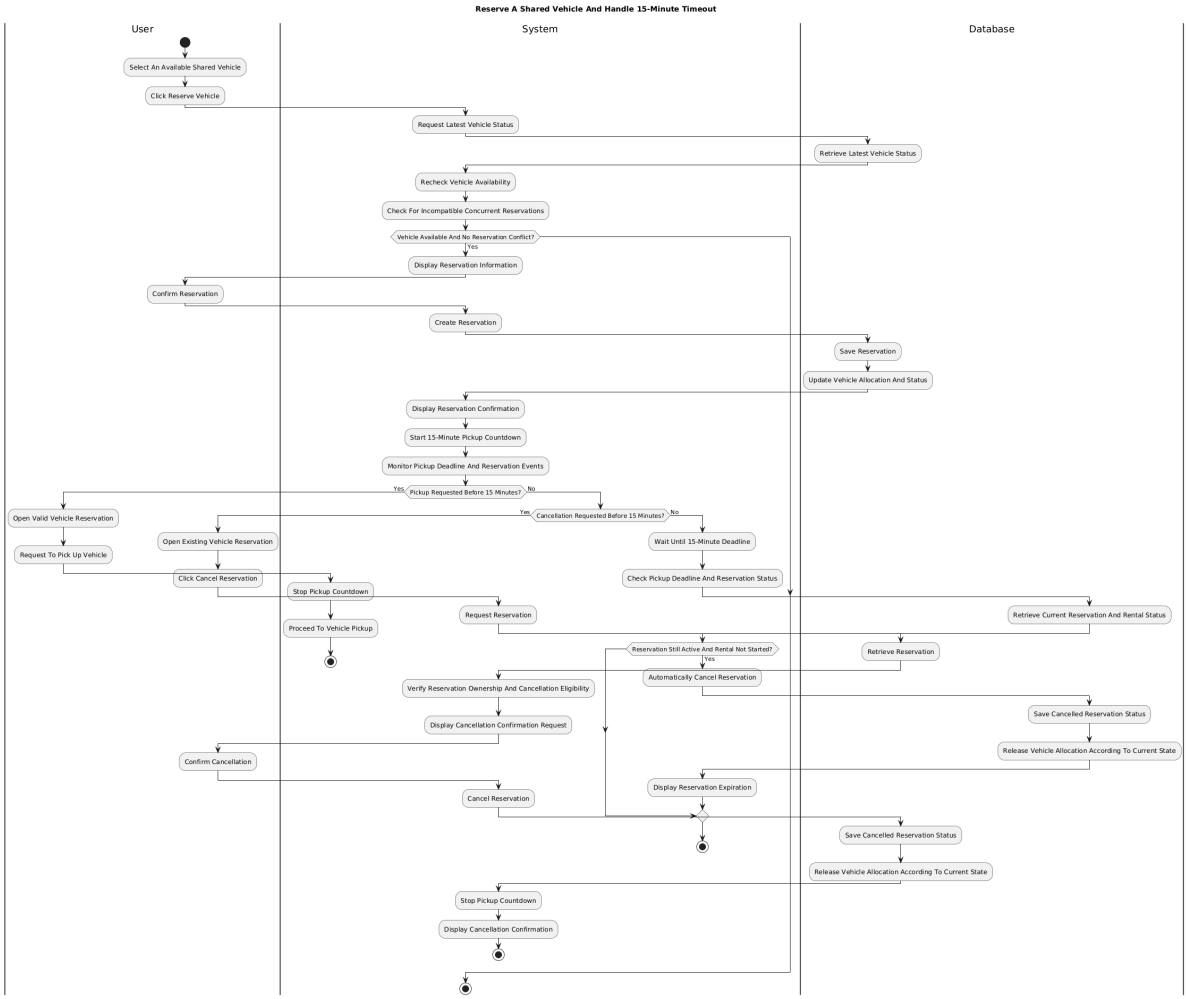


Figure 49: Activity diagram for vehicle reservation and hold timeout (UC-R03).

3.2.2 UC-R03, UC-R04, and UC-R05 — Complete a Shared-Vehicle Rental Journey

The student reserves and unlocks a vehicle, uses it for a trip, and returns it to a Hub. The system records the state changes and closes the trip after return checks are complete.

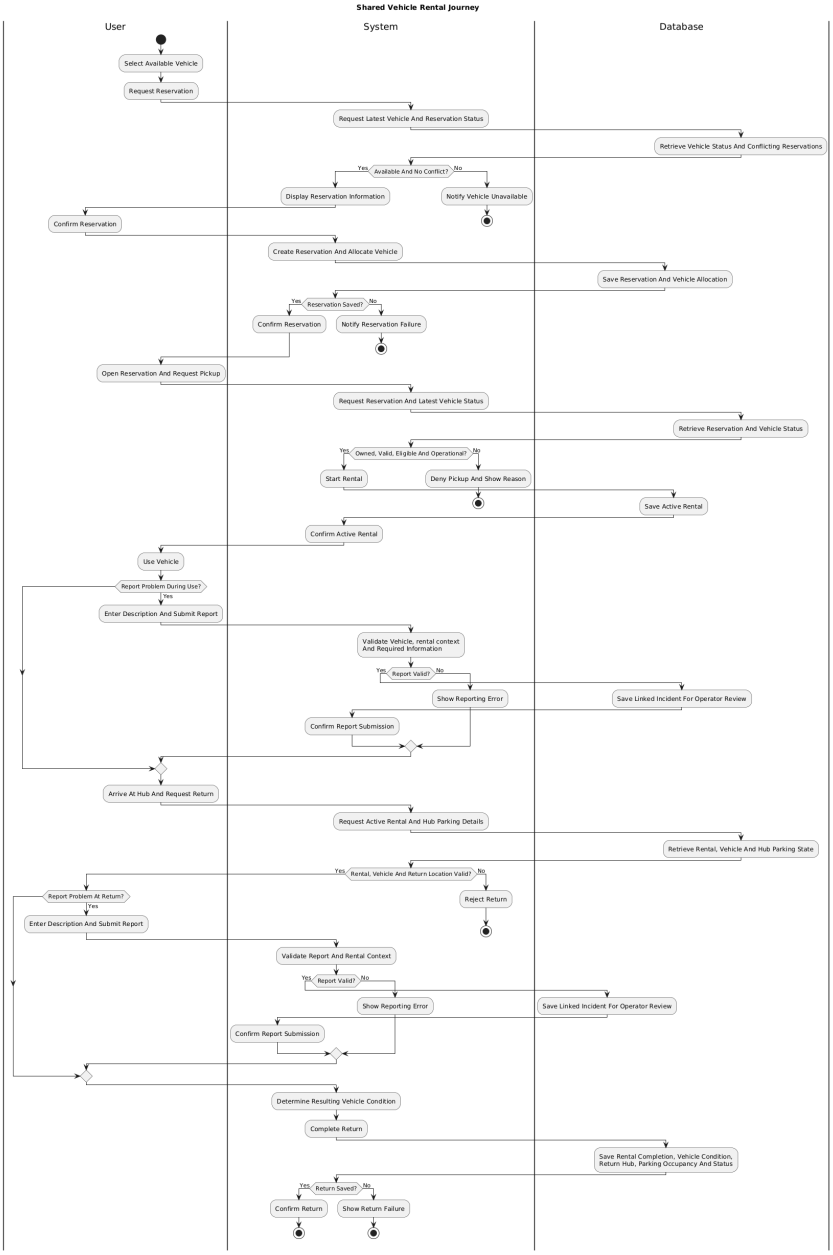


Figure 50: Activity diagram for the shared-vehicle rental journey (UC-R03, UC-R04, UC-R05).

3.3 Parking and Private EV

3.3.1 UC-P01 — Reserve a Parking Space

The student chooses a Hub and a parking interval. The system checks capacity in each 15-minute block and either records the complete reservation or returns an unavailable result.

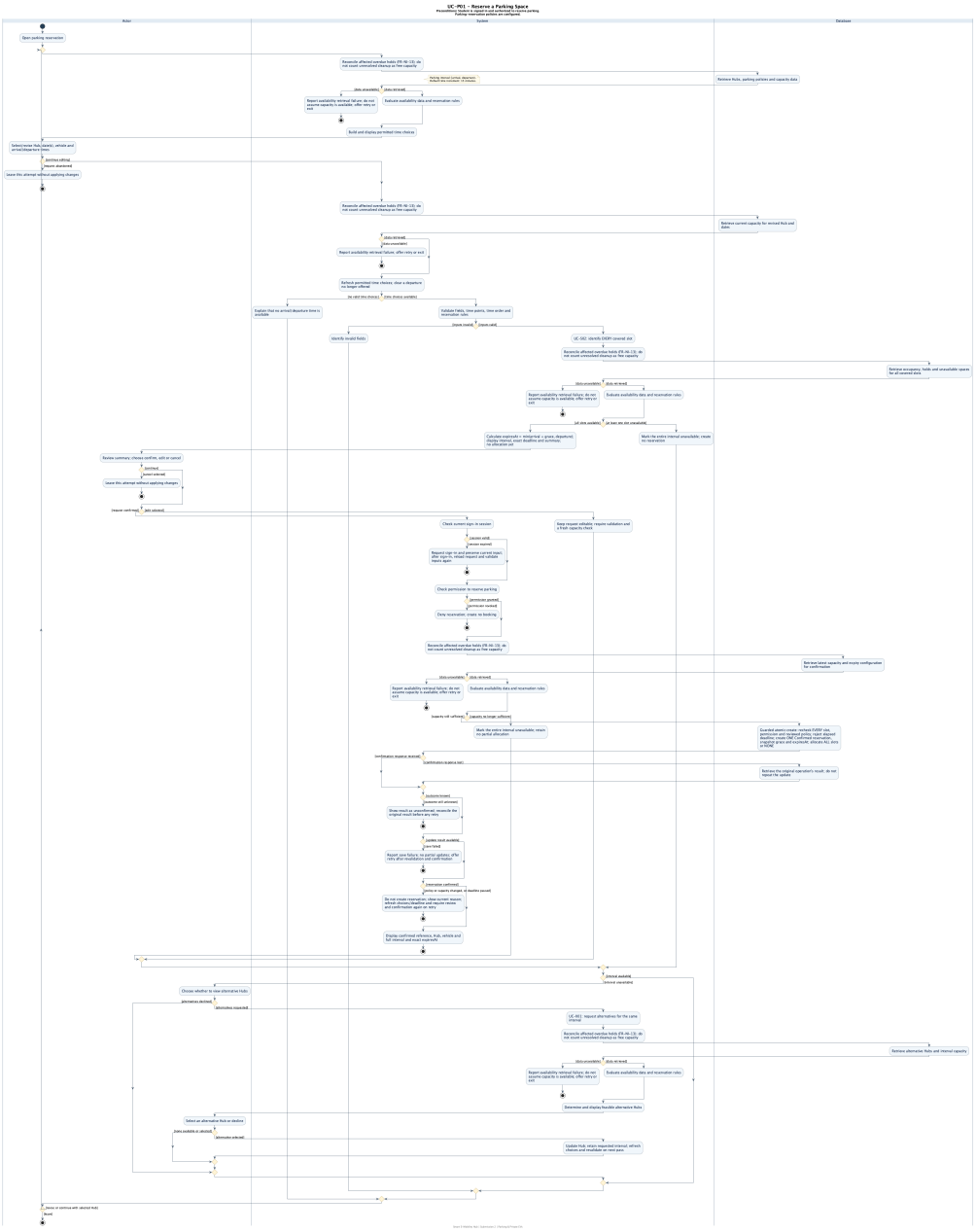


Figure 51: Activity diagram for reserving a parking space (UC-P01).

3.3.2 UC-P02 — Submit a Private-EV Charging Request

The student selects a parking booking and provides charging requirements. The system validates the booking and charging details before recording the request.

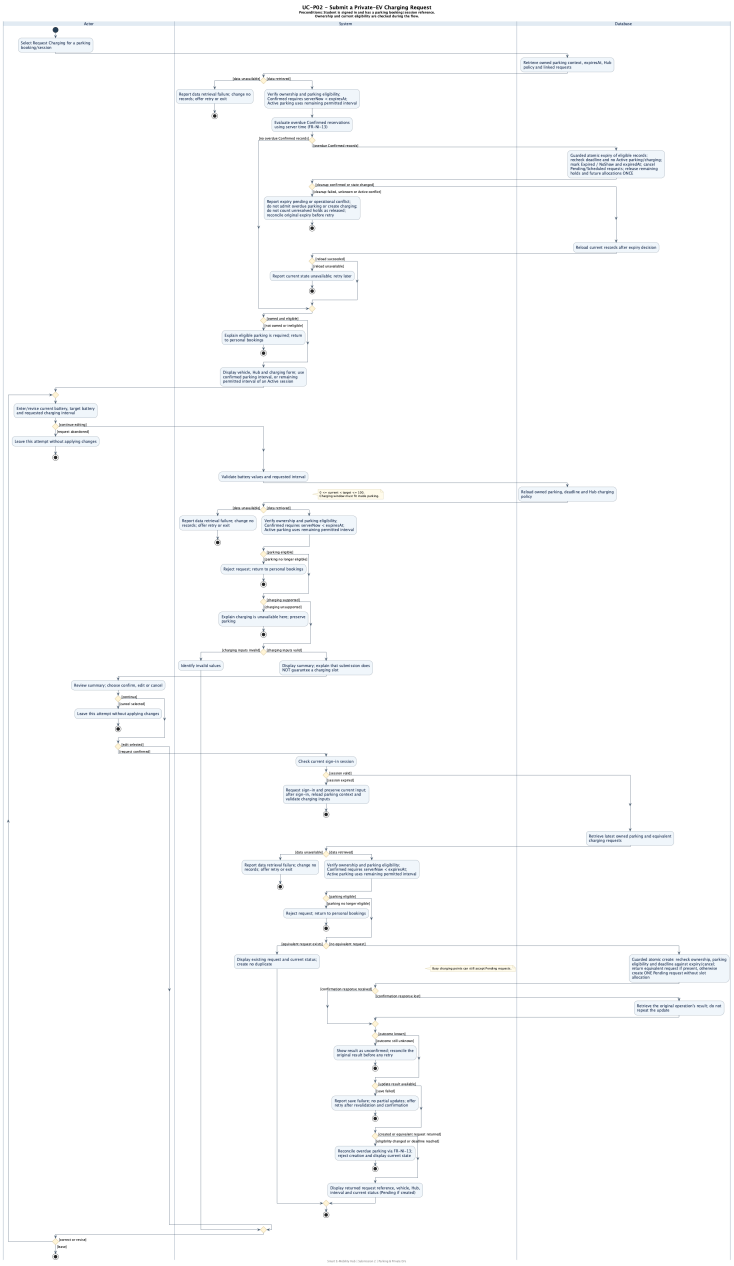


Figure 52: Activity diagram for submitting a private-EV charging request (UC-P02).

3.3.3 UC-P03 — View Parking and Charging Bookings

The student opens the personal booking list. The system retrieves parking and charging records and presents their state and available actions.

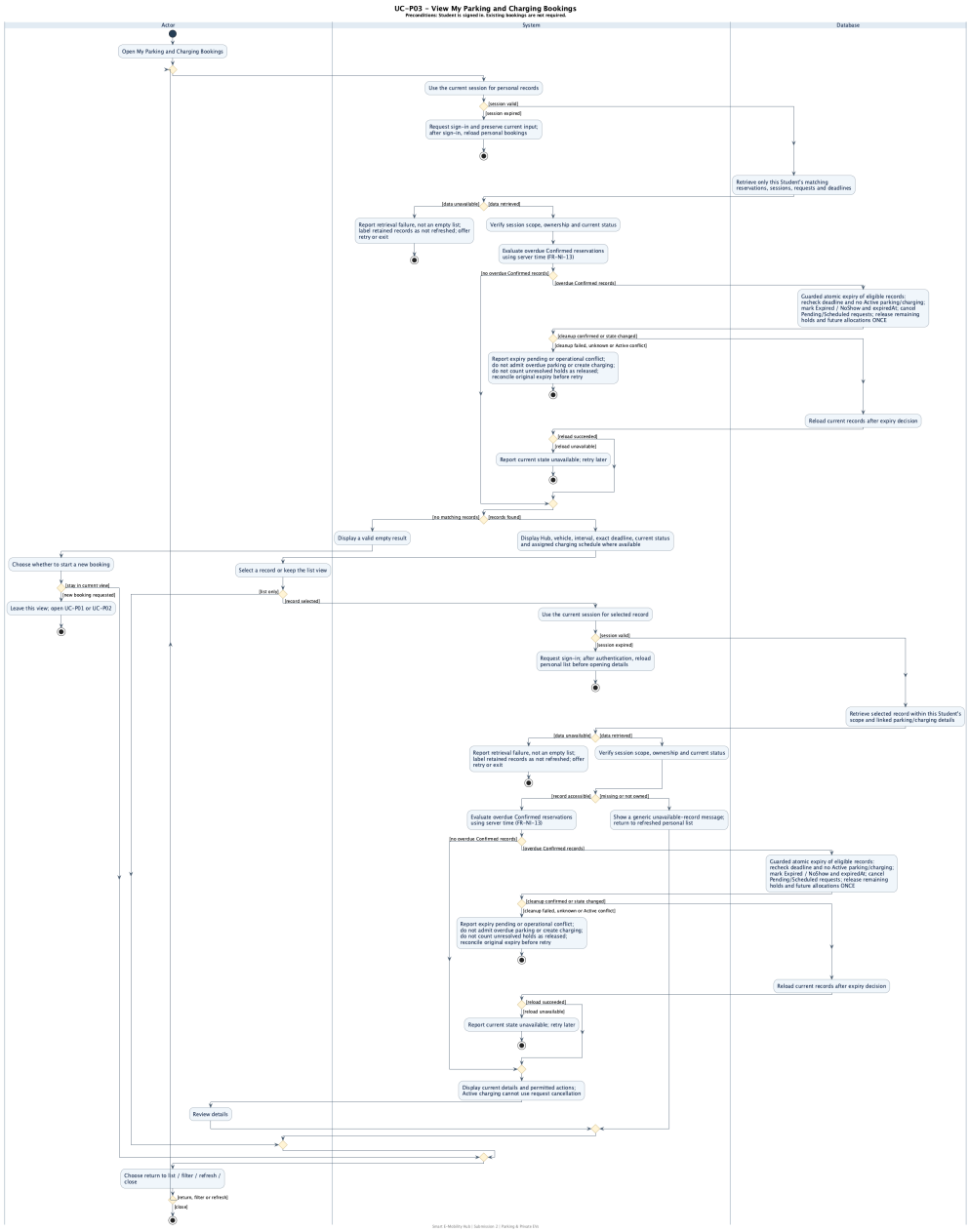


Figure 53: Activity diagram for viewing parking and charging bookings (UC-P03).

3.3.4 UC-P04 — Cancel a Parking Reservation

The student requests cancellation of a reservation. The system checks its current state and applicable cancellation conditions, then records the cancellation and releases capacity when allowed.

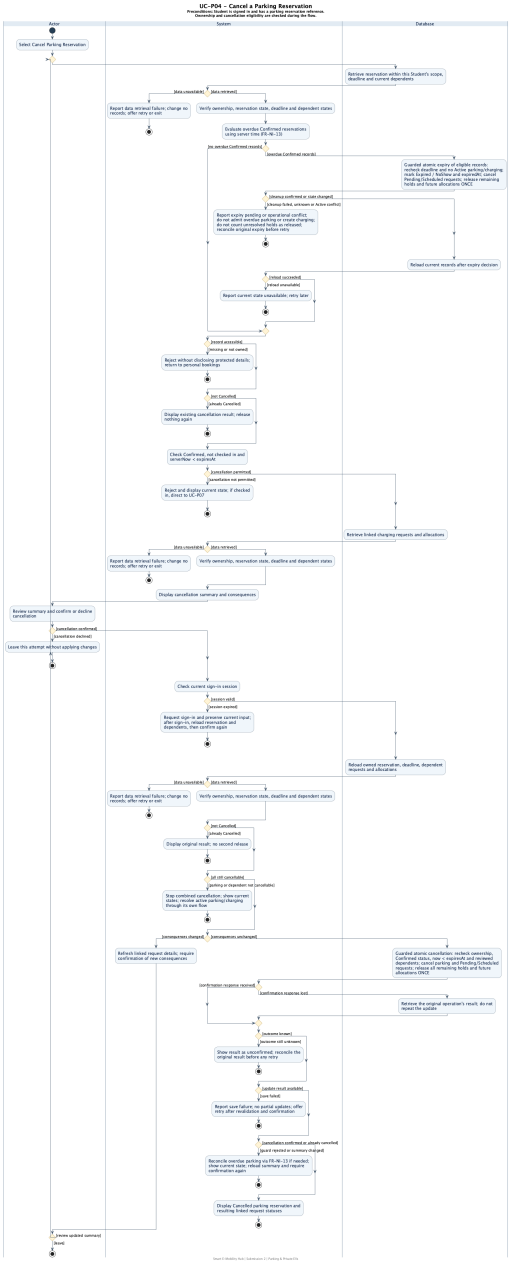


Figure 54: Activity diagram for cancelling a parking reservation (UC-P04).

3.3.5 UC-P05 — Cancel a Charging Request

The student requests cancellation of a charging request. The system validates whether the request can be cancelled and updates the request and associated schedule accordingly.

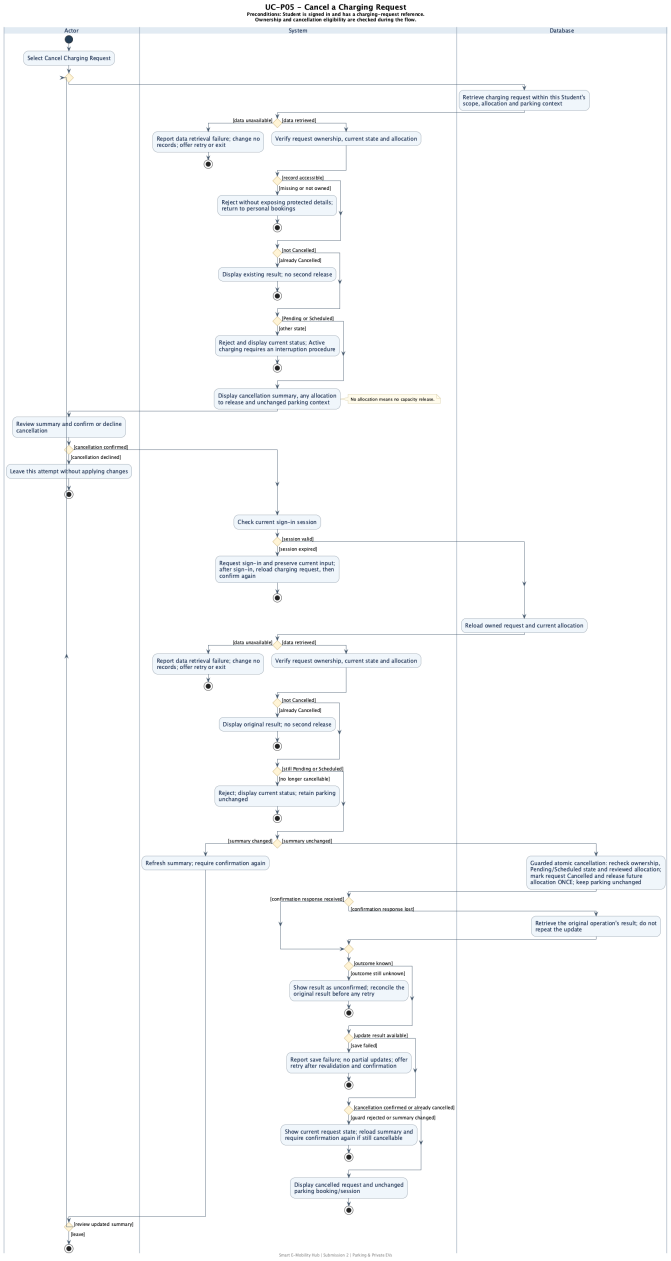


Figure 55: Activity diagram for cancelling a charging request (UC-P05).

3.3.6 UC-P06 — Check In a Private EV

The student presents the parking ticket at the Hub. The system validates the reservation, records the vehicle's arrival, and updates actual occupancy.

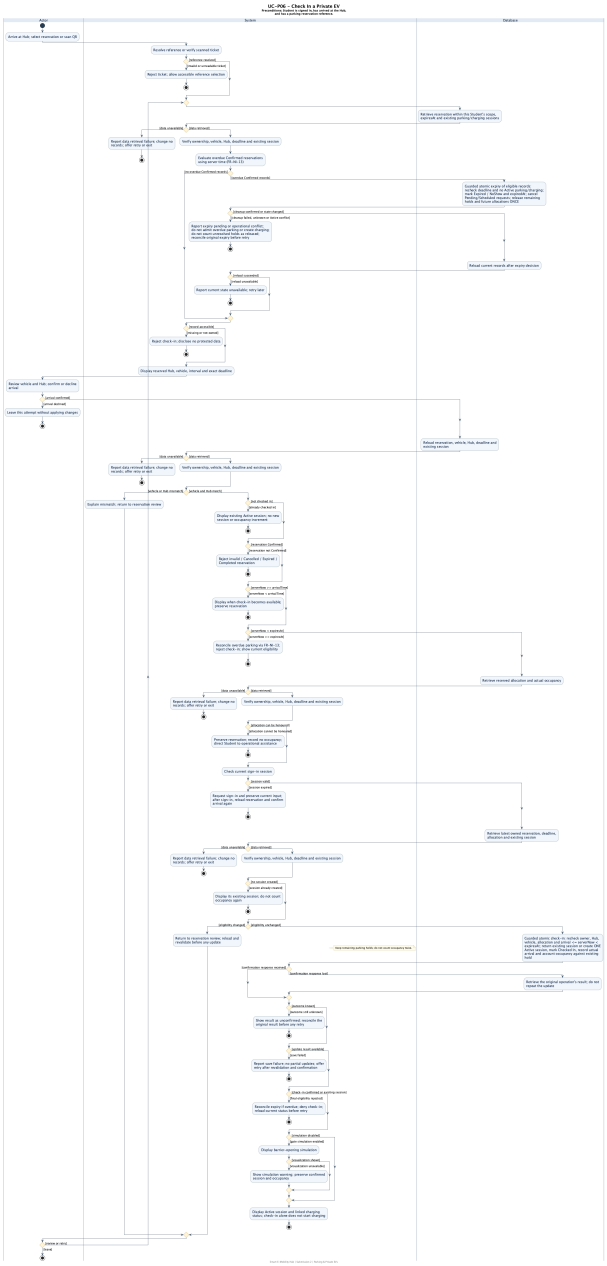


Figure 56: Activity diagram for checking in a private EV (UC-P06).

3.3.7 UC-P07 — Check Out a Private EV

The student checks out when leaving the Hub. The system records departure, handles active charging or overstay conditions, and releases the occupied space.

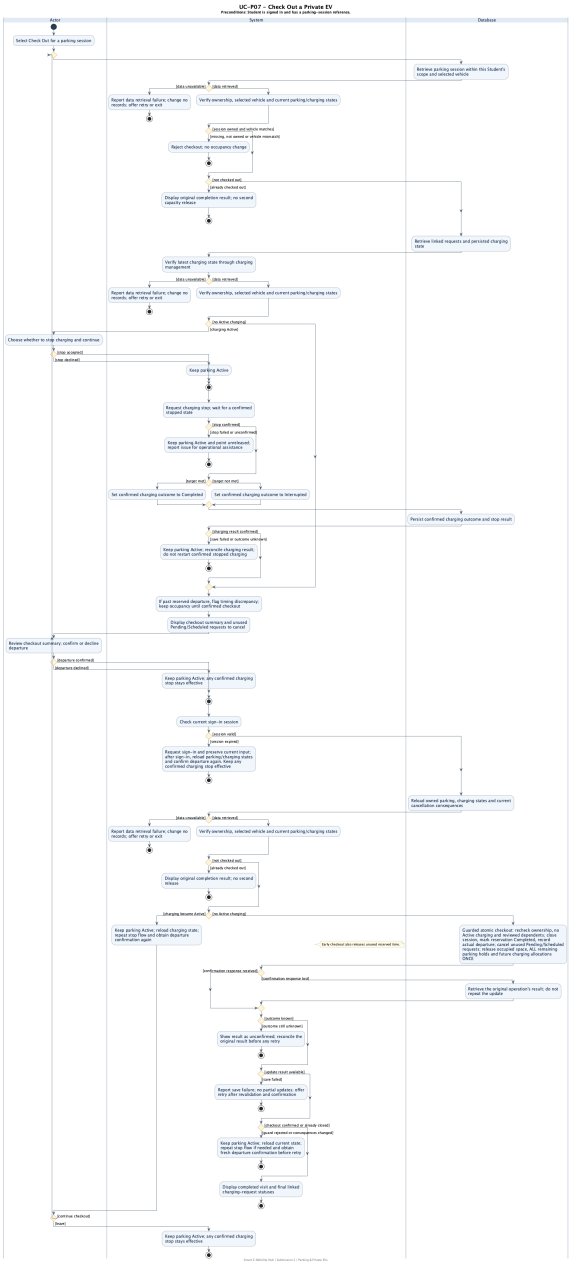


Figure 57: Activity diagram for checking out a private EV (UC-P07).

3.4 Fleet Monitoring and Incident Handling

3.4.1 UC-F01 — Monitor the Mobility Hub Network

The operator opens the dashboard and the system checks access before retrieving Hub, vehicle, charging, and incident information. It calculates network indicators and freshness, then presents the overview and alerts.

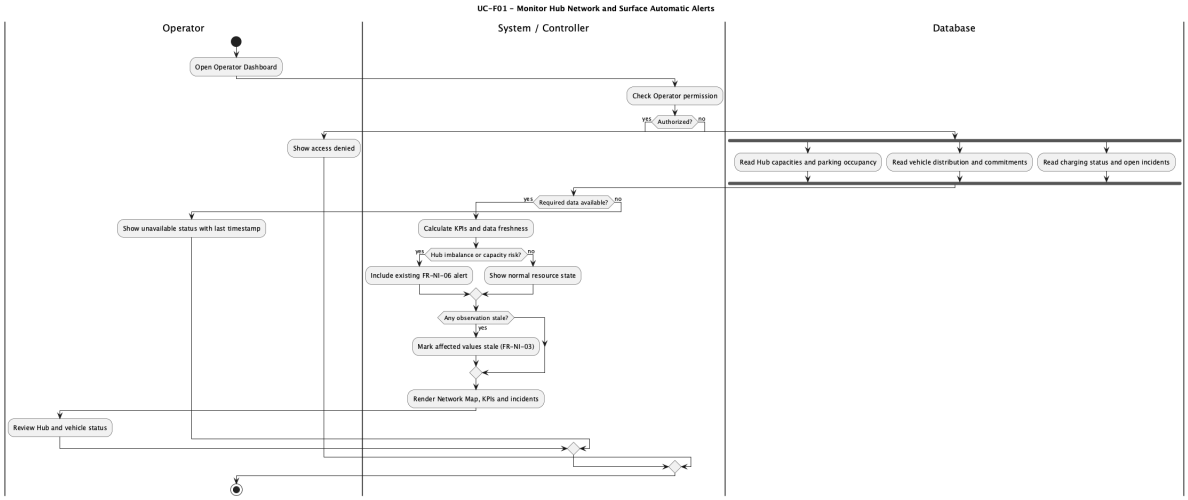


Figure 58: Activity diagram for monitoring the Hub network (UC-F01).

3.4.2 UC-F04 and UC-F05 — Plan and Complete Vehicle Redistribution

The operator proposes a transfer and reviews feasibility before creating an assignment. Later, confirmed arrivals update the order and Hub occupancy while unconfirmed vehicle movements remain unchanged.

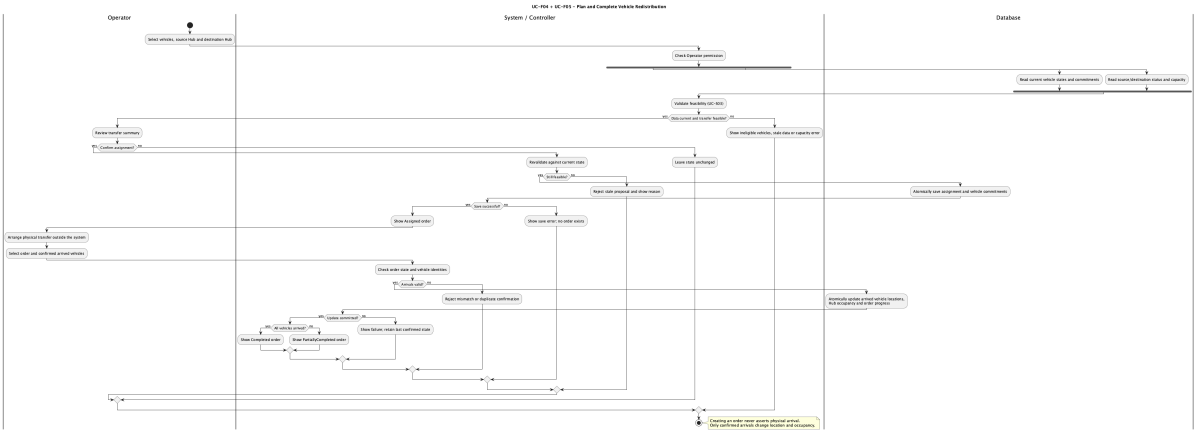


Figure 59: Activity diagram for planning and completing vehicle redistribution (UC-F04, UC-F05).

3.4.3 UC-I02 — Handle a Vehicle Failure

The system restricts unsafe allocation while the operator investigates the incident and addresses affected commitments. Vehicle eligibility is restored only after repair evidence and safety checks are recorded.

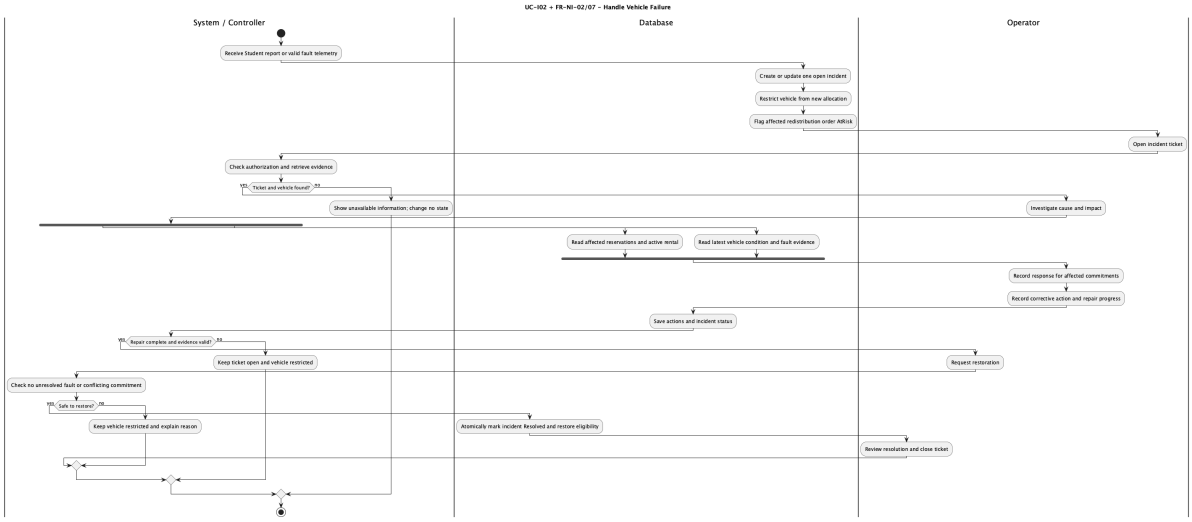


Figure 60: Activity diagram for handling, isolating, and restoring a vehicle (UC-I02).

3.5 Smart Charging Scheduling

3.5.1 UC-C02 — Generate a Charging Schedule

The system loads valid requests and available ports, then processes requests by arrival time. For identical arrival timestamps, the lower-battery vehicle is considered first before the system assigns feasible slots or leaves requests unscheduled.

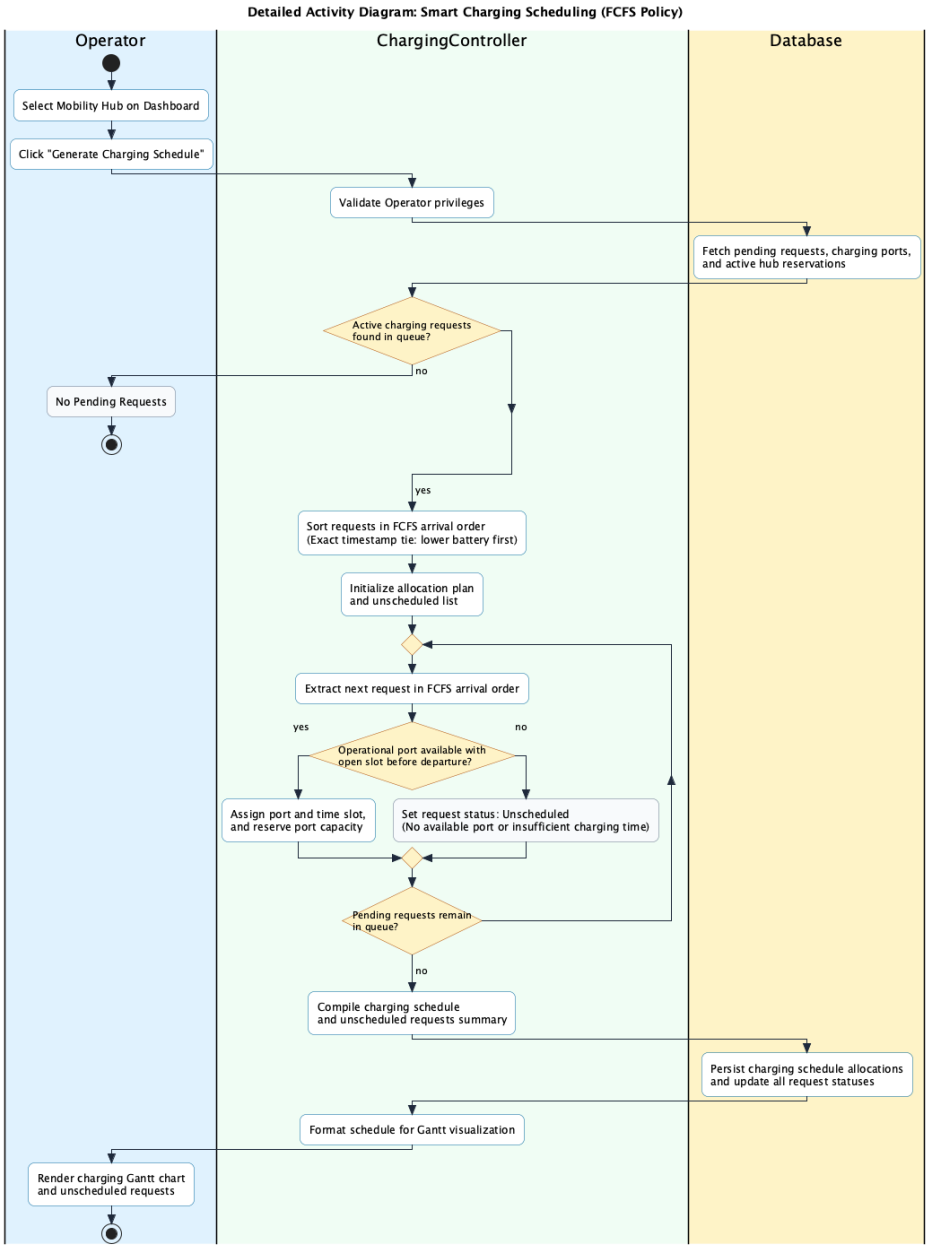


Figure 61: Activity diagram for generating an FCFS charging schedule (UC-C02).

3.5.2 UC-C03 — Adjust a Charging Schedule

The operator selects a request and proposes a new port or time interval. The system checks availability and conflicts before saving the schedule change.

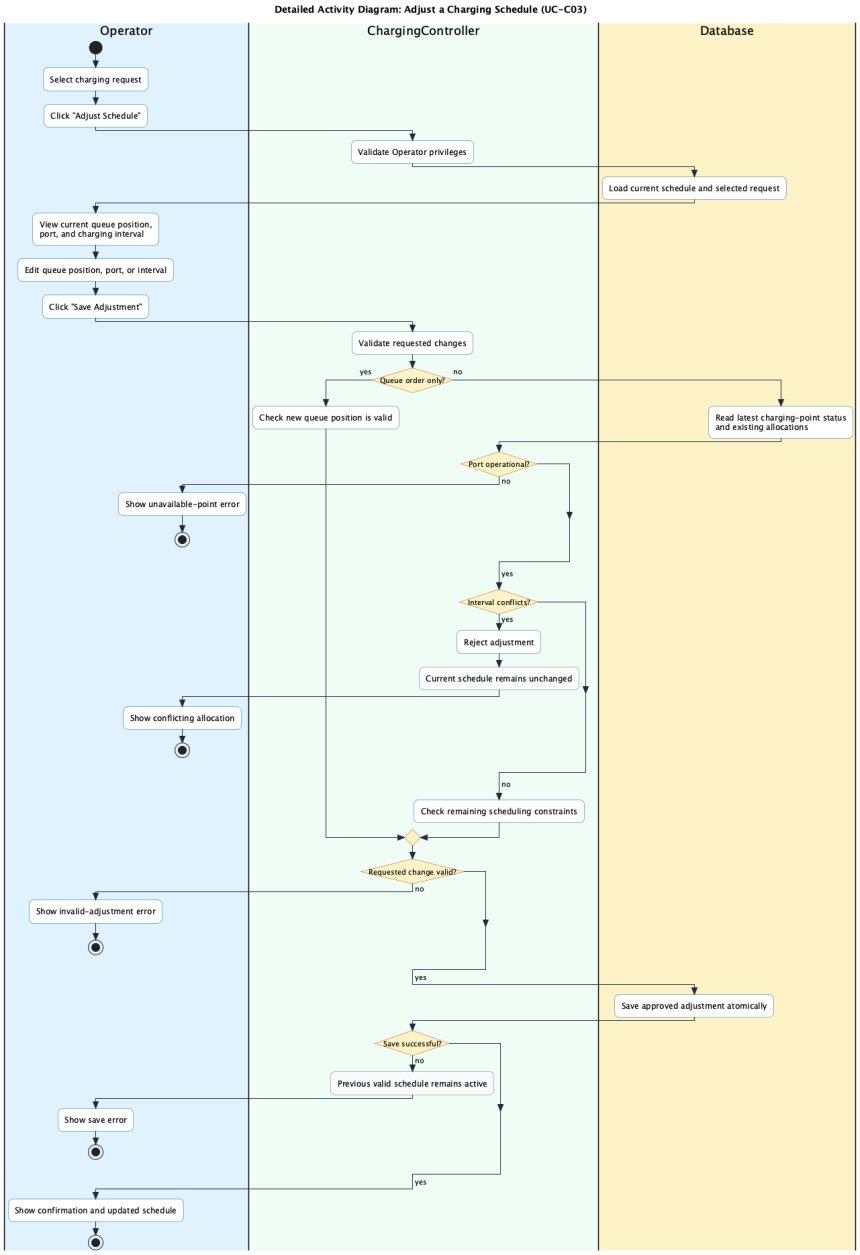


Figure 62: Activity diagram for adjusting a schedule and checking conflicts (UC-C03).

3.6 What-if Simulation

3.6.1 UC-W02 — Run a What-if Simulation

The operator runs a validated scenario against a timestamped network snapshot. The system simulates changes in an isolated state, evaluates their effects, and stores recommendations without modifying live operations.

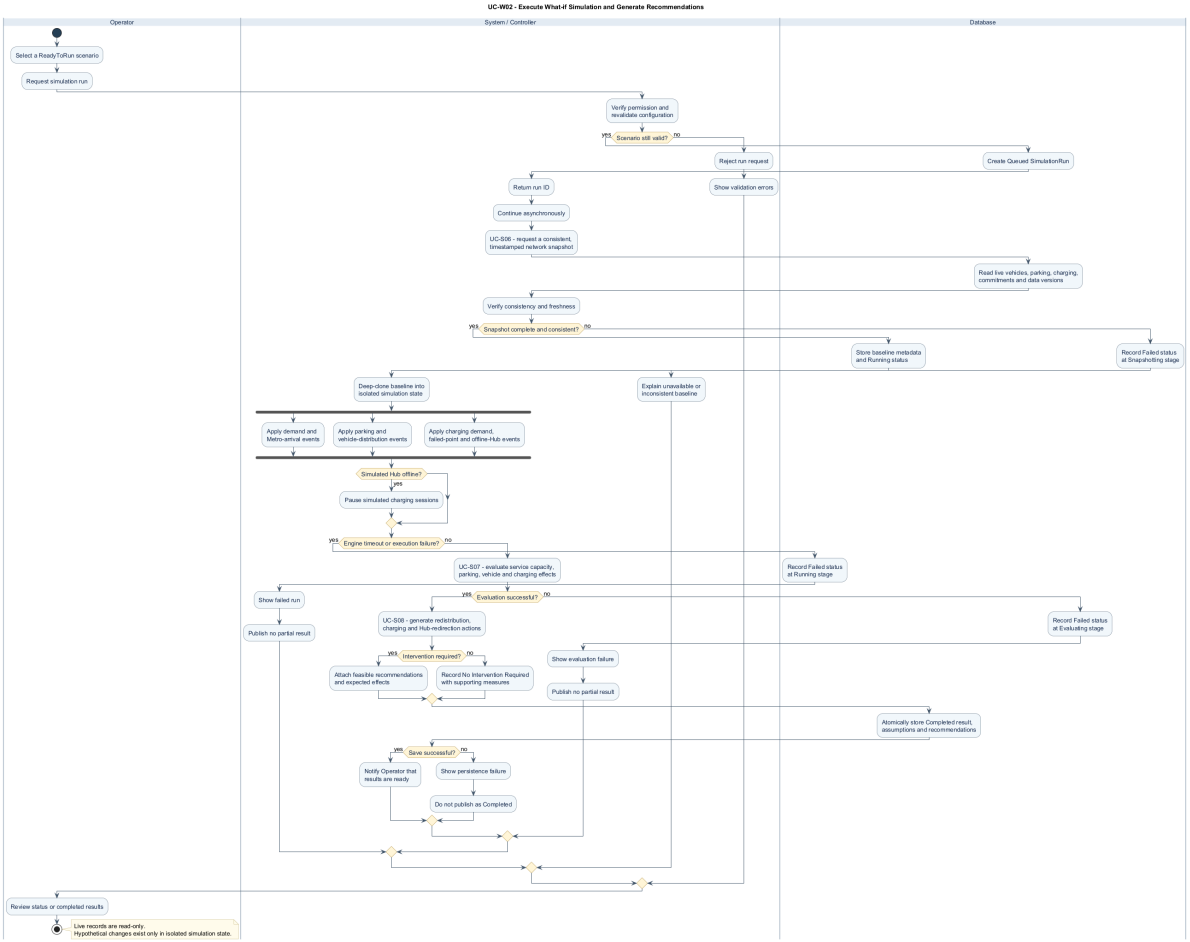


Figure 63: Activity diagram for running an isolated what-if simulation (UC-W02).

3.6.2 UC-W05 — Compare What-if Scenarios

The operator selects completed runs and the system checks whether their measures and context are comparable. It presents compatible differences with warnings or explains why a meaningful comparison cannot be produced.

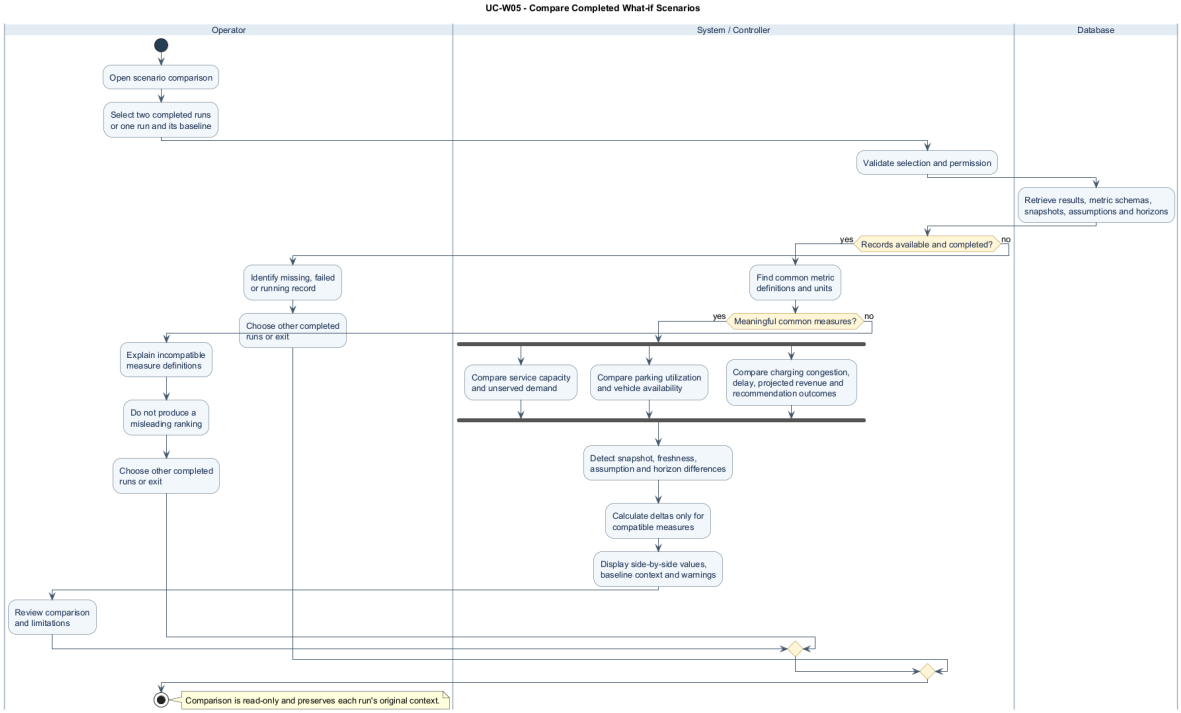


Figure 64: Activity diagram for comparing completed what-if scenarios (UC-W05).

3.7 External Telemetry

3.7.1 UC-T01 — Update Vehicle Telemetry

A telemetry event reports vehicle condition and status. The system validates and records current observations while avoiding duplicate or stale updates.

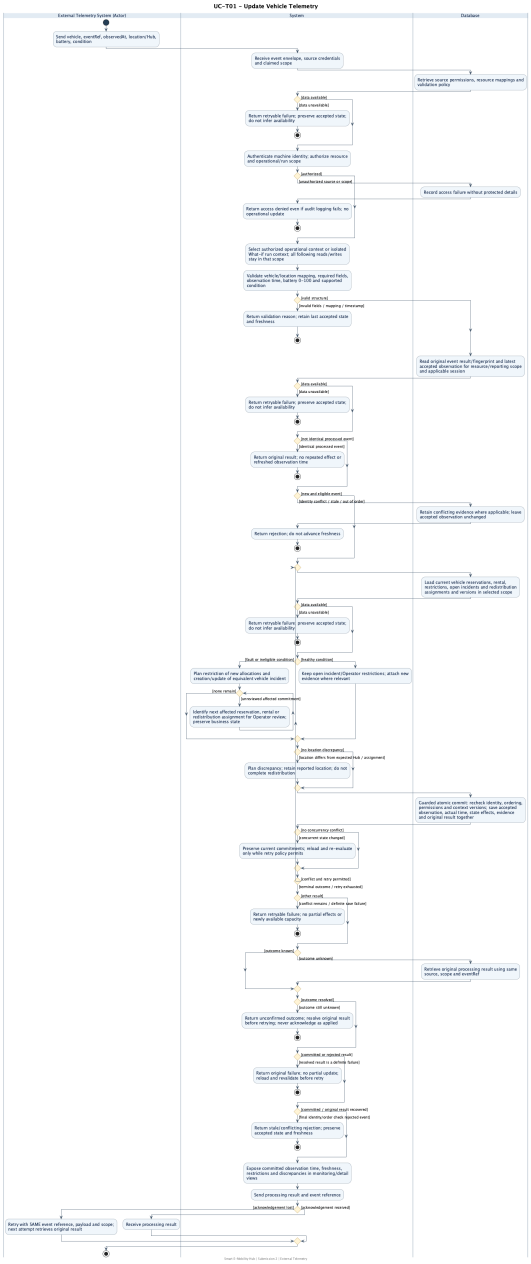


Figure 65: Activity diagram for updating vehicle telemetry (UC-T01).

3.7.2 UC-T02 — Update Parking Occupancy

A parking sensor reports an occupancy change. The system validates the event and reconciles the updated count with reservation and check-in records.

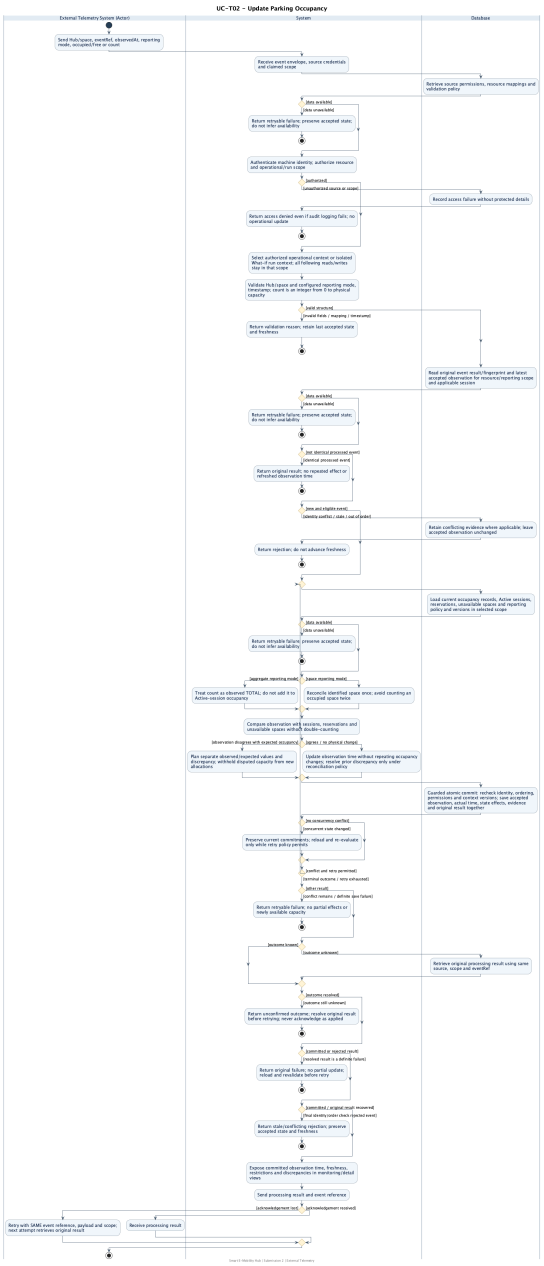


Figure 66: Activity diagram for updating parking occupancy (UC-T02).

3.7.3 UC-T03 — Update Charging-Point Status

A charging point reports a status or session change. The system validates the observation and publishes updated availability for charging workflows.

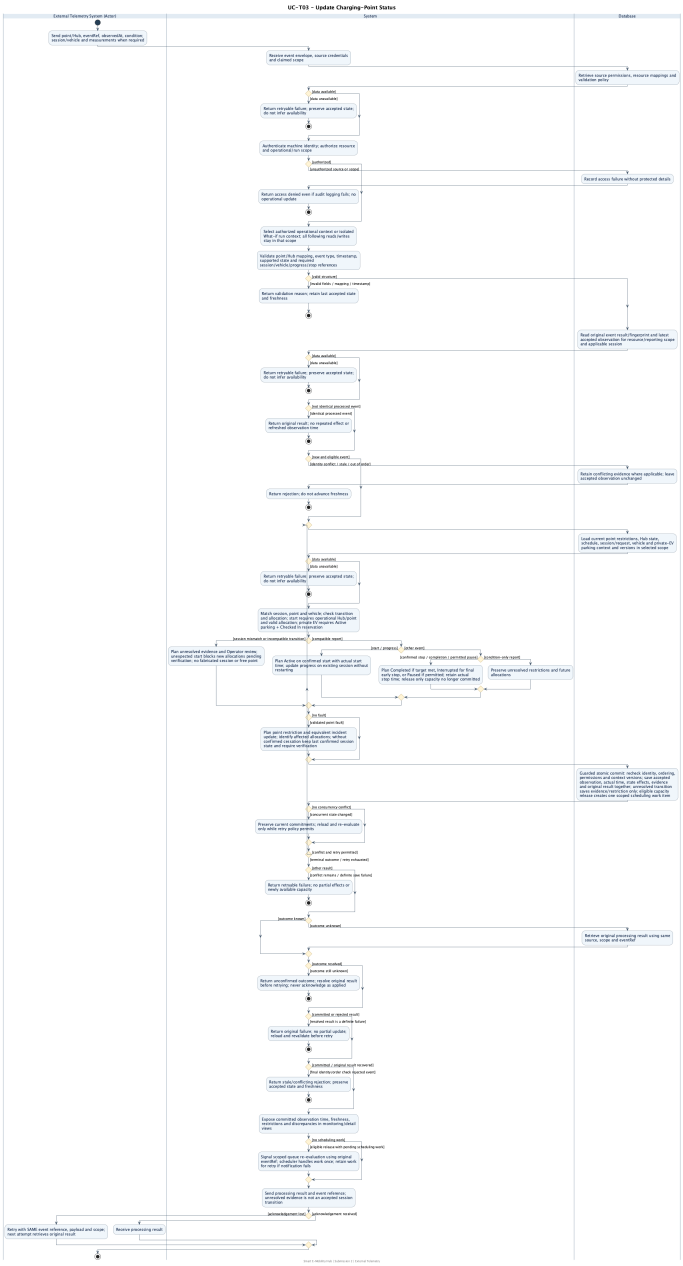


Figure 67: Activity diagram for updating charging-point status (UC-T03).

4 State-Chart Diagrams

4.1 Authentication and Hub & Identity

4.1.1 MobilityHub — Operational Lifecycle

A MobilityHub changes state as its utilization and operational availability change. The model covers normal operation, peak load, maintenance, and offline conditions with guarded recovery.

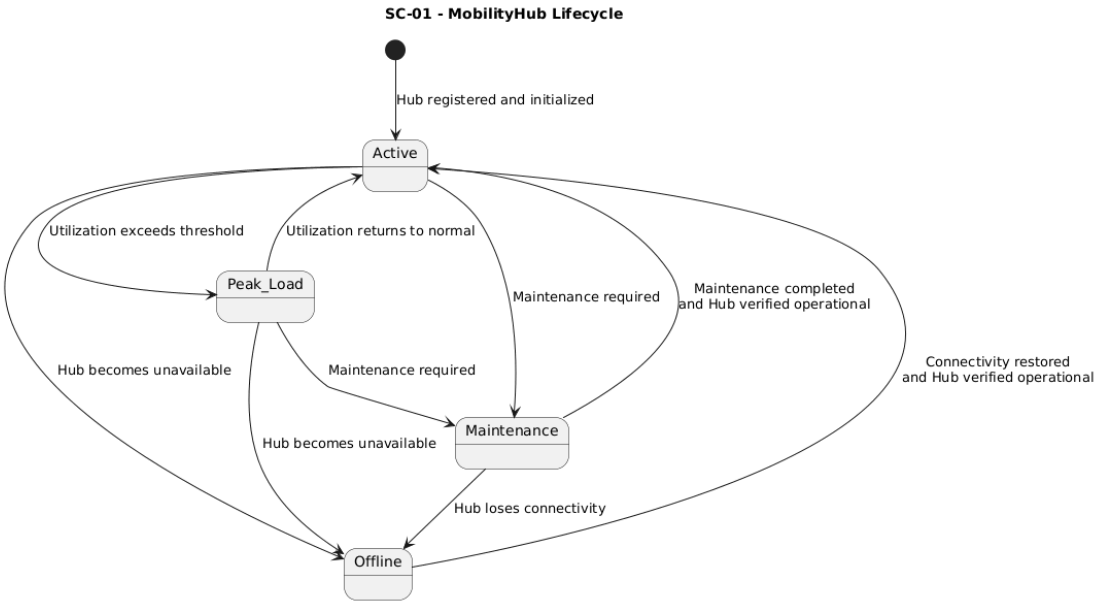


Figure 68: MobilityHub operational lifecycle.

4.2 Vehicle Rental

4.2.1 SharedVehicle — Vehicle Lifecycle

A shared vehicle moves among available, reserved, in-use, maintenance, and redistribution states. Transitions represent reservation, pickup, return, repair, and operational movement.

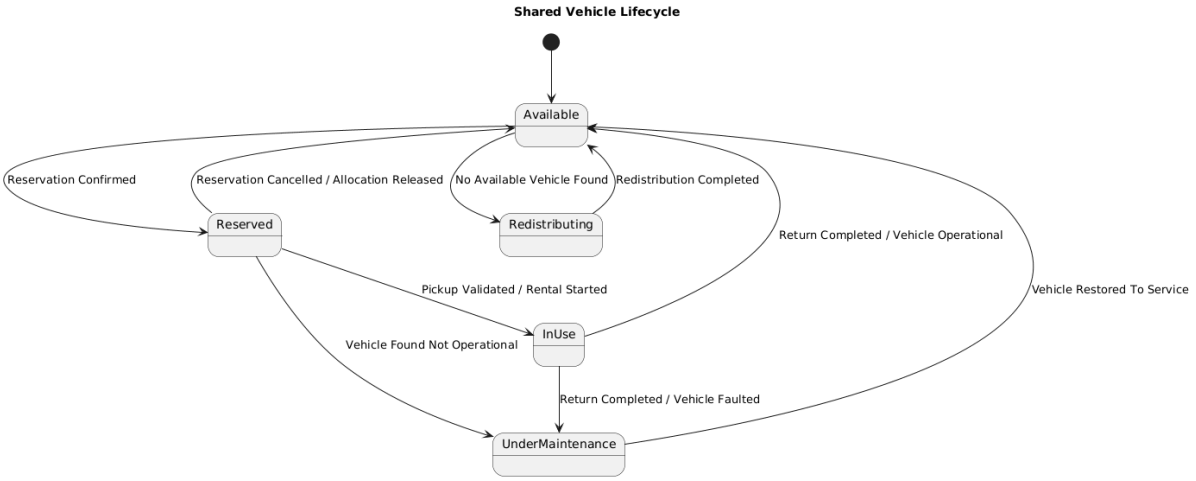


Figure 69: SharedVehicle lifecycle.

4.3 Parking and Private EV

4.3.1 ParkingReservation — Reservation Lifecycle

A parking reservation progresses from creation through confirmation and check-in to completion or another terminal outcome. The model captures cancellation and overstay transitions alongside normal use.

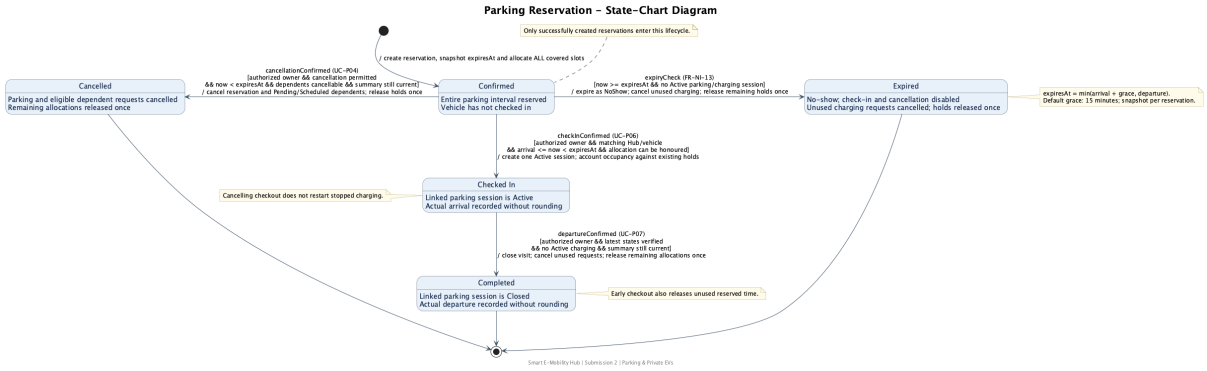


Figure 70: ParkingReservation lifecycle.

4.4 Fleet Monitoring and Incident Handling

4.4.1 IncidentTicket — Incident Lifecycle

An incident ticket advances from reporting through investigation and corrective action to resolution and closure. The vehicle remains restricted until the recovery conditions are verified.

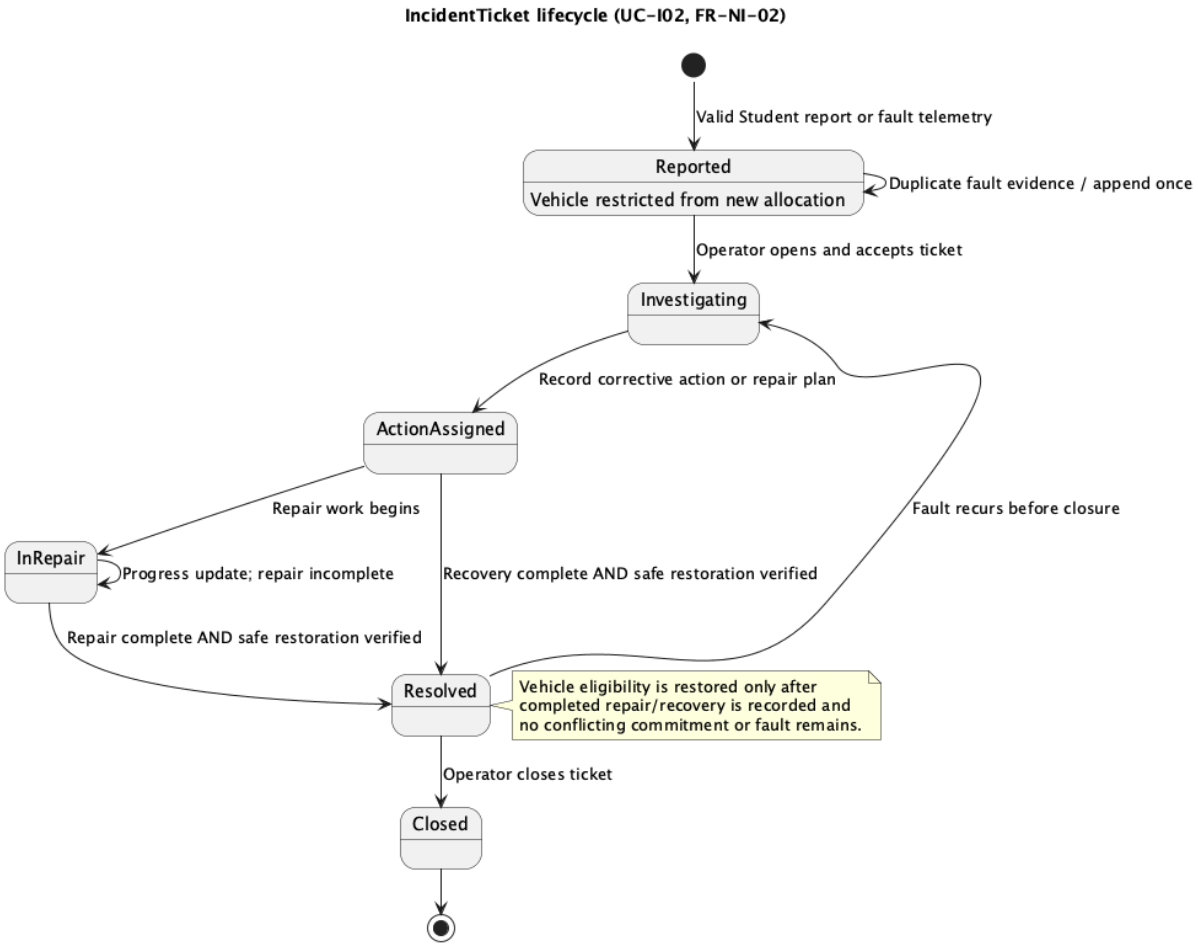


Figure 71: IncidentTicket lifecycle.

4.4.2 RedistributionOrder — Transfer Lifecycle

A redistribution order records assignment, transfer progress, partial arrival, completion, cancellation, and risk handling. Vehicle location and Hub occupancy change only for confirmed arrivals.

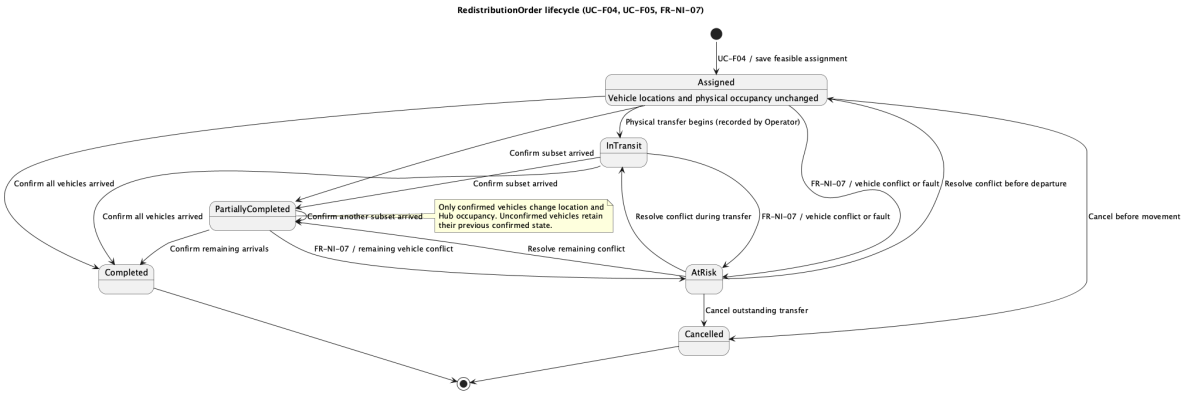


Figure 72: RedistributionOrder lifecycle.

4.5 Smart Charging Scheduling

4.5.1 ChargingRequest — Charging Request Lifecycle

A charging request progresses from pending through scheduling and active charging to completion or interruption. The lifecycle connects queue management, FCFS schedule generation, and manual adjustment.

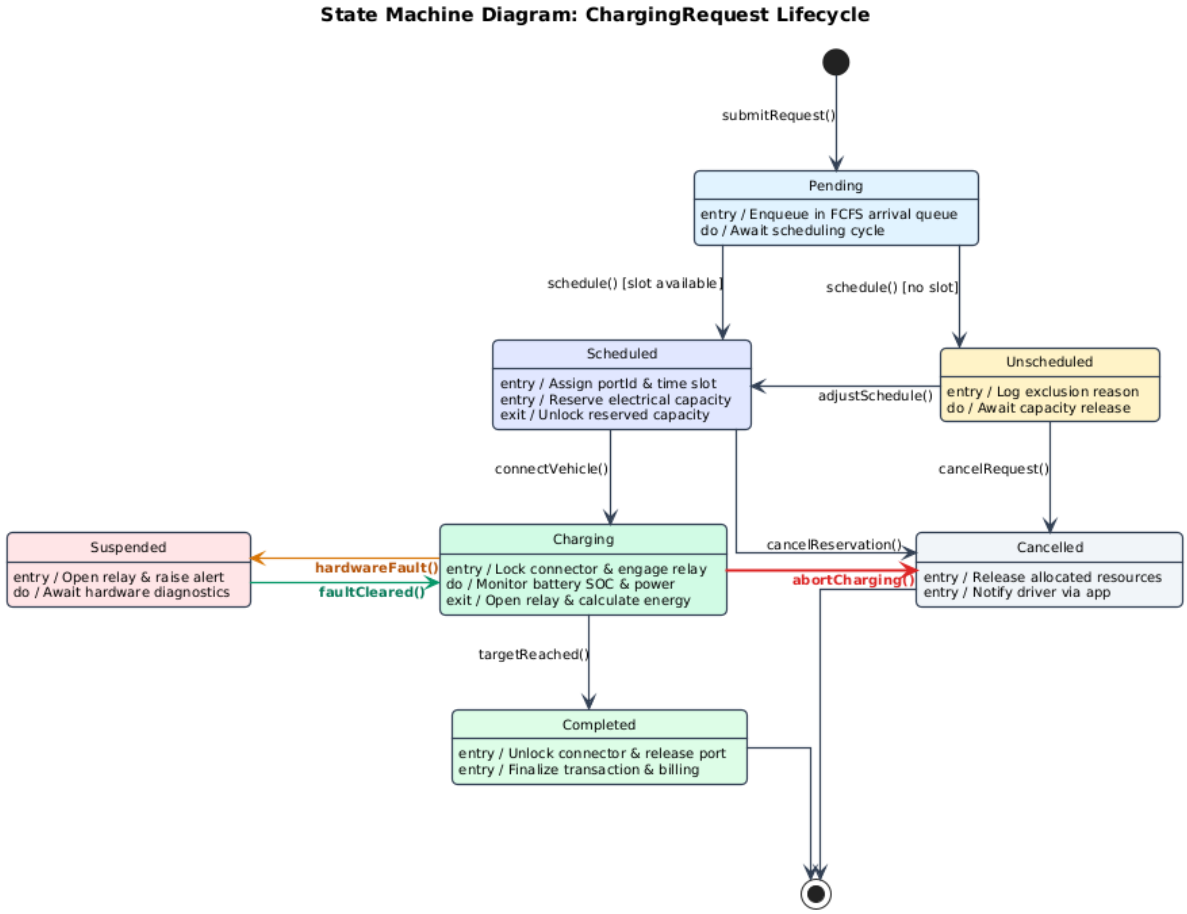


Figure 73: ChargingRequest lifecycle.

4.6 What-if Simulation

4.6.1 SimulationRun — Simulation Lifecycle

A simulation run moves through queueing, snapshot acquisition, execution, evaluation, and recommendation generation. Completed and failed runs retain their context for later review or archiving.

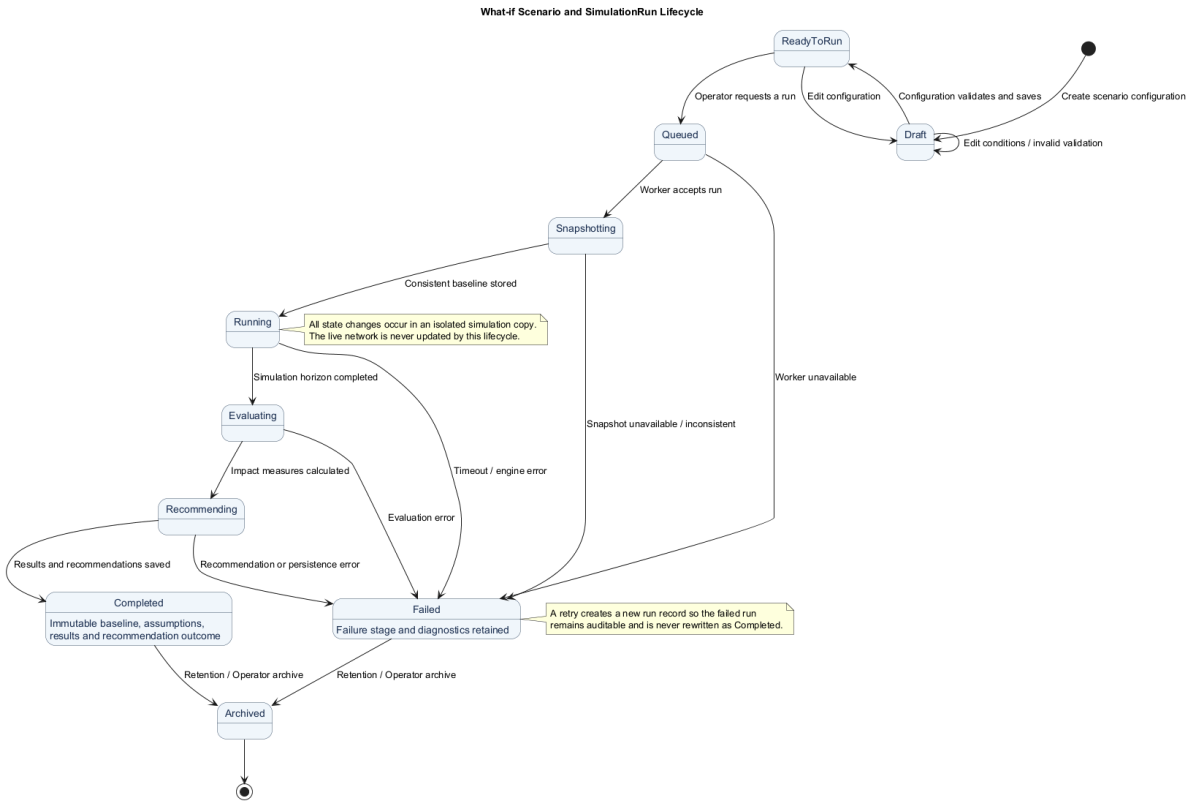


Figure 74: SimulationRun lifecycle.